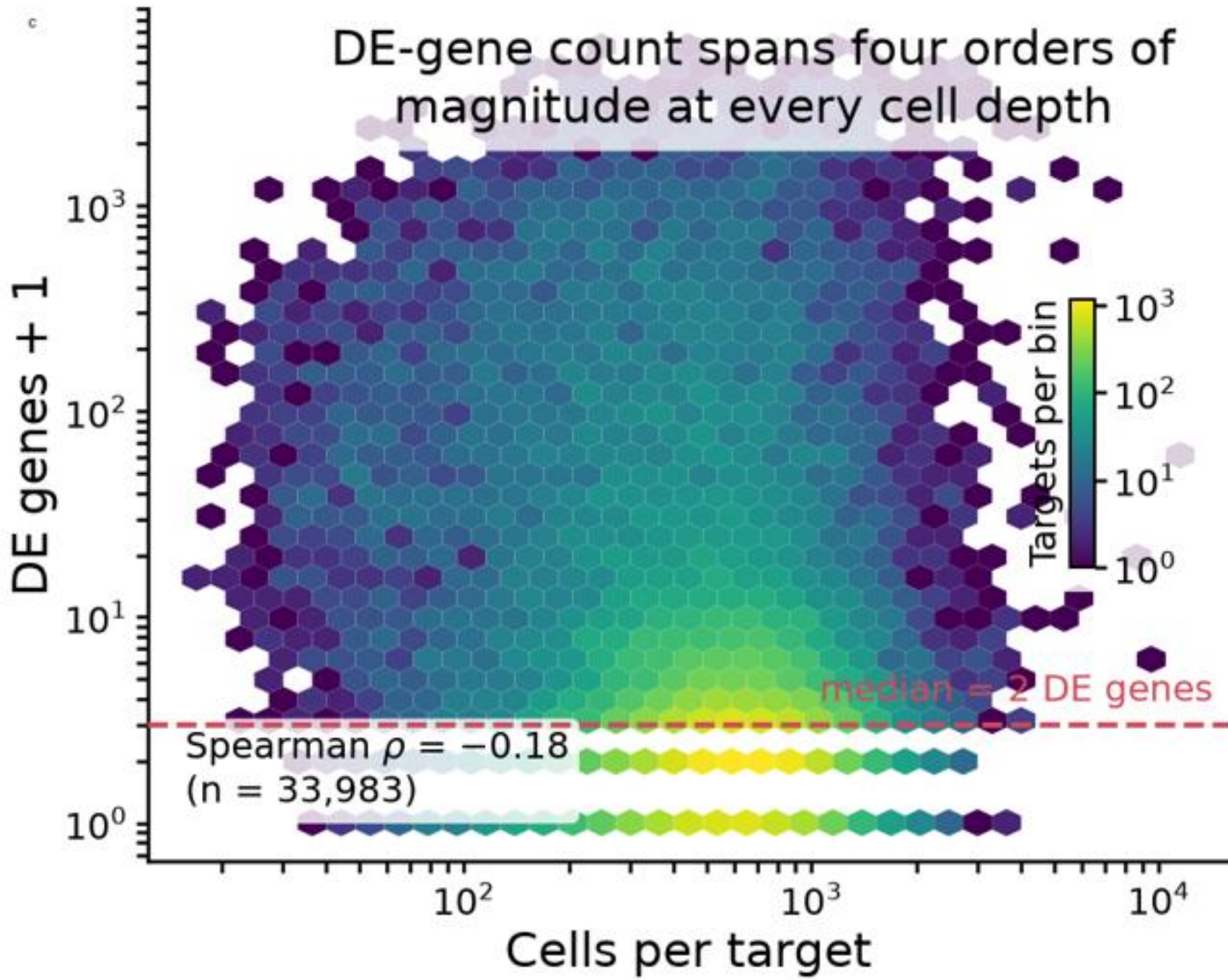


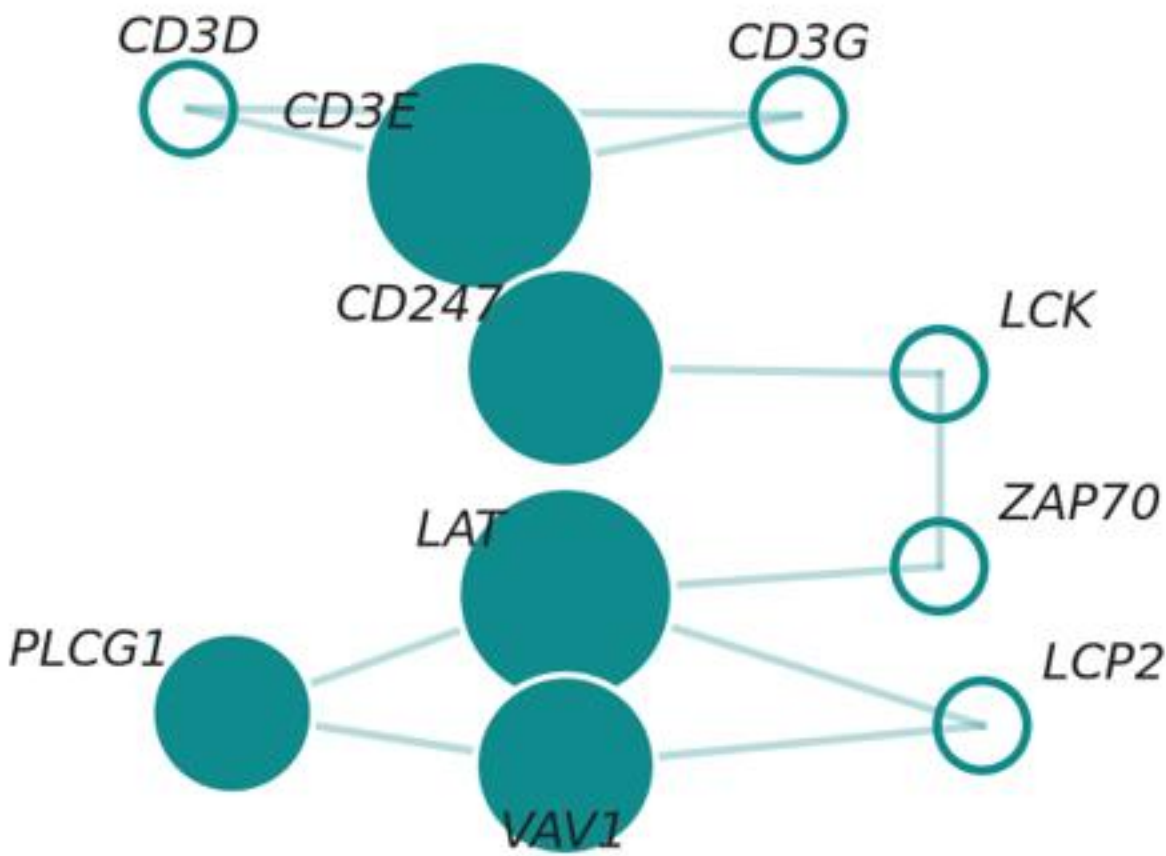
source: DE_stats, genome-scale CRISPRi Perturb-seq in primary human CD4+ T cells

Curation gate distills 33,983 profiles to 2,131 (6.3%)

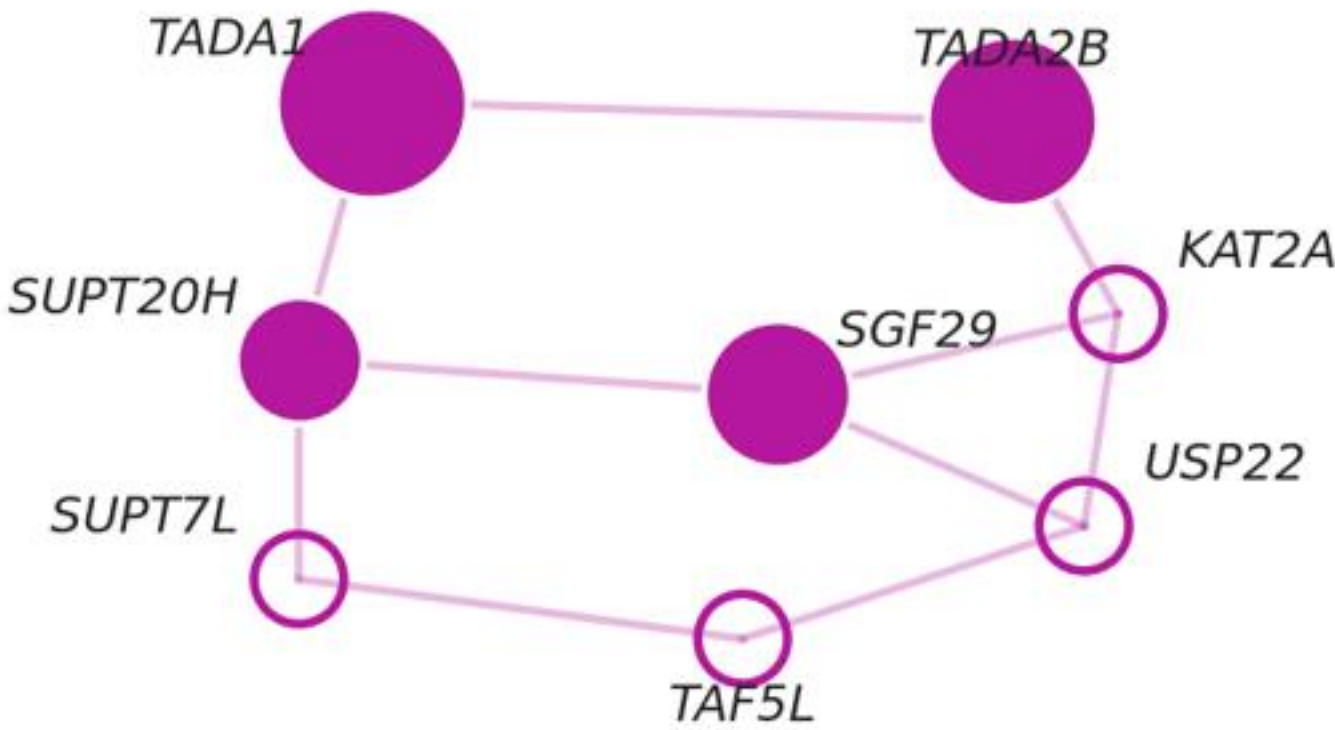
across four filters → 1,235 unique targets



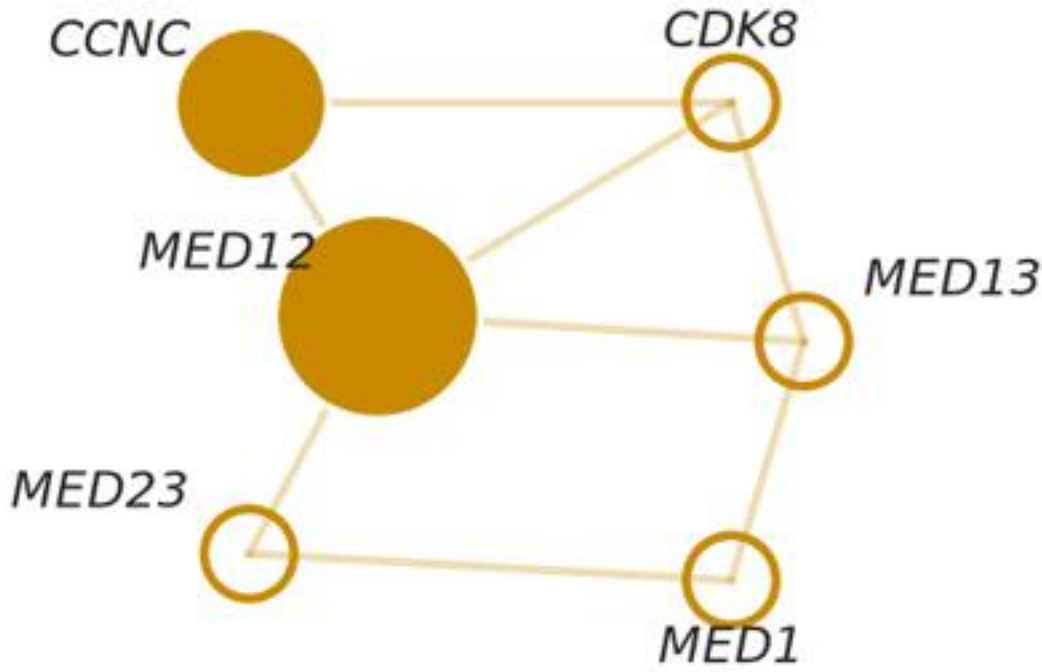
TCR proximal signalling



SAGA complex



Mediator (CDK)

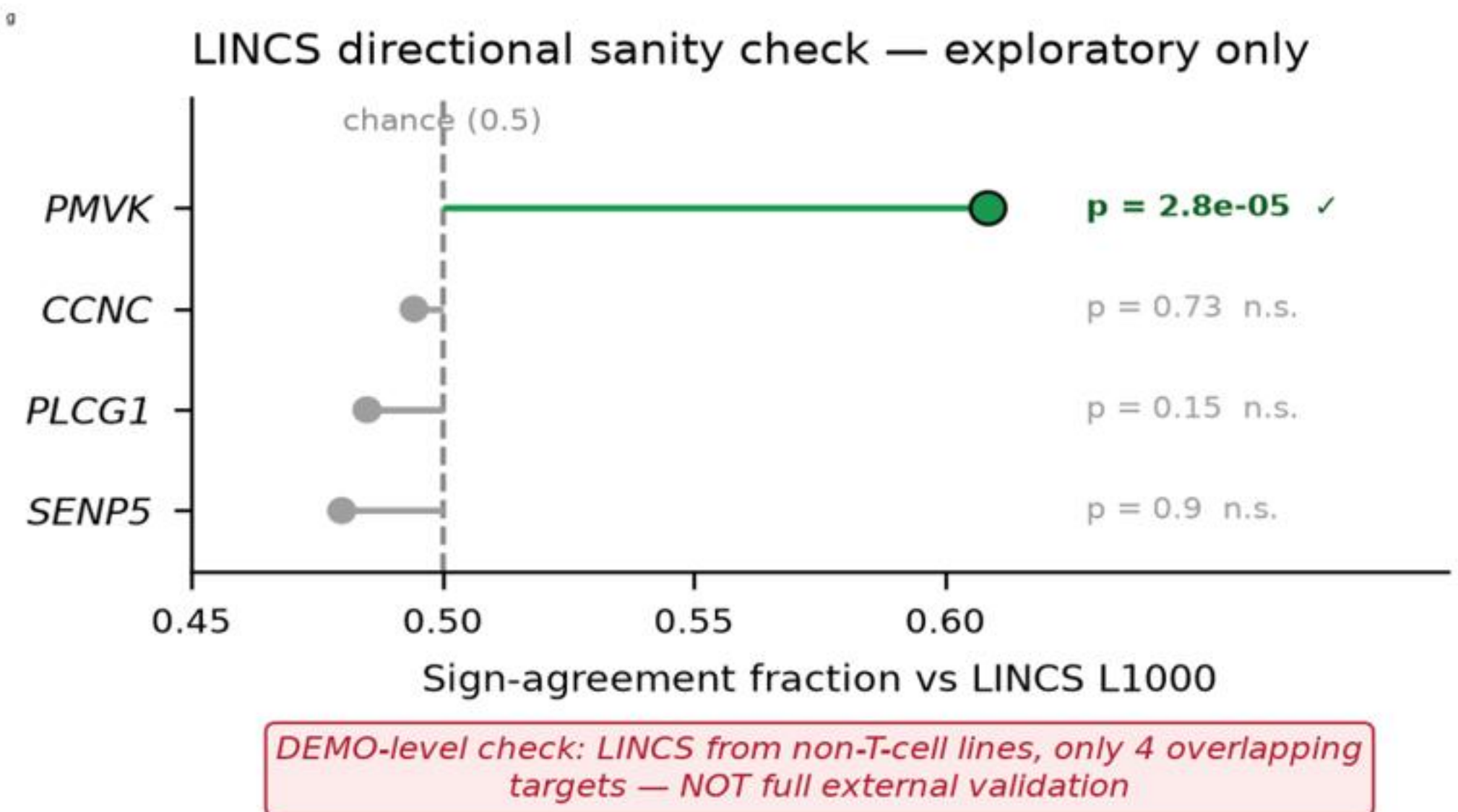
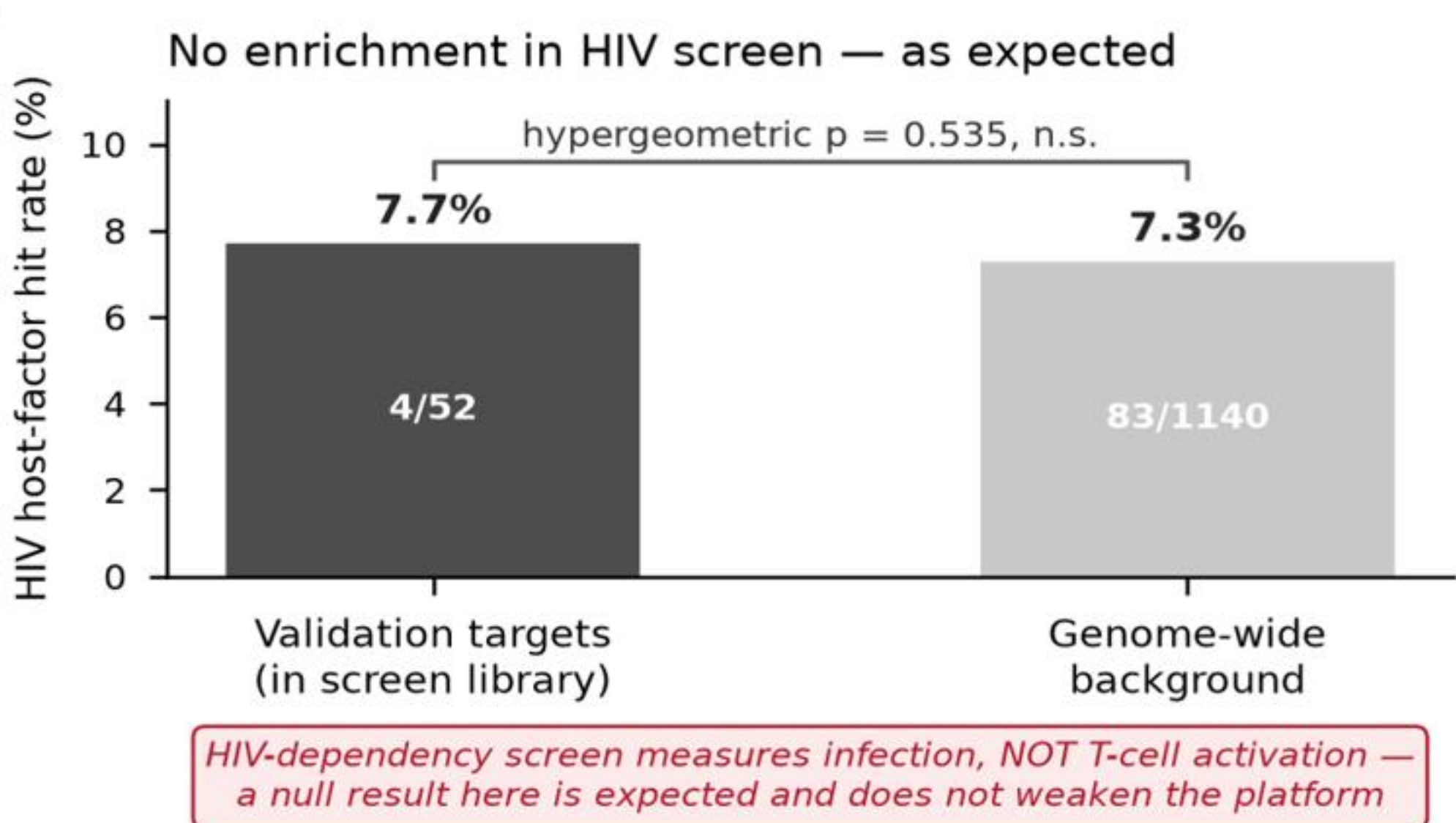
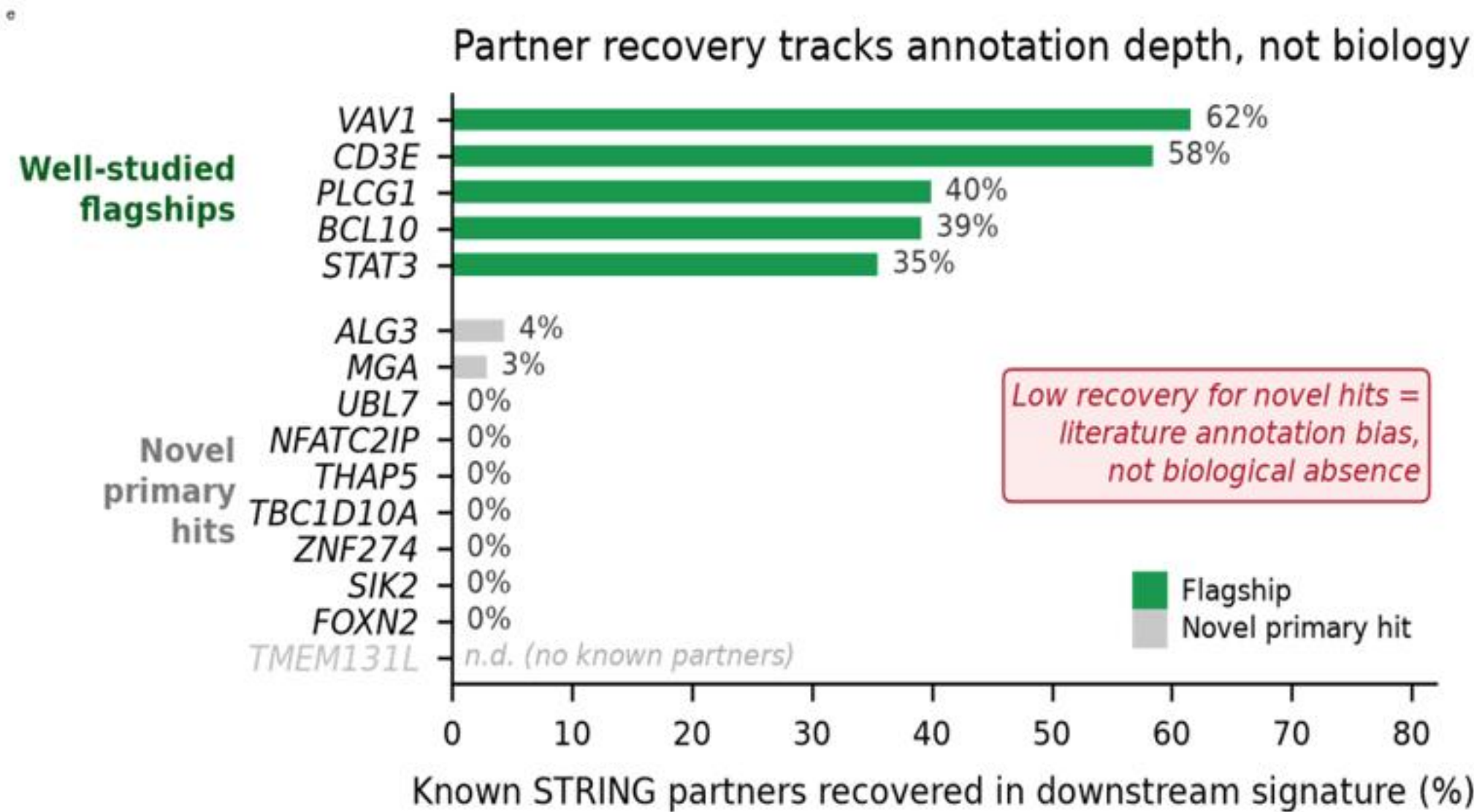
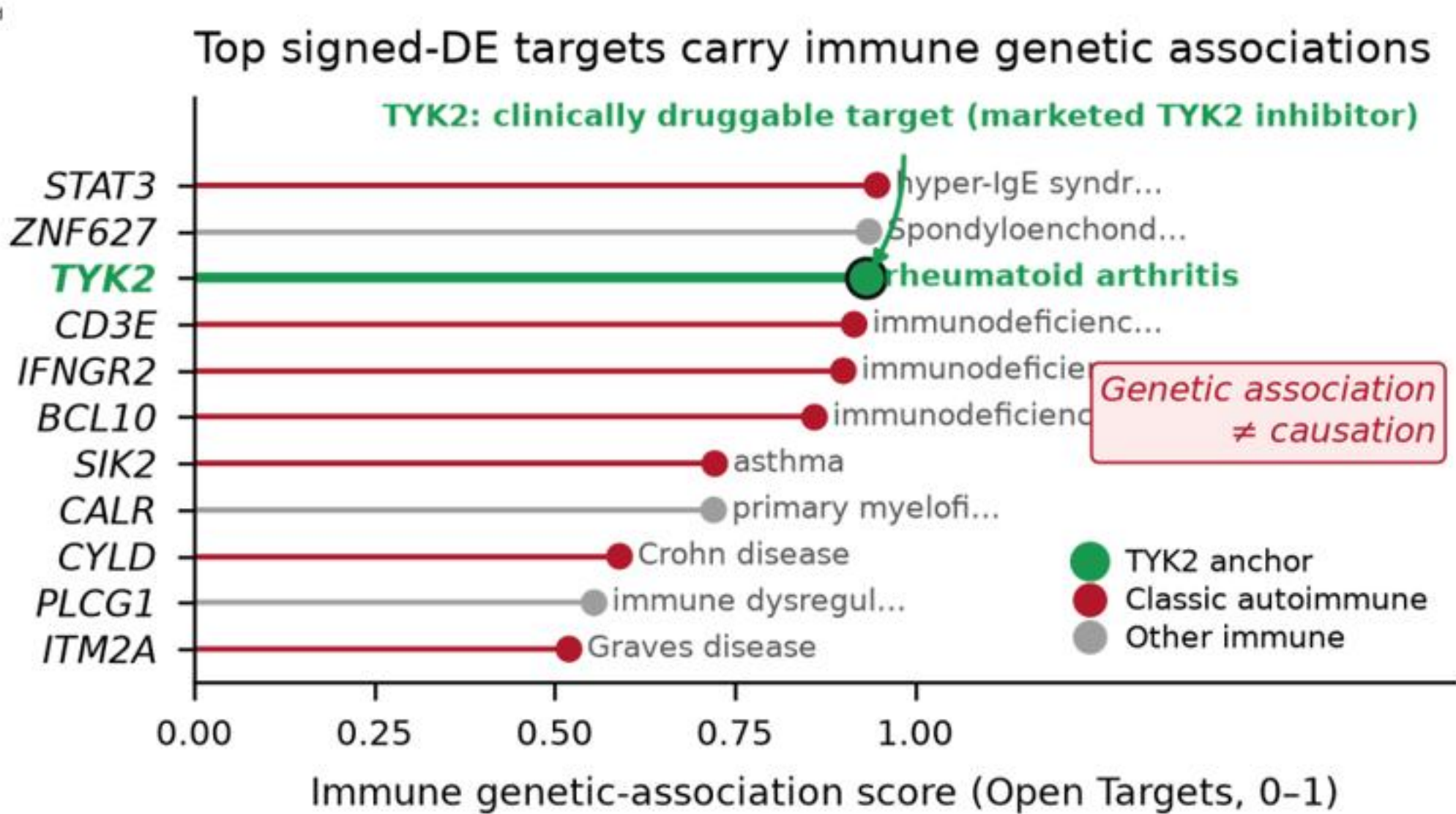
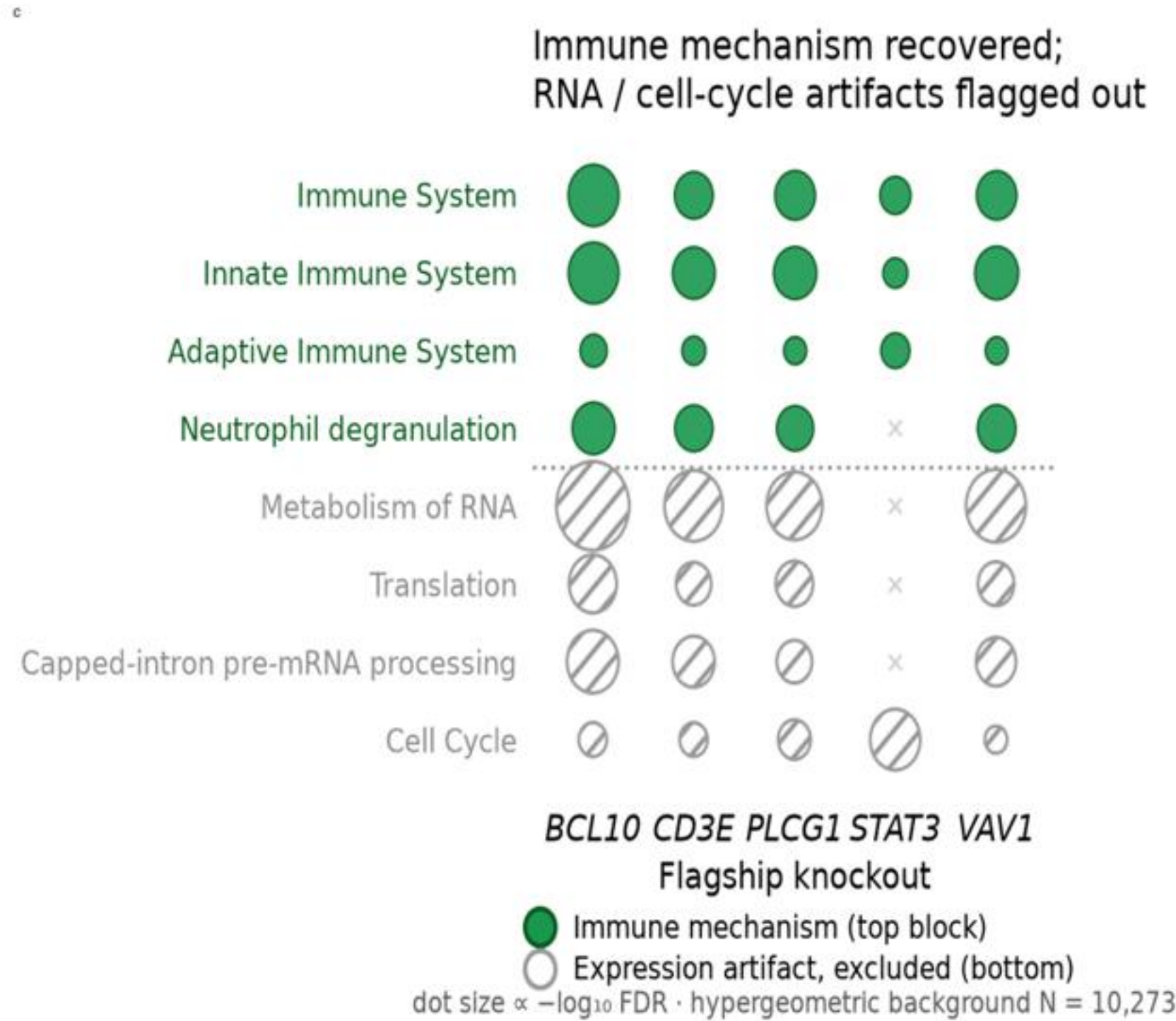
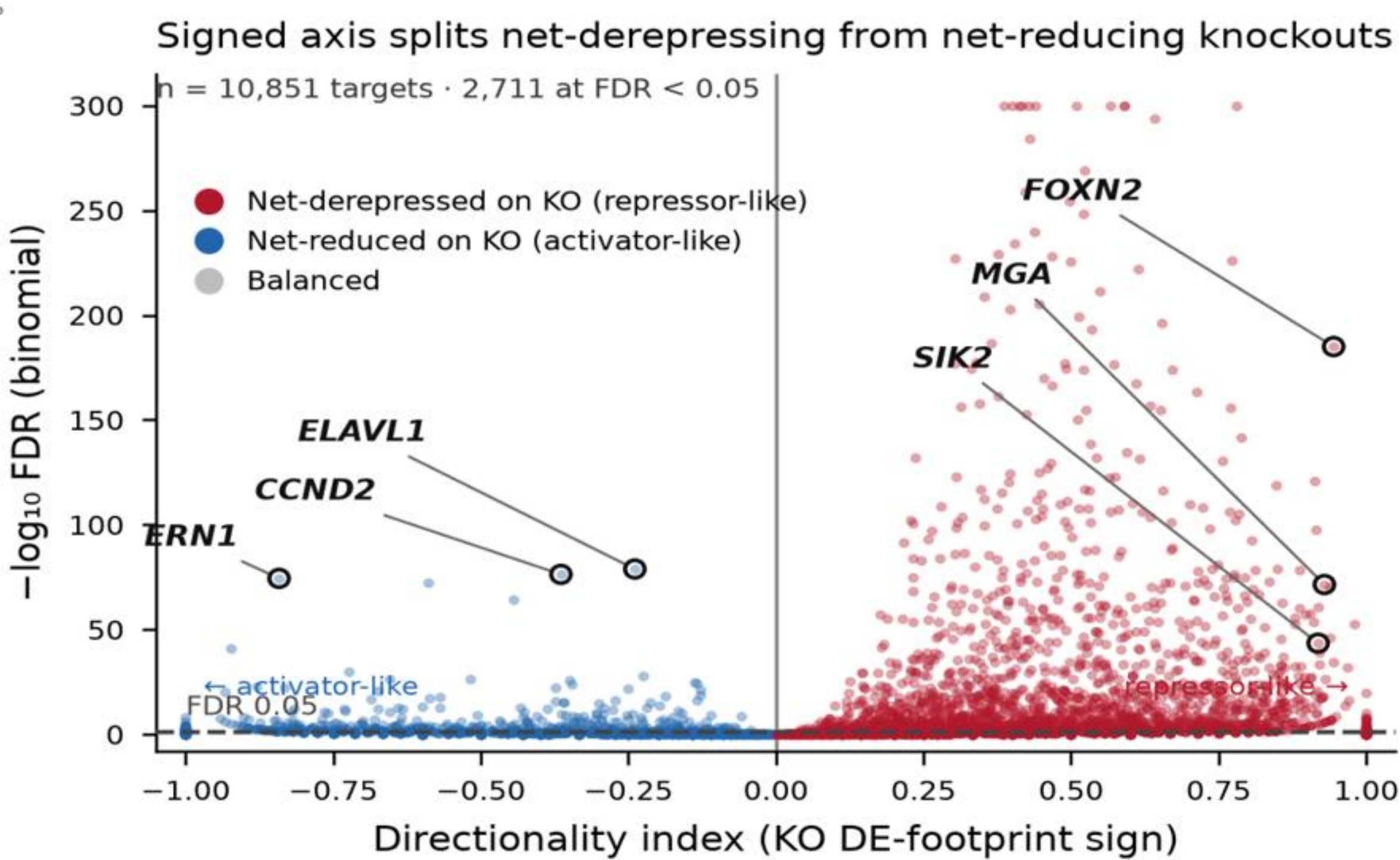
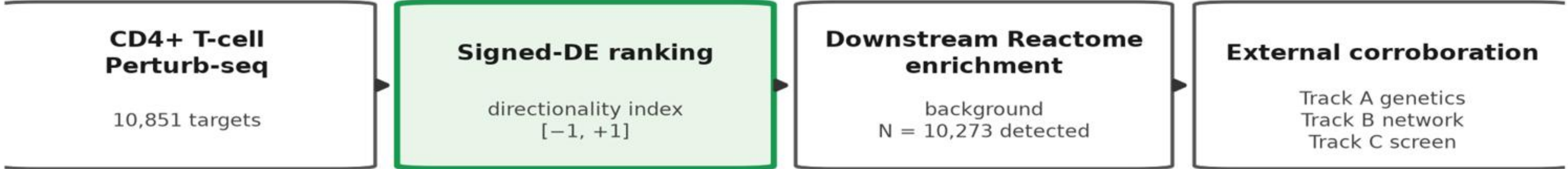


● shortlist hit (size $\propto$ mean on-target effect) ○ pathway context (not screened)

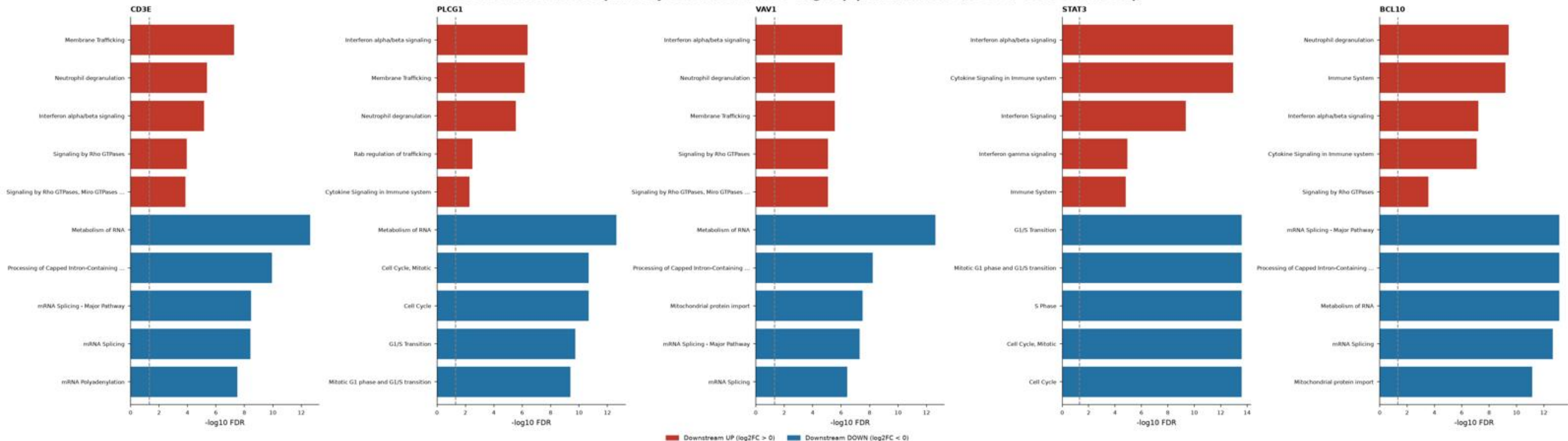
TCR-signalling genes are silent at rest and surge on stimulation



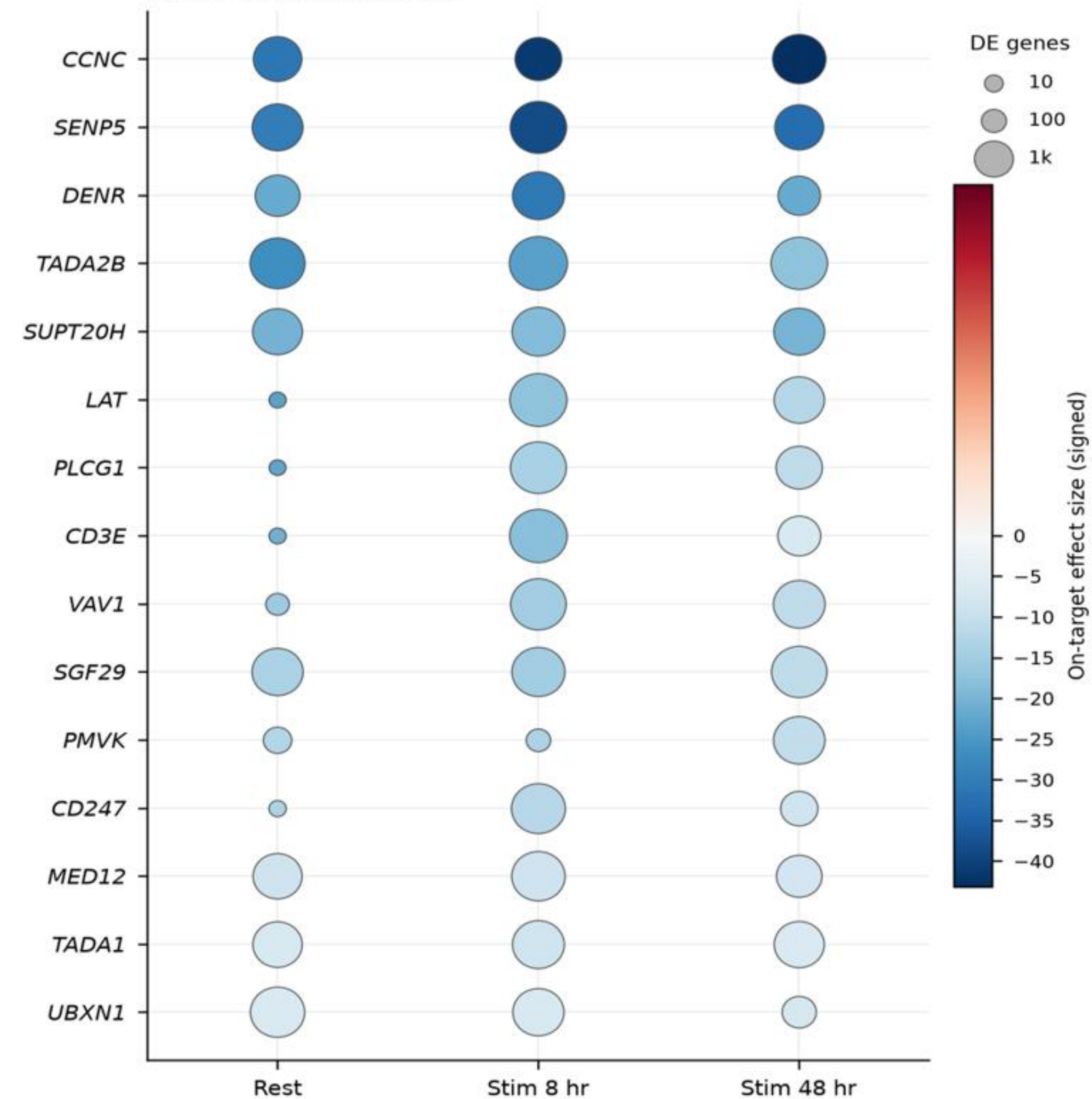
CD4+ T-cell Perturb-seq target-discovery platform: signed axis → downstream mechanism → external corroboration



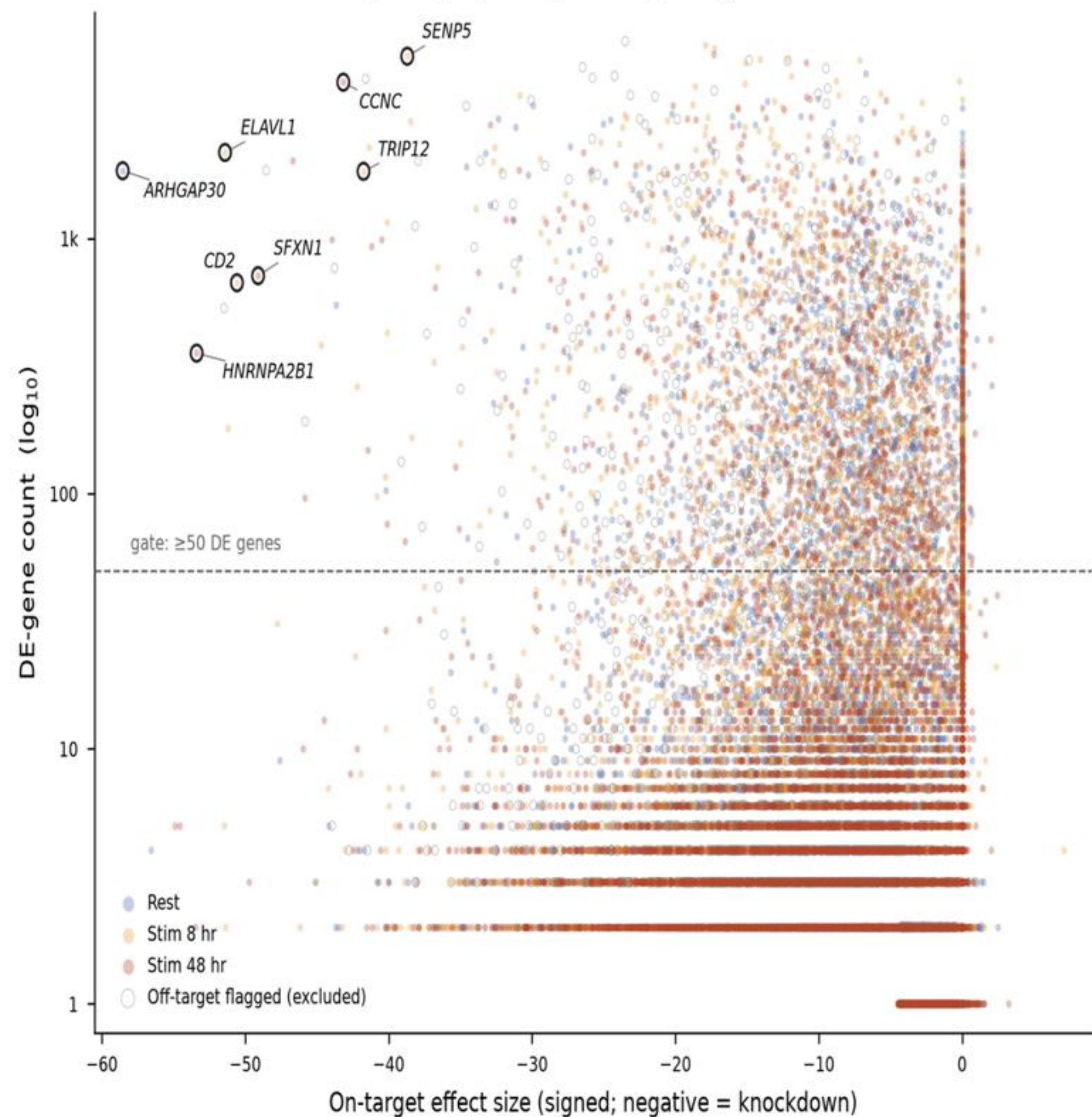
Downstream Reactome pathway enrichment for 5 flagship perturbations (CD4+ T-cell Perturb-seq)



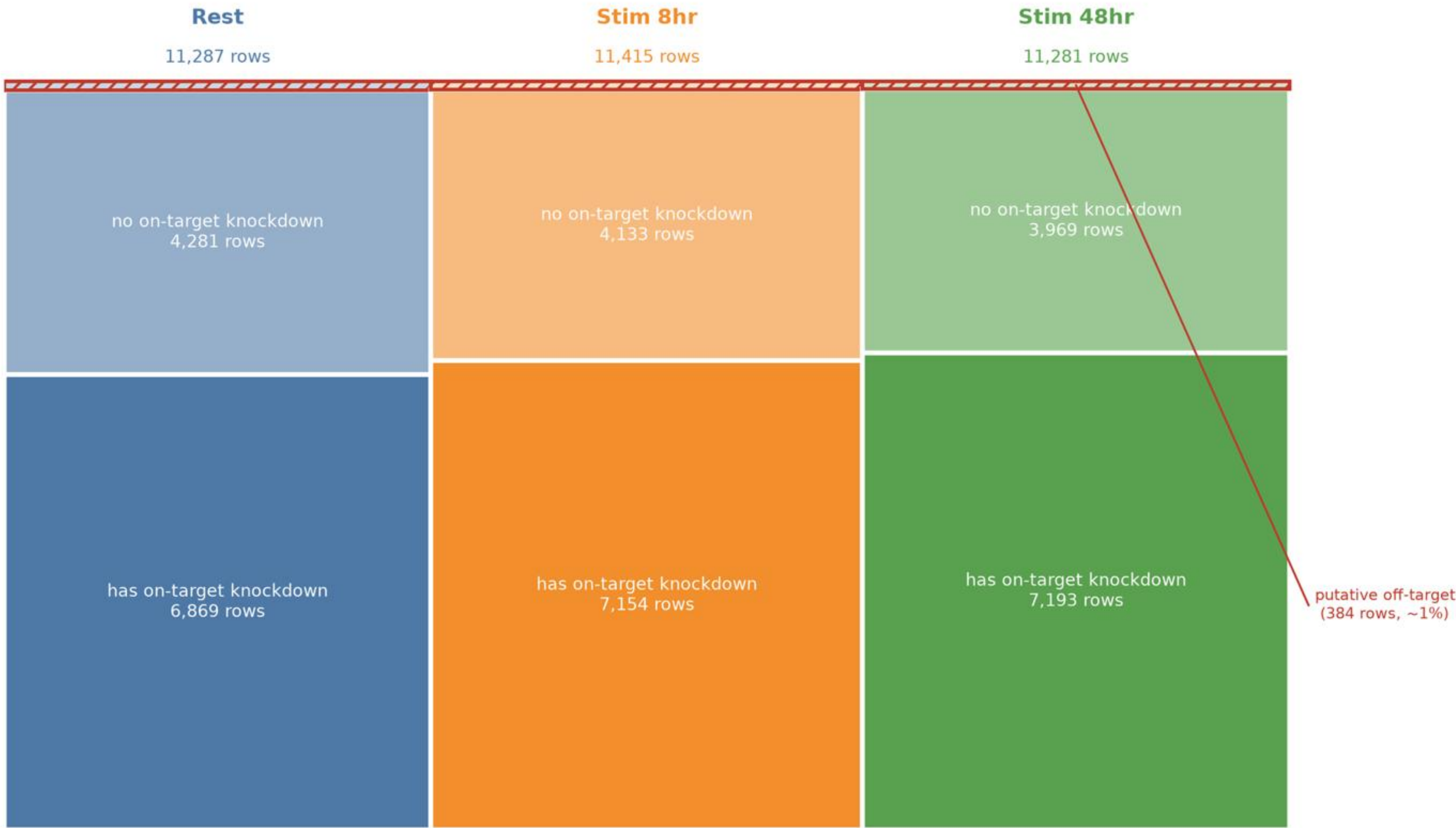
Knockdown effect and breadth across stimulation states



Pseudo-volcano: data has no p-value, so y is DE-gene count, not significance



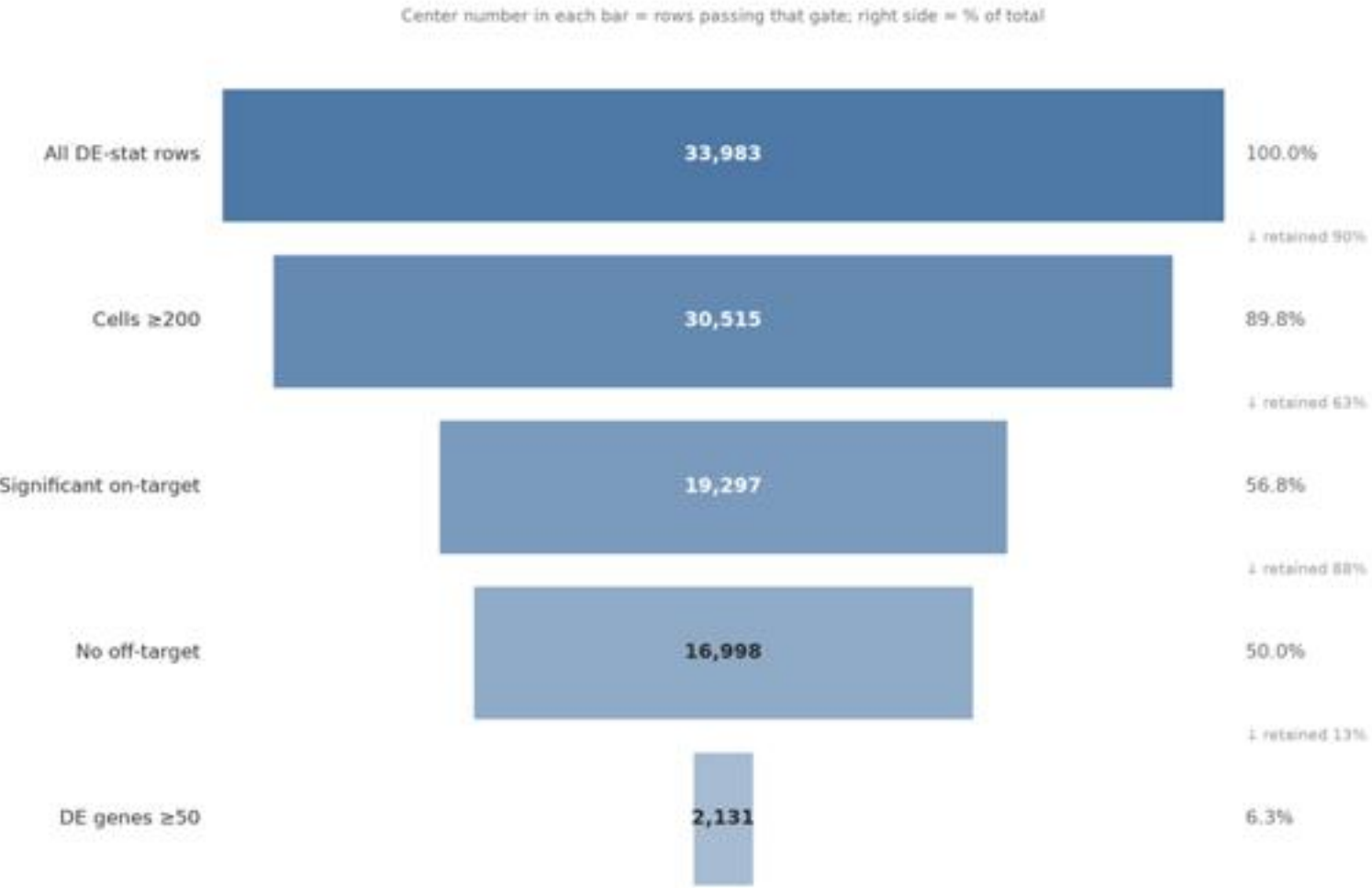
On-target knockdown is the dominant effect category across all three conditions (~62% of DE-stat rows)



Whole figure n=33,983 DE-stat rows; rectangle area $\propto$ row count, outer columns by culture condition, inner by on-target effect category; hatched red top band = putative off-target (384 rows overall, ~1%)

b

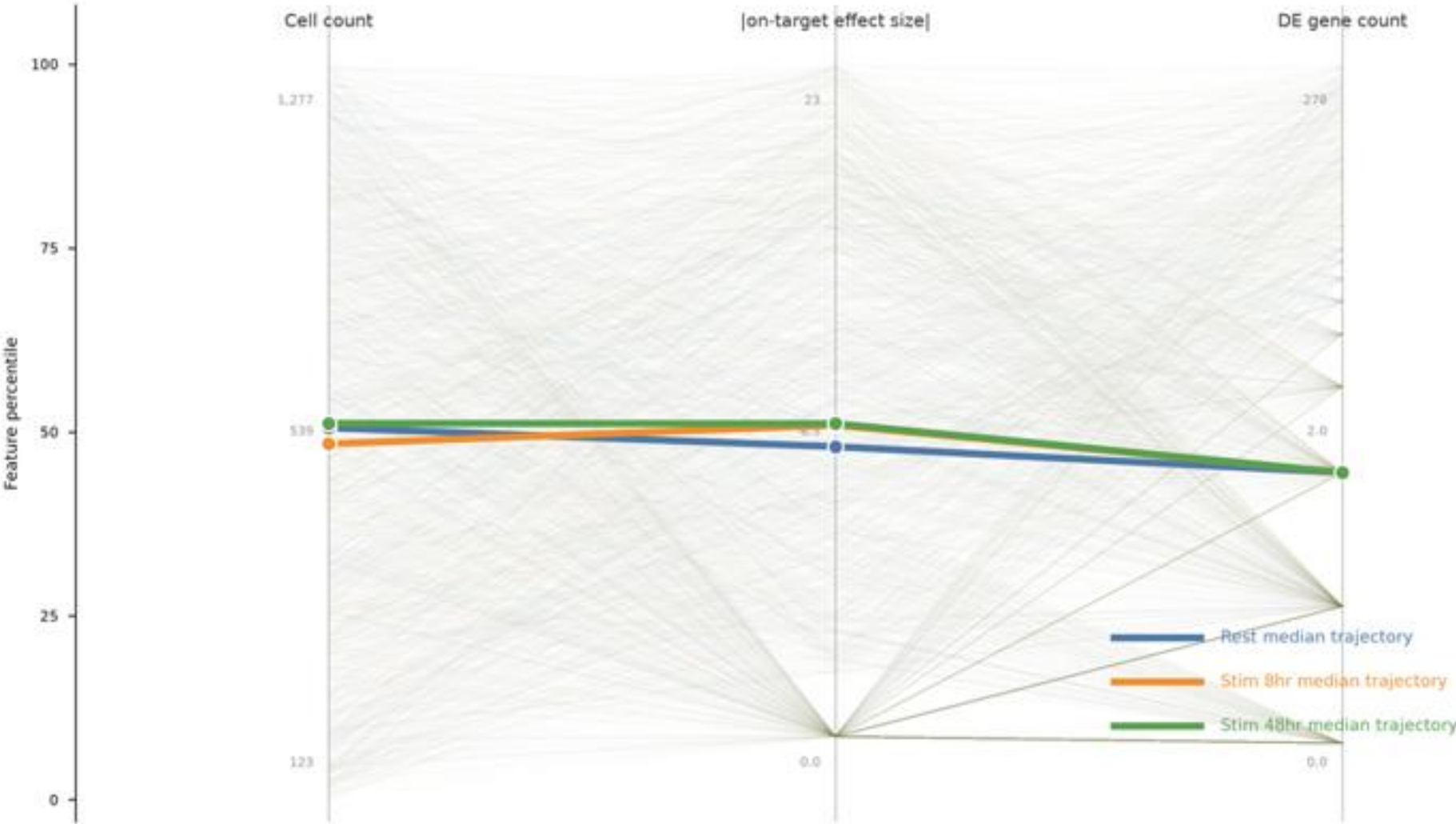
Filtering funnel: DE genes ≥ 50 is the strictest gate (16,998→2,131, only 13% retained)



Gates applied in sequence: cells $\geq 200 \rightarrow$ significant on-target $\rightarrow$ no off-target $\rightarrow$ DE genes ≥ 50 (n=33,983 starting rows)

c

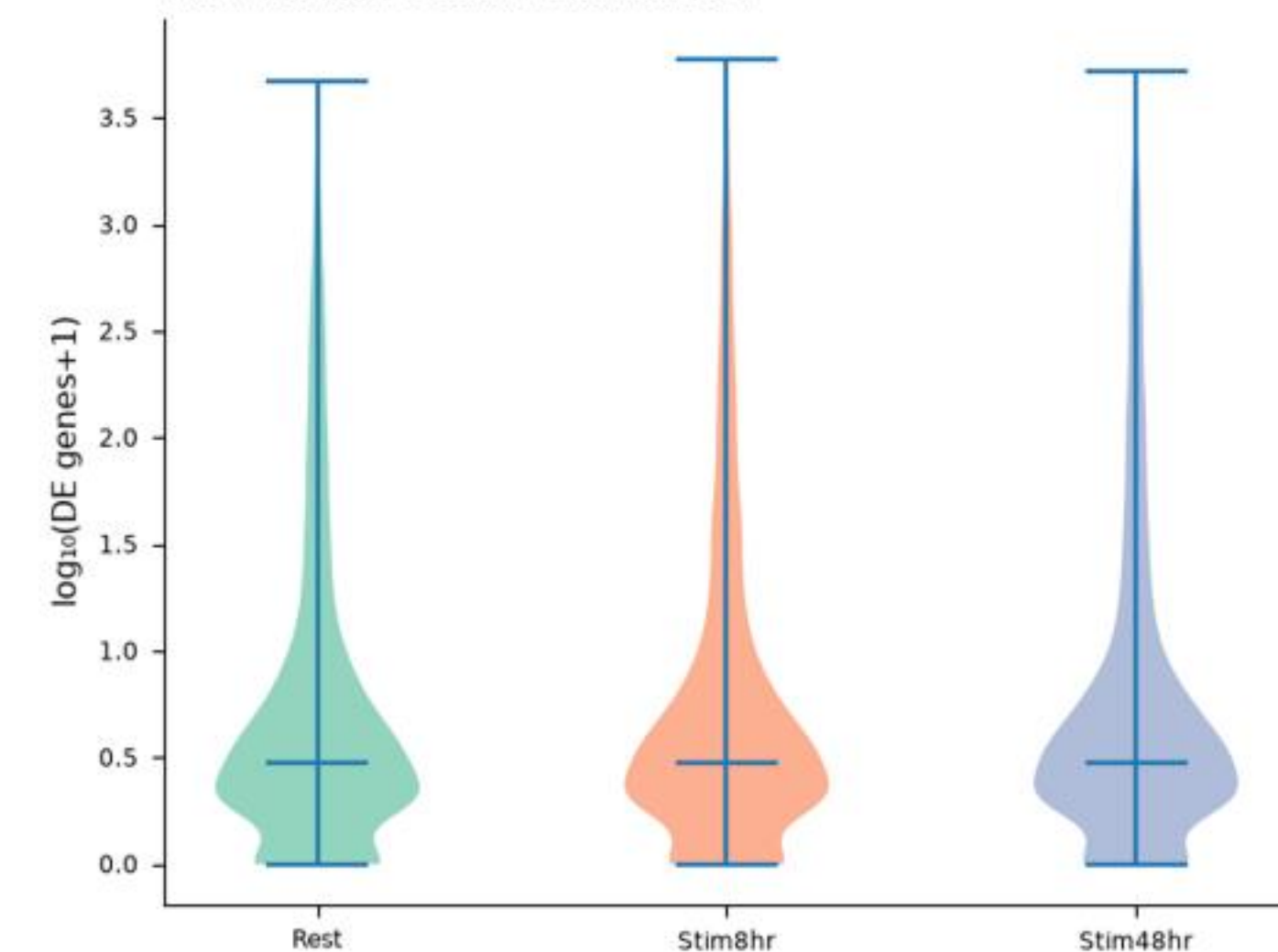
Target feature distributions overlap across the three conditions: culture condition does not change effect size or DE breadth



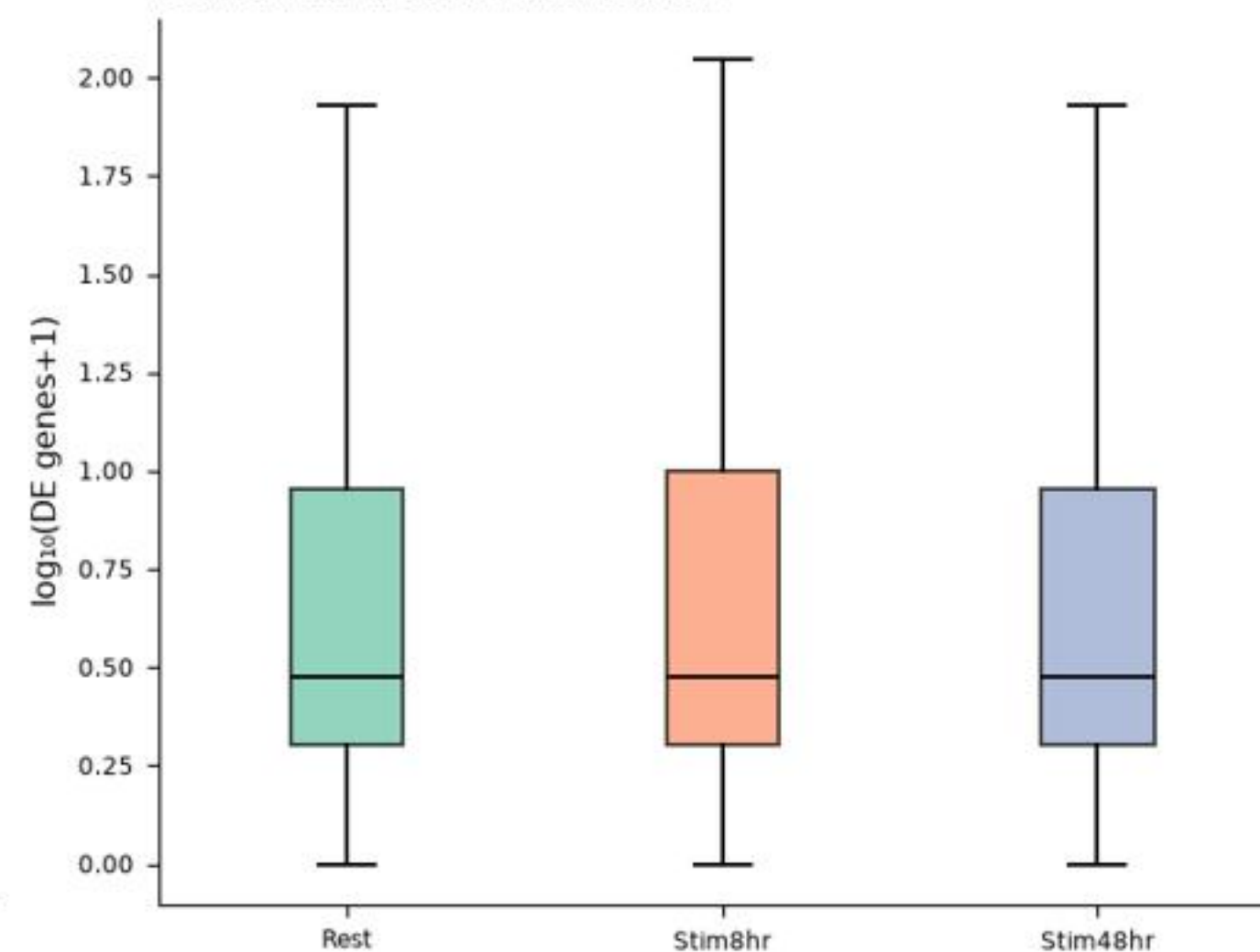
Each thin line = one DE-stat row (500 rows sampled per condition, $\alpha=0.025$); thick line = per-condition median trajectory; axes are feature percentiles, annotated with 5/50/95 percentile true values (all three axes n=33,983 rows, no missing)

Chart gallery — Batch 1: distributions

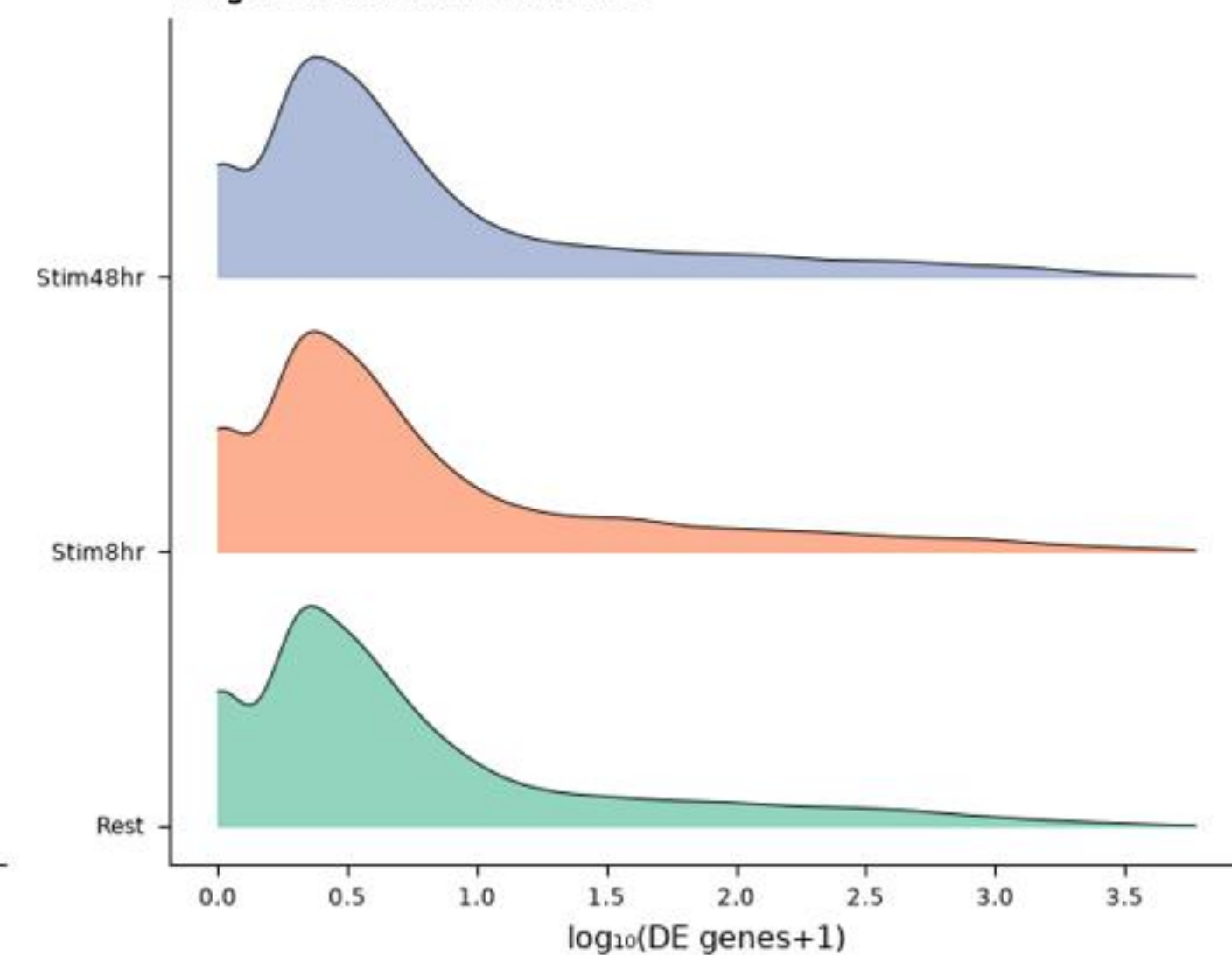
Violin: effect-breadth distribution



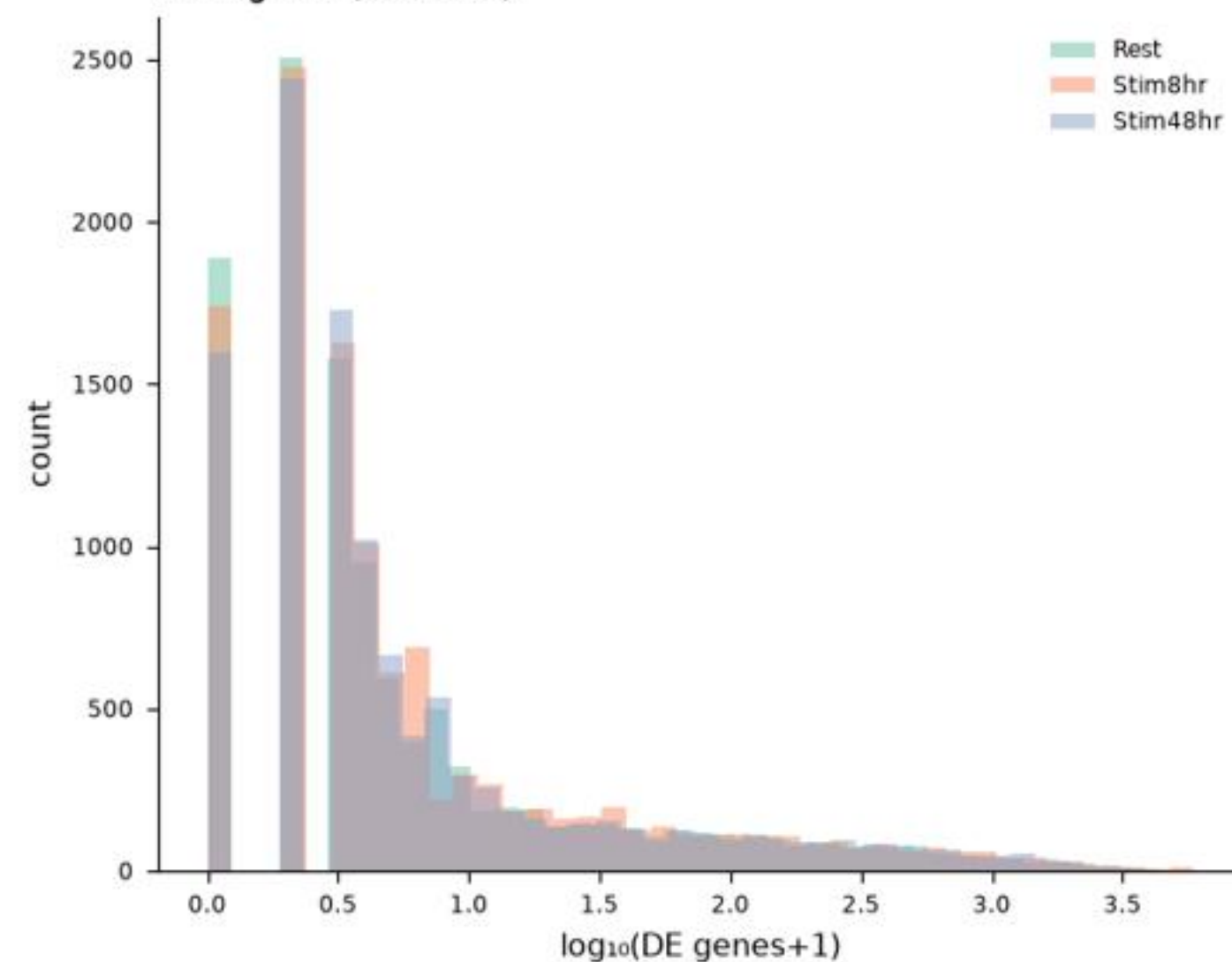
Box: effect-breadth distribution



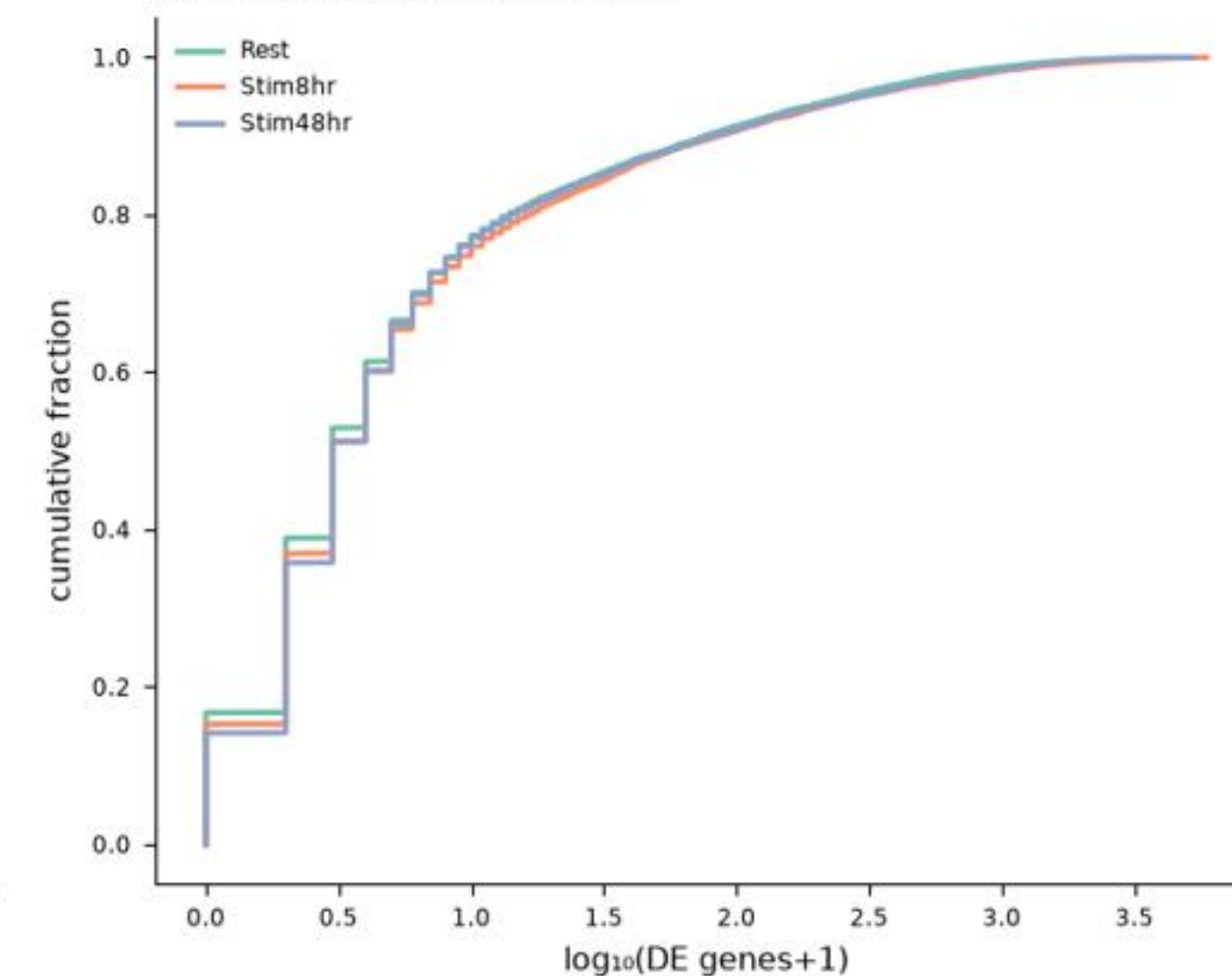
Ridgeline: stacked densities



Histogram (overlaid)



ECDF: cumulative distribution



Strip plot (jittered, n=400/cond)

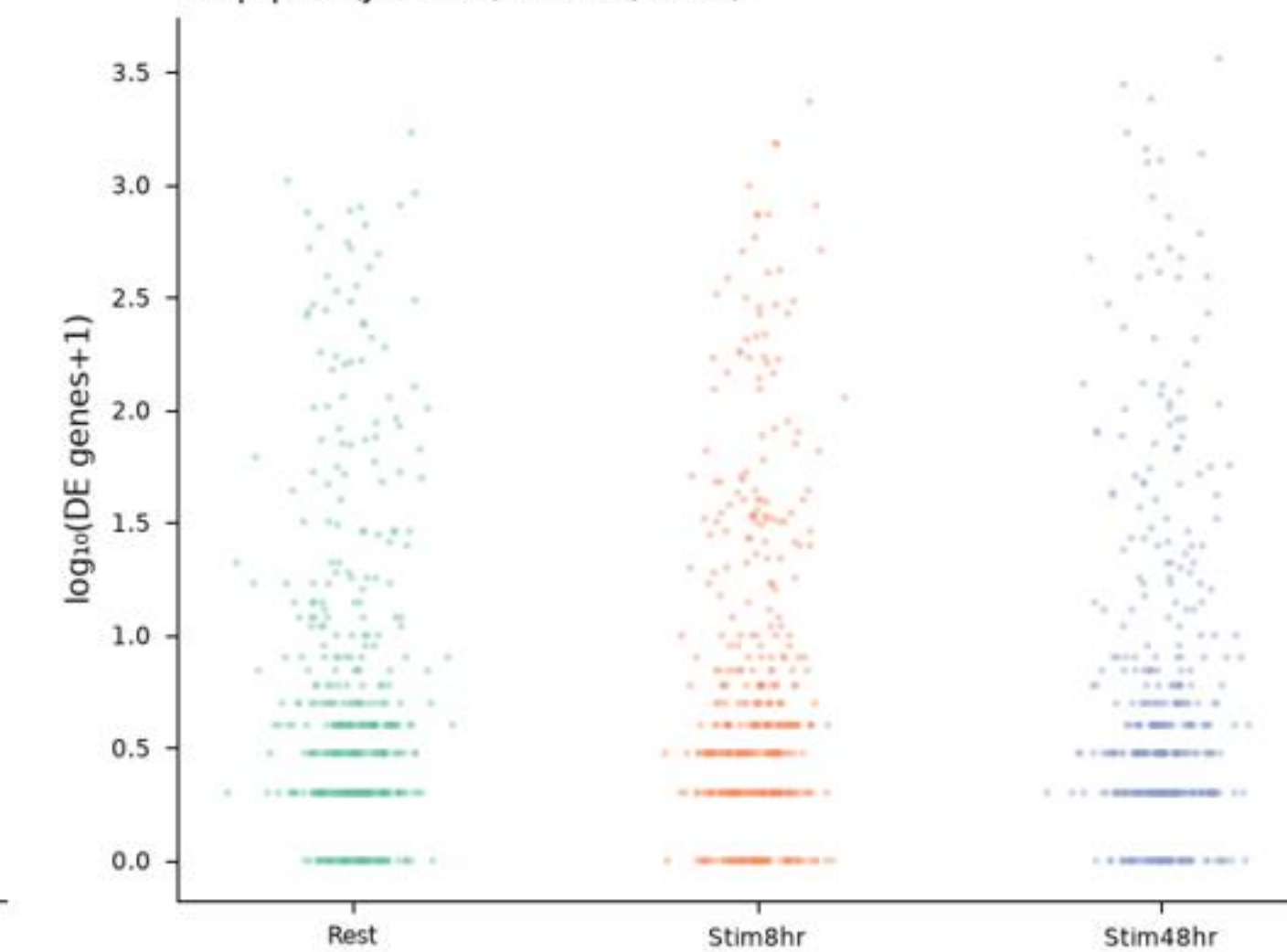


Chart gallery — Batch 2: correlation & matrices

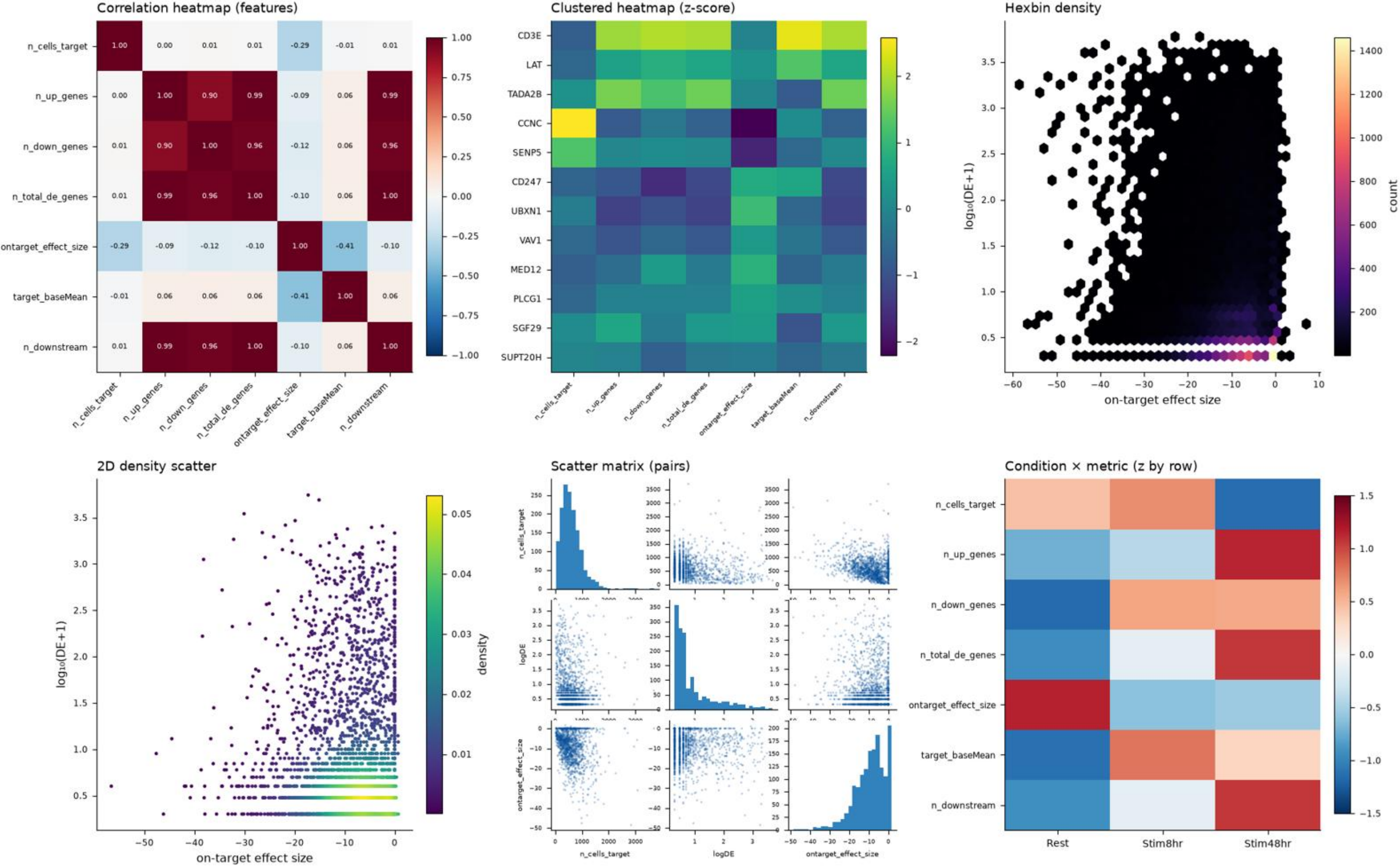


Chart gallery — Batch 3: ranking & composition

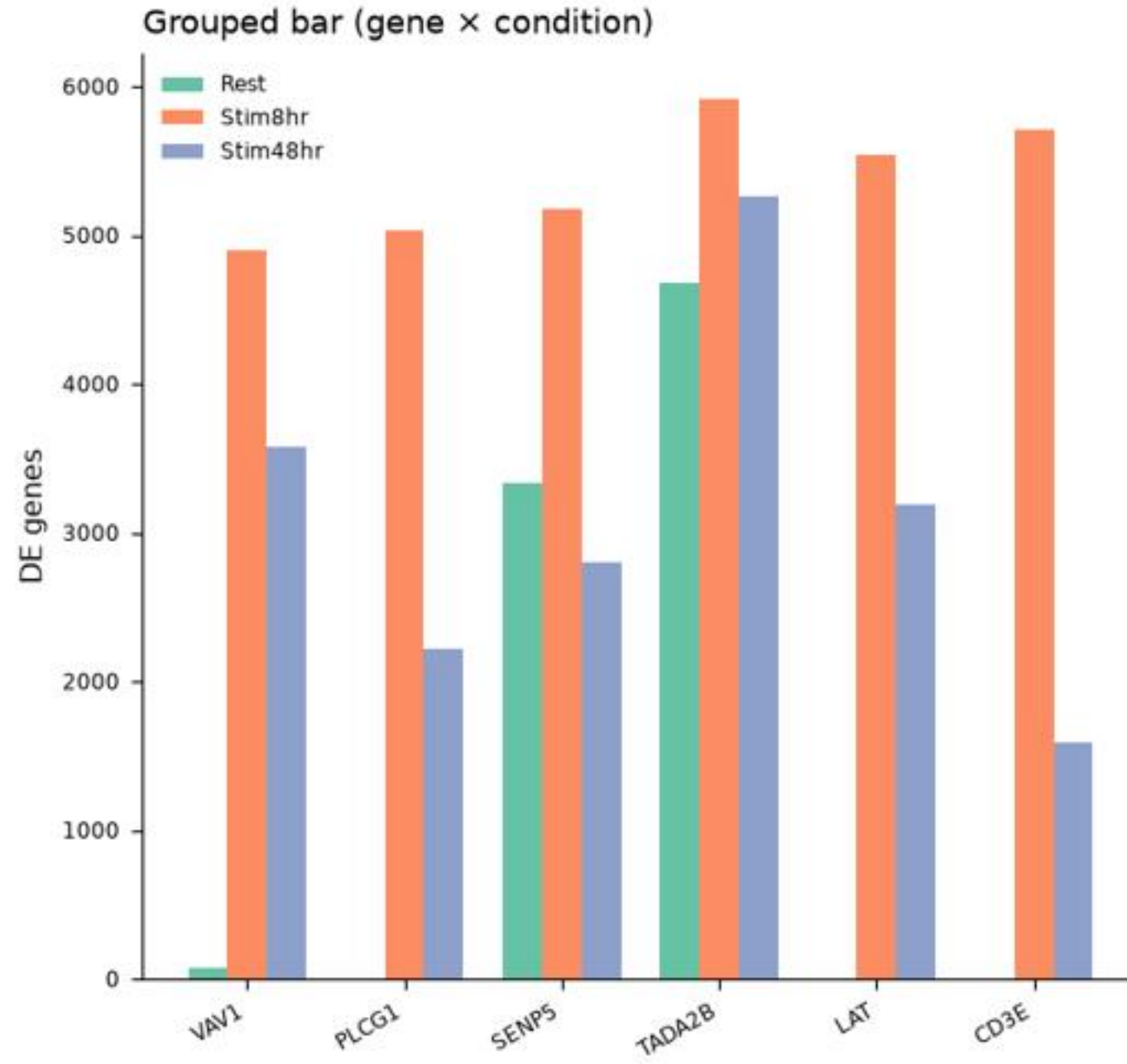
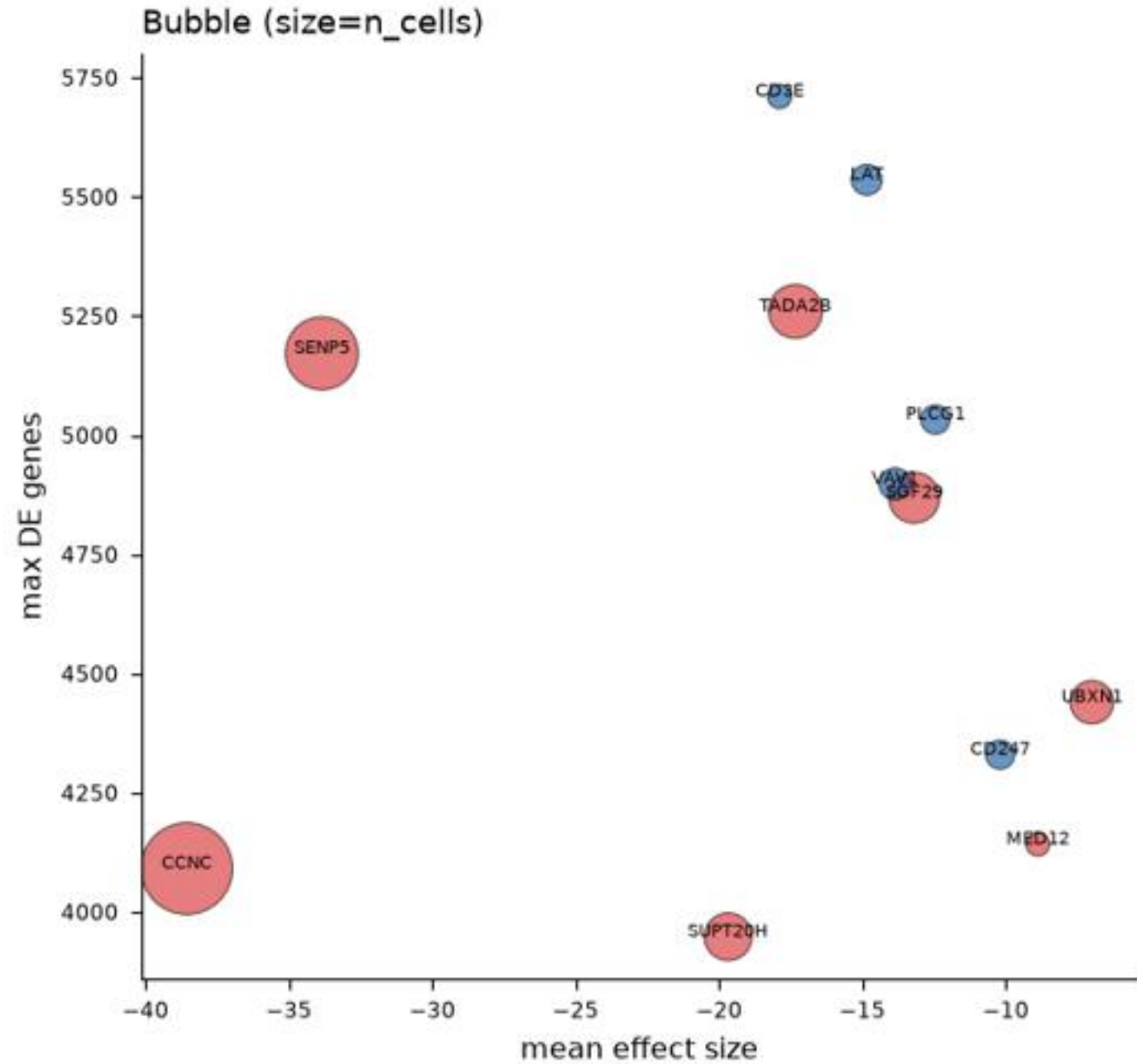
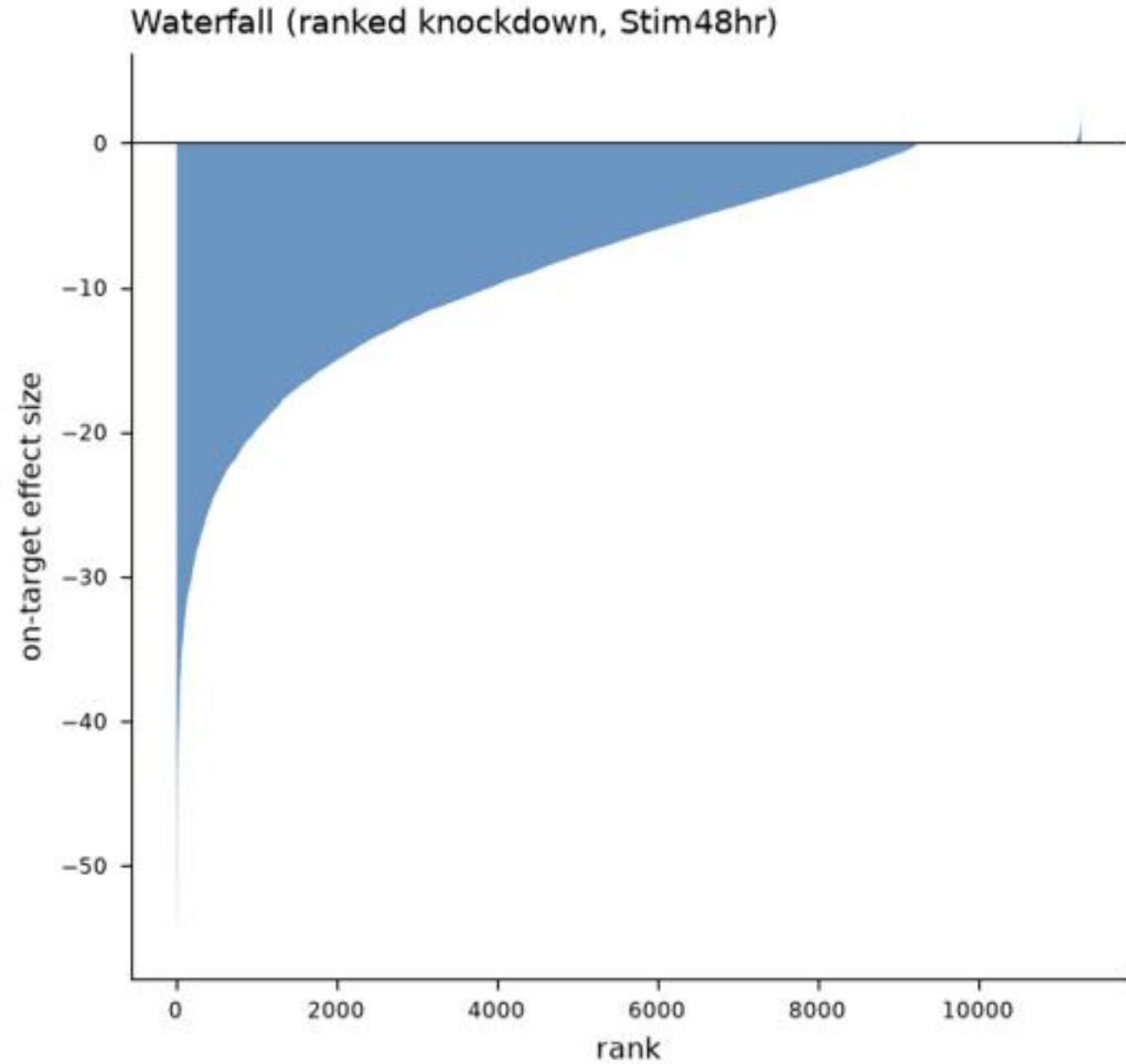
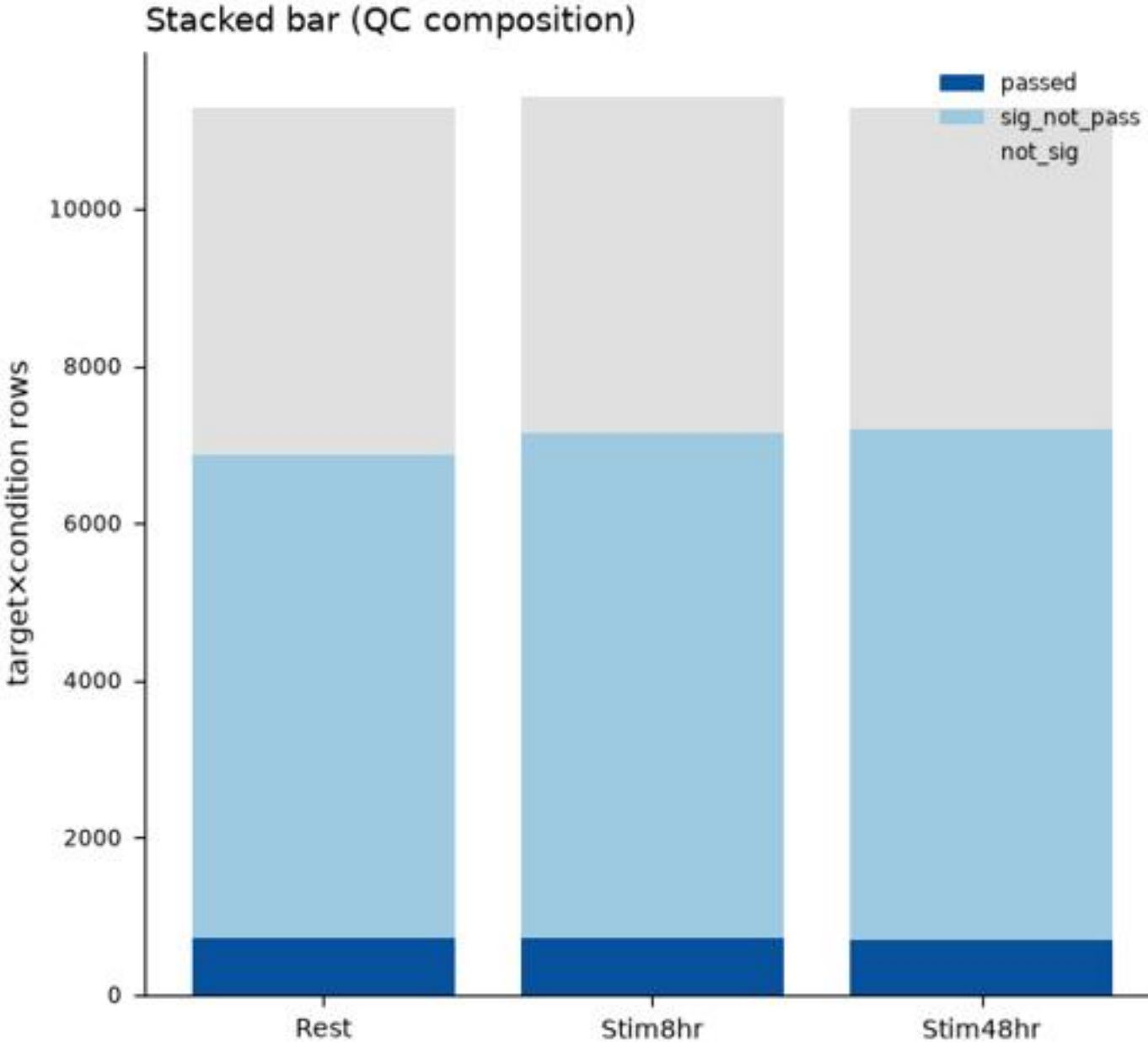
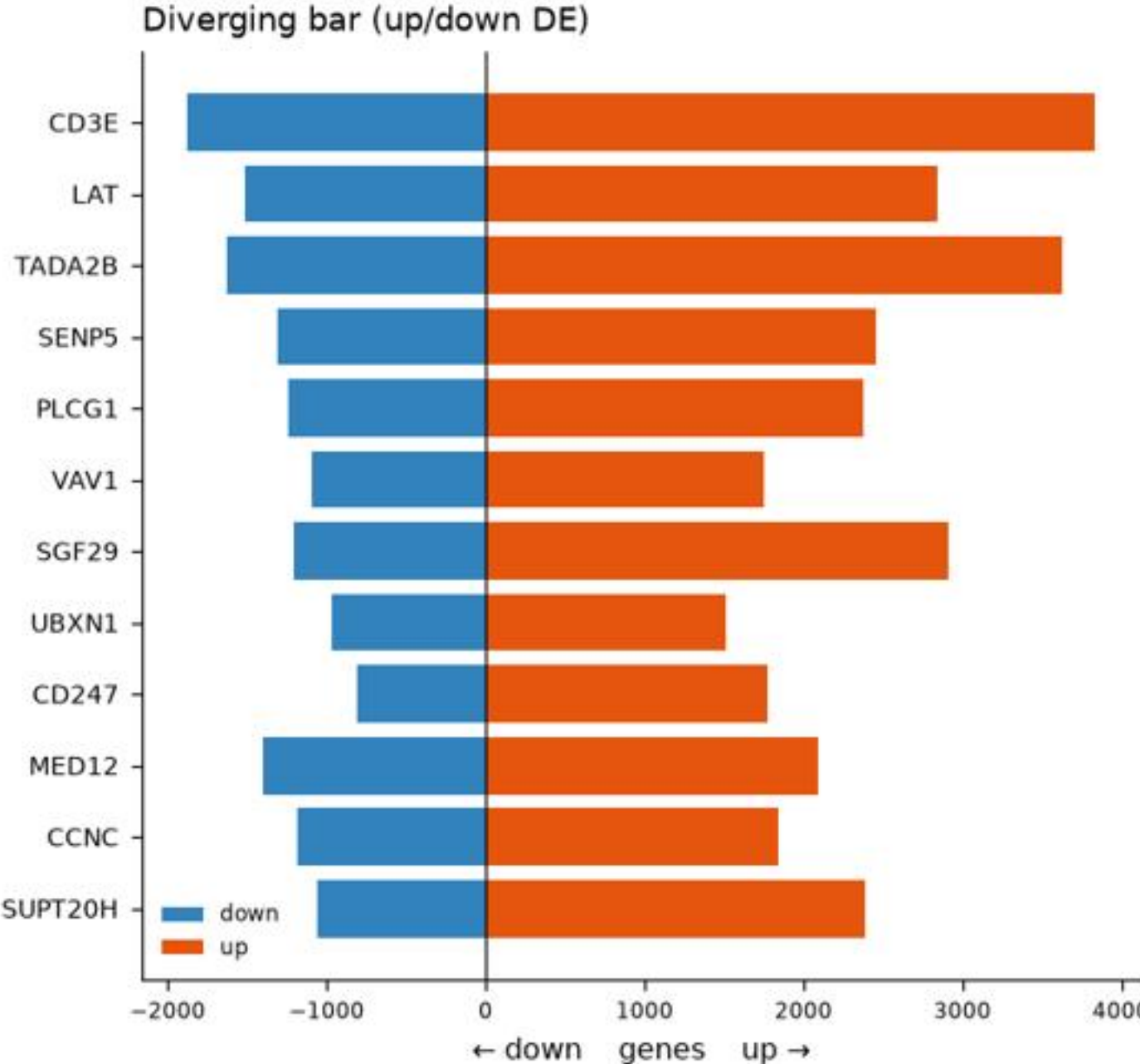
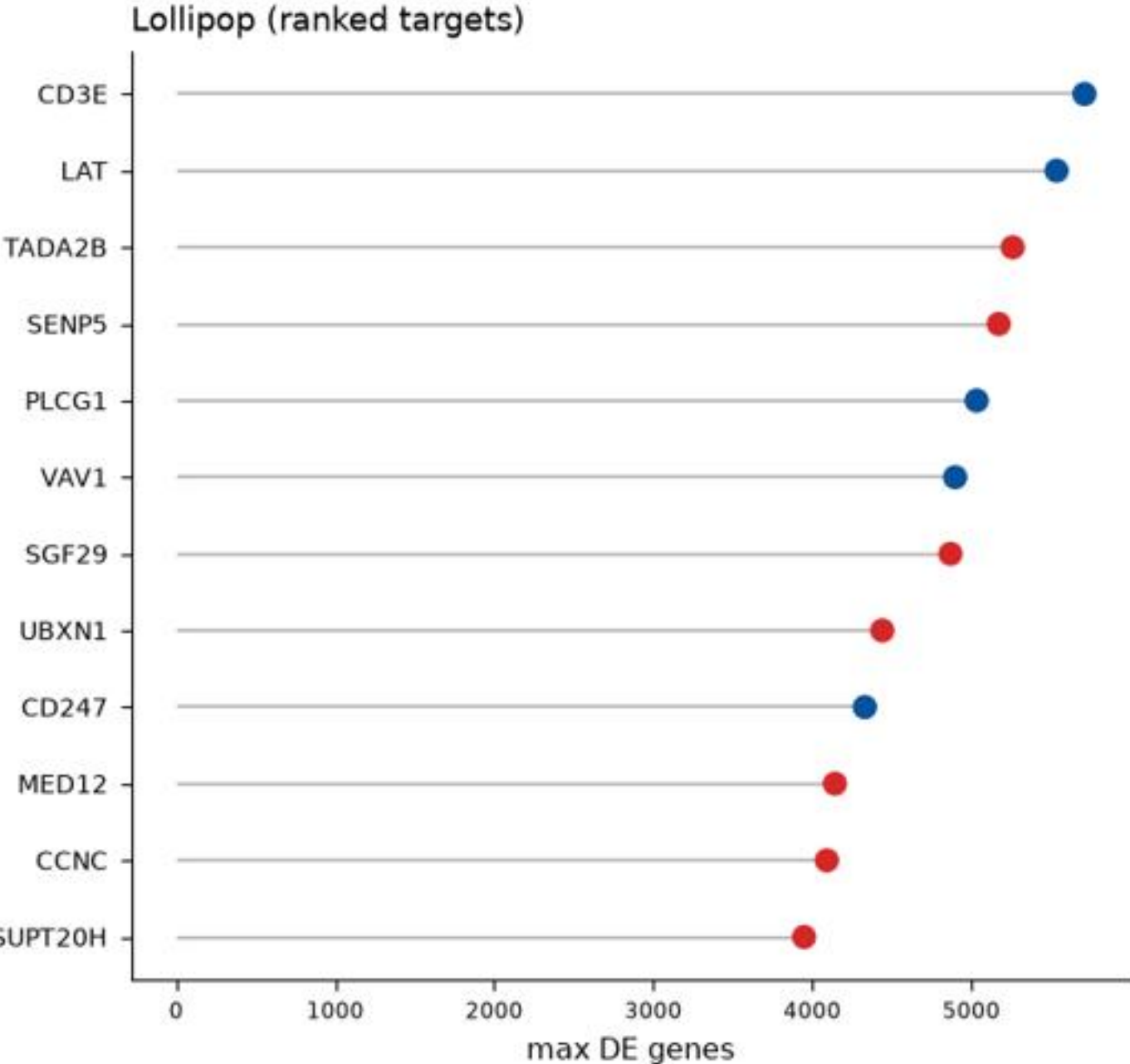


Chart gallery — Batch 4: dimensionality reduction & circular

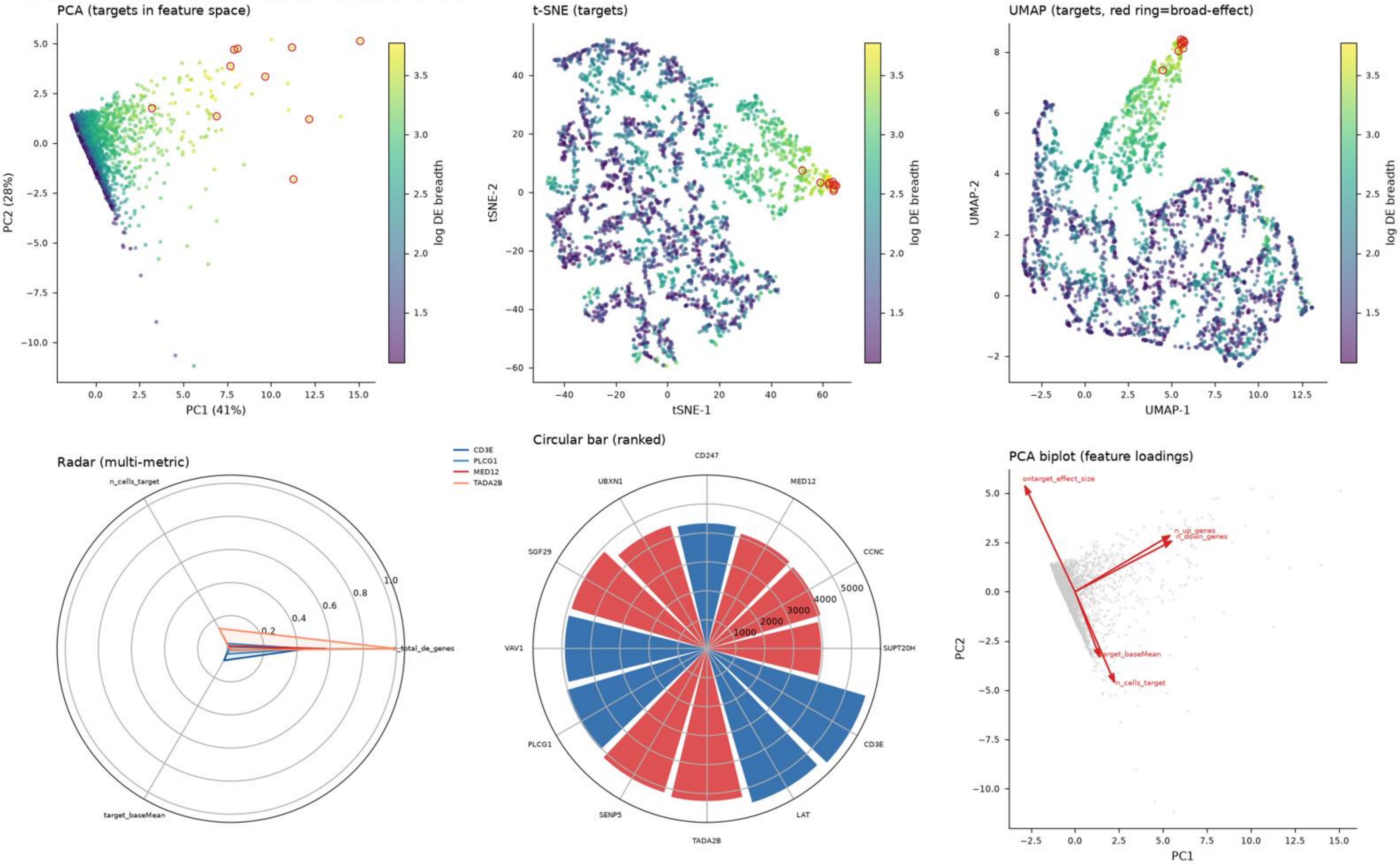
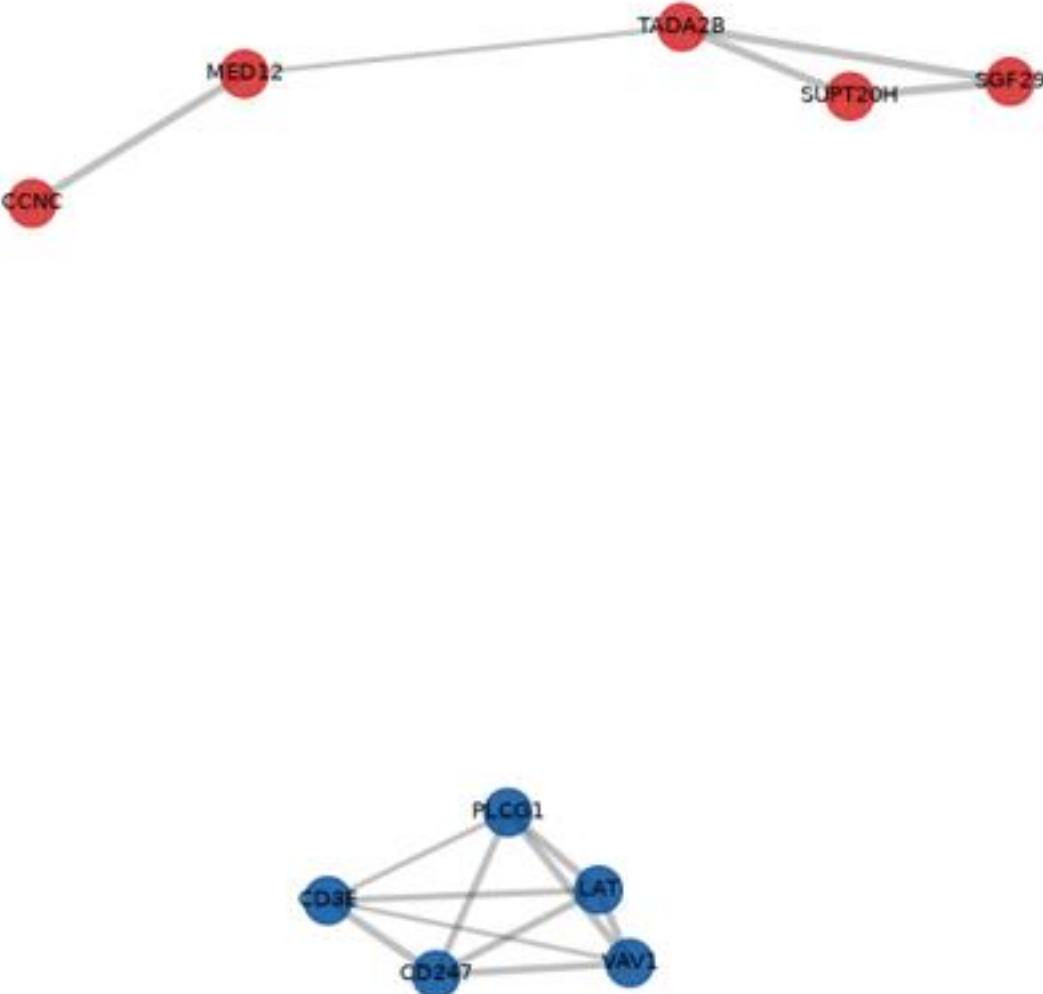
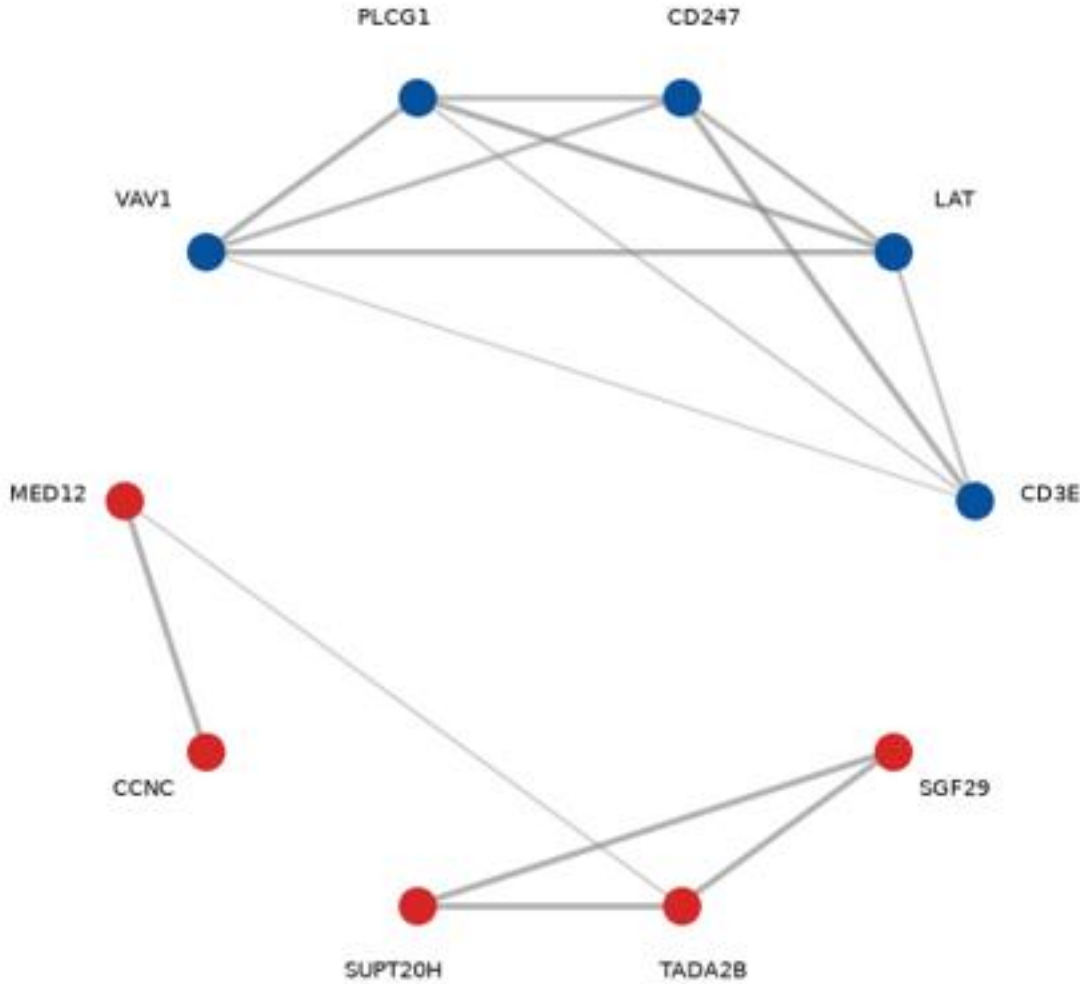


Chart gallery — Batch 5: networks & relationships

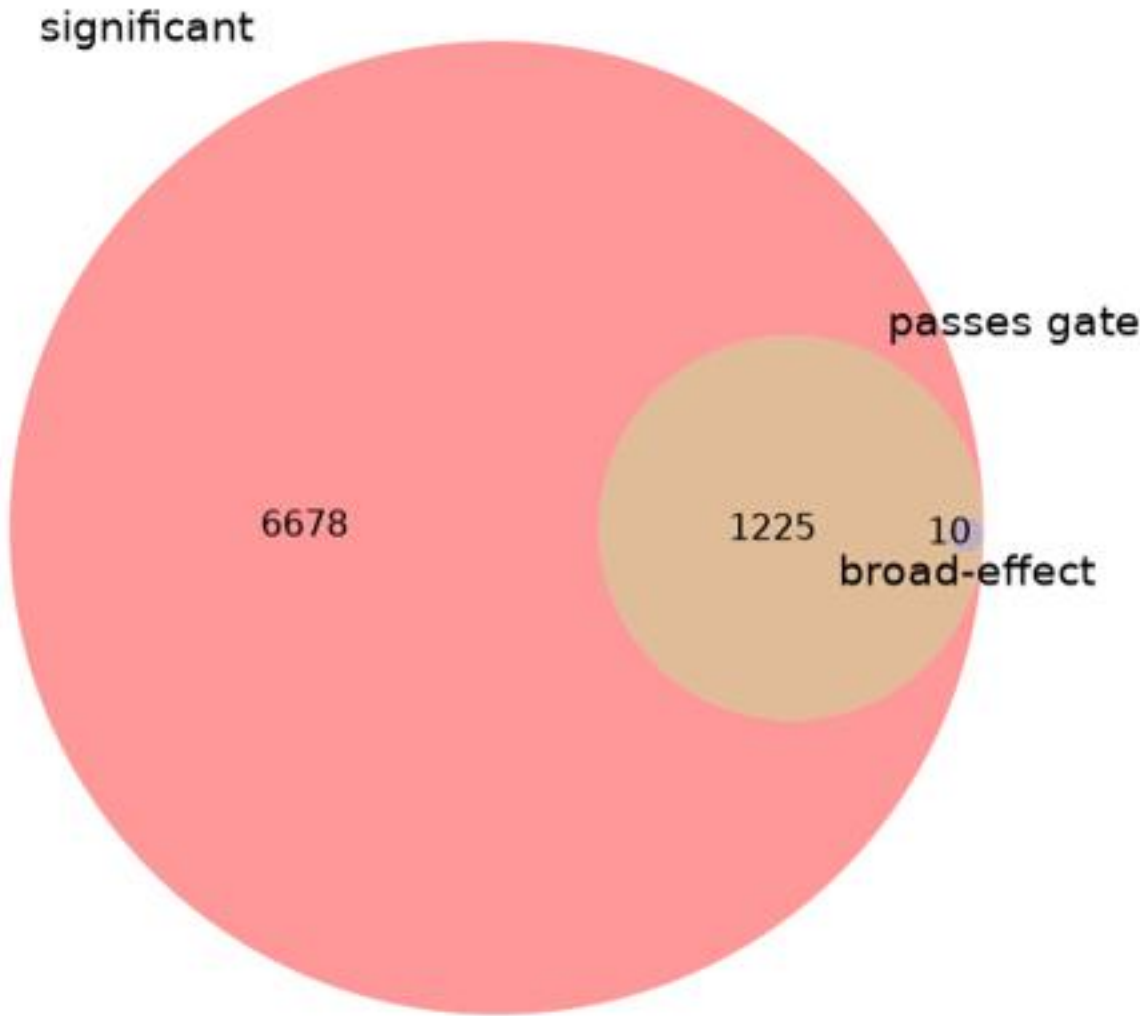
STRING network (10 genes)



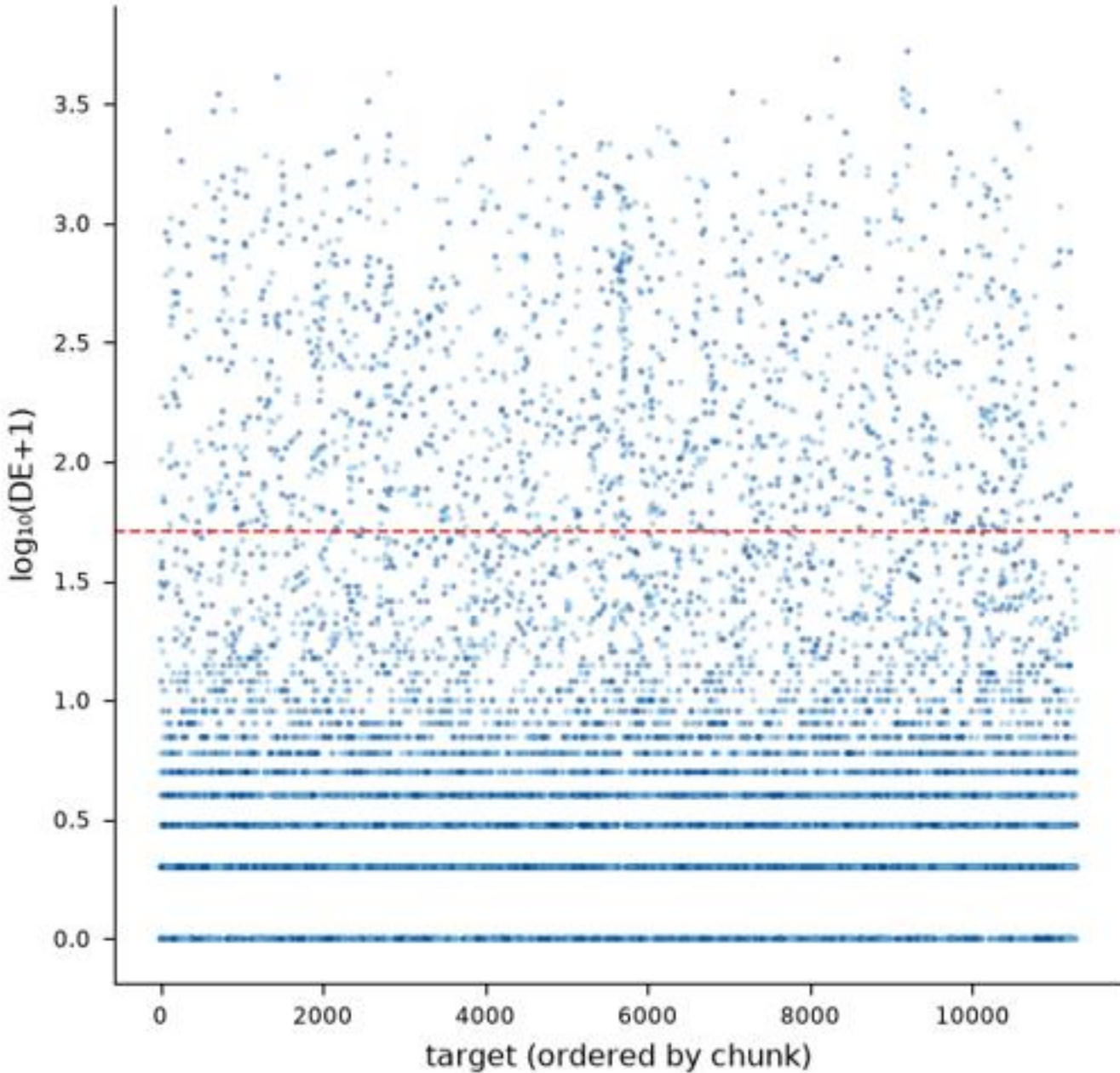
Chord/circular network



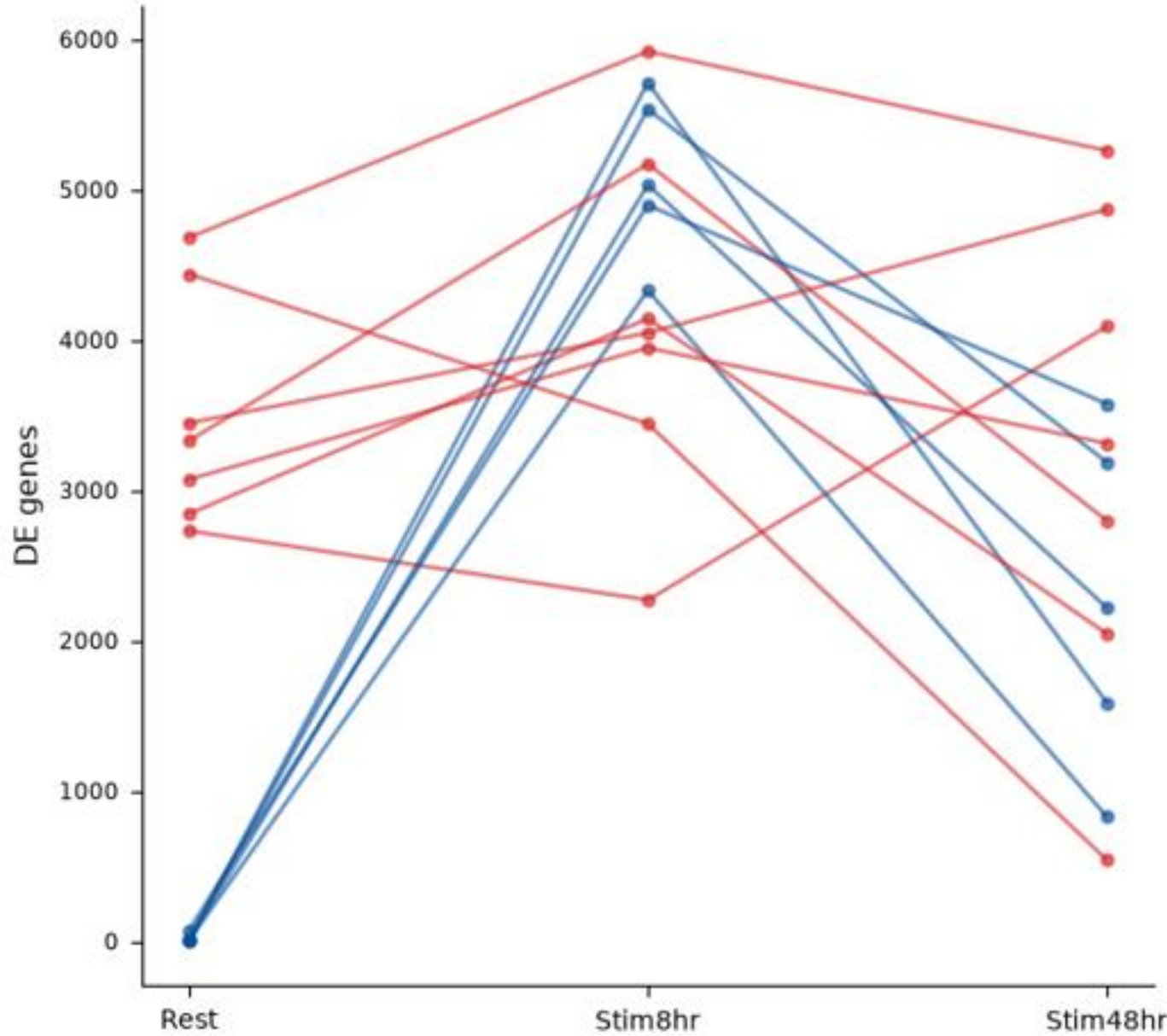
Venn (evidence set overlap)



Manhattan-style (chunk-alternating)



Slope/parallel (across conditions)



Stacked area (up/down by condition)

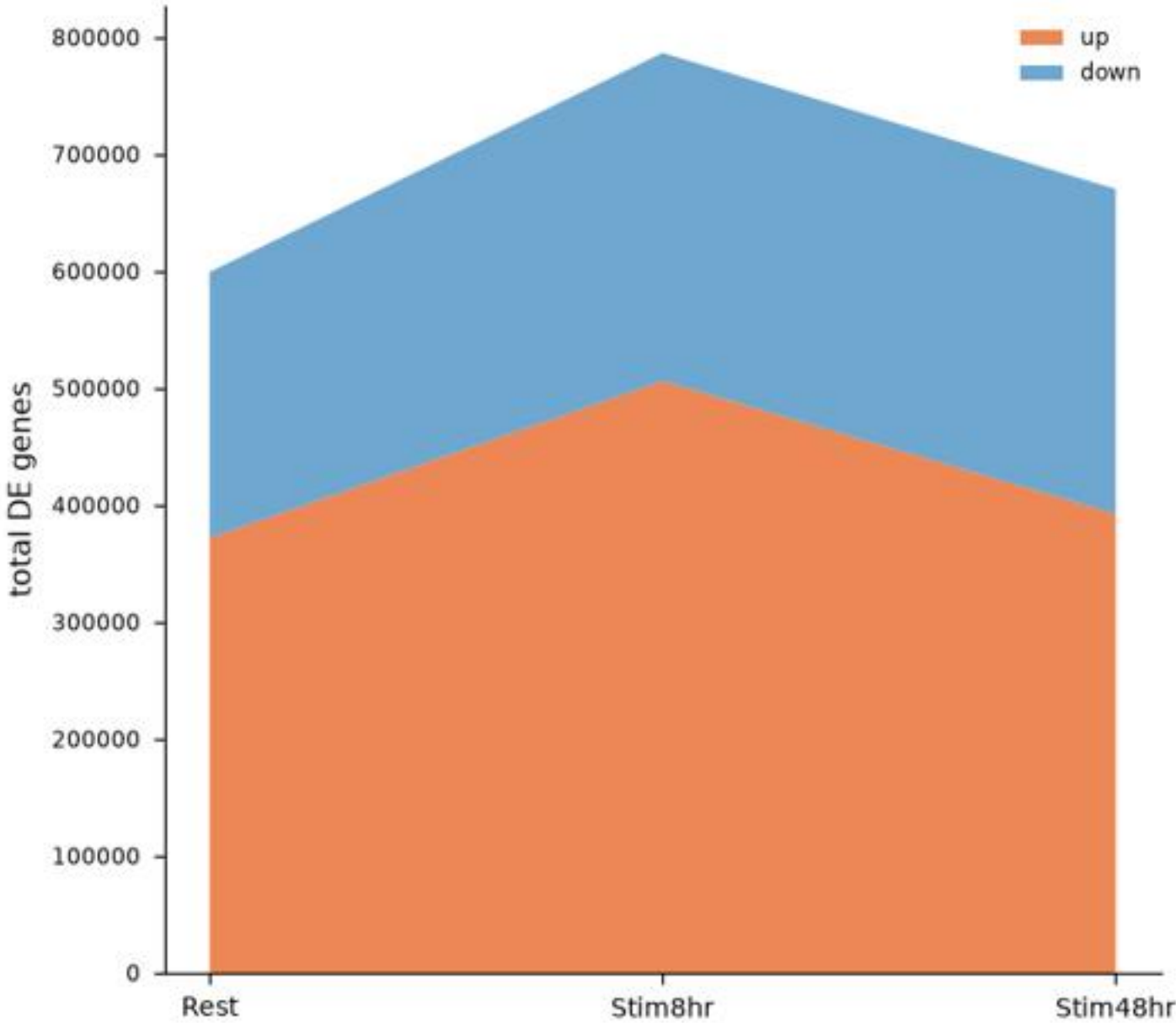
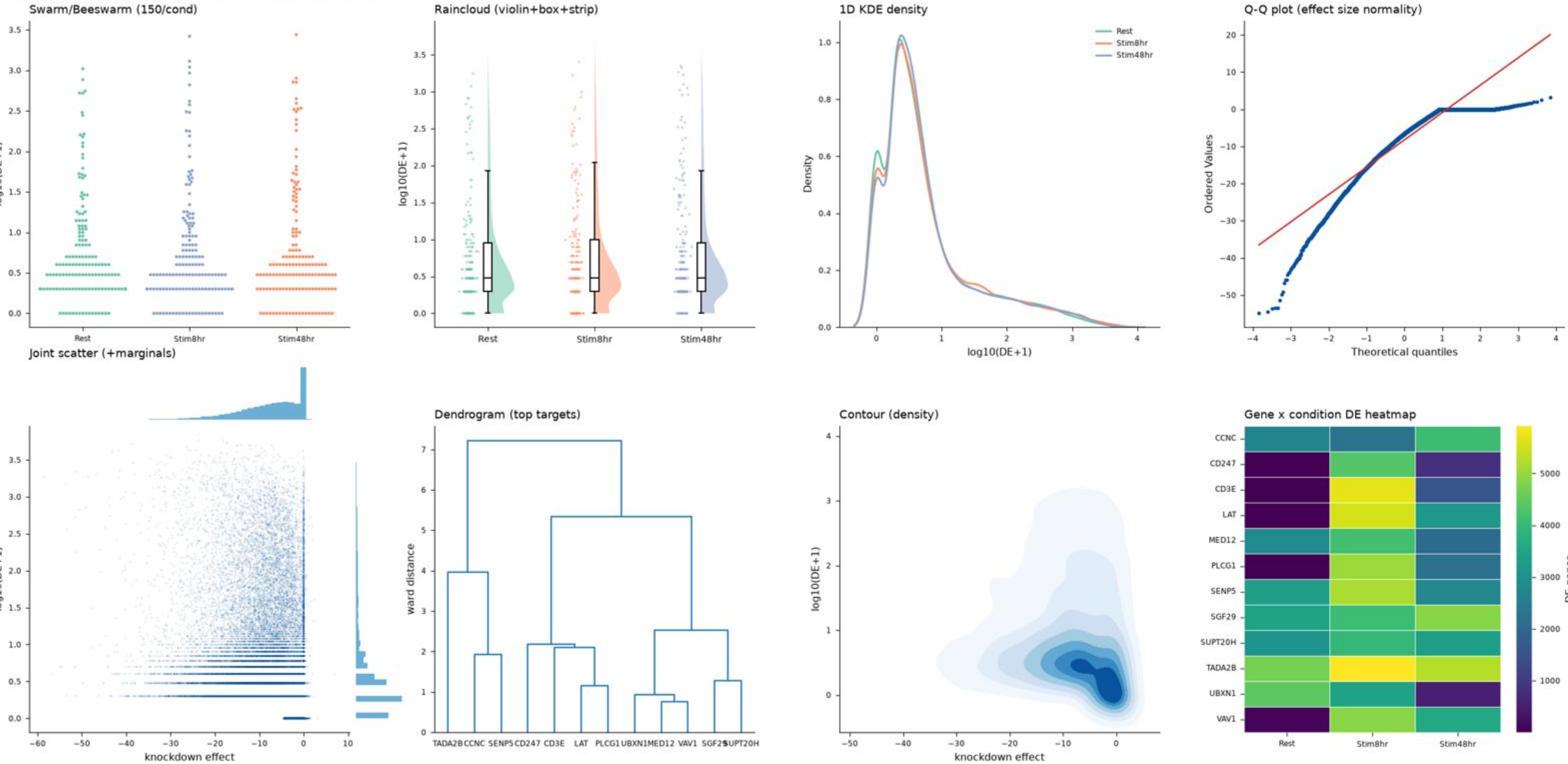
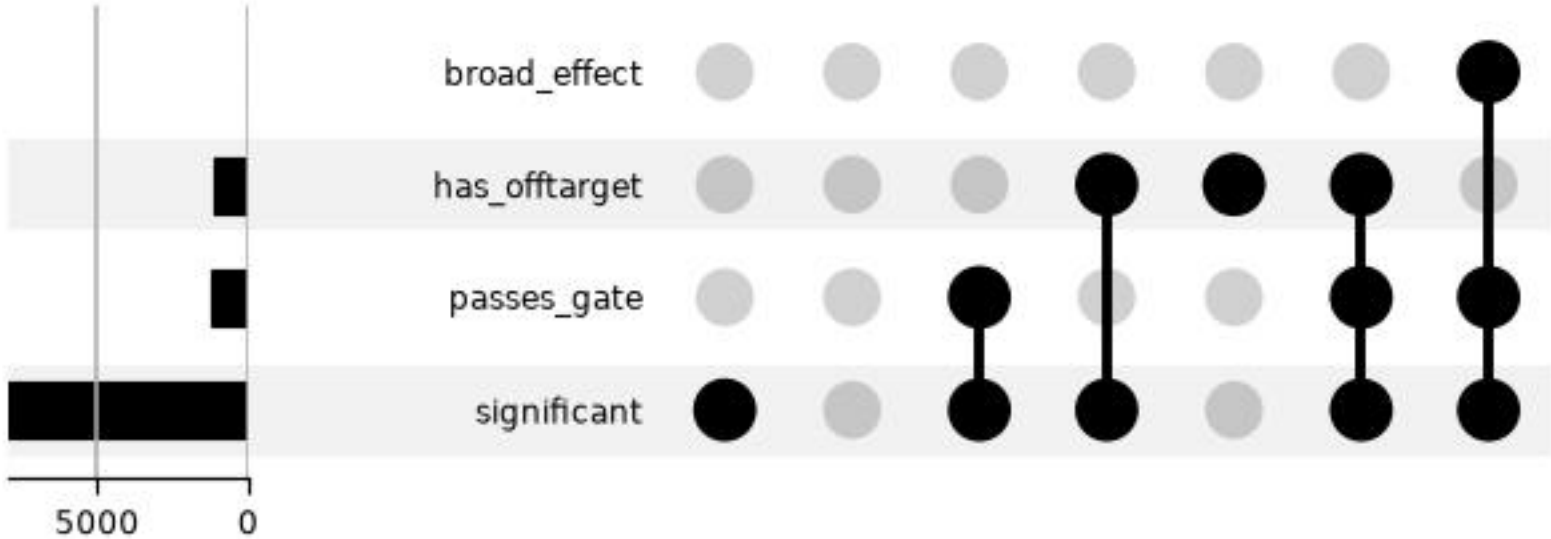
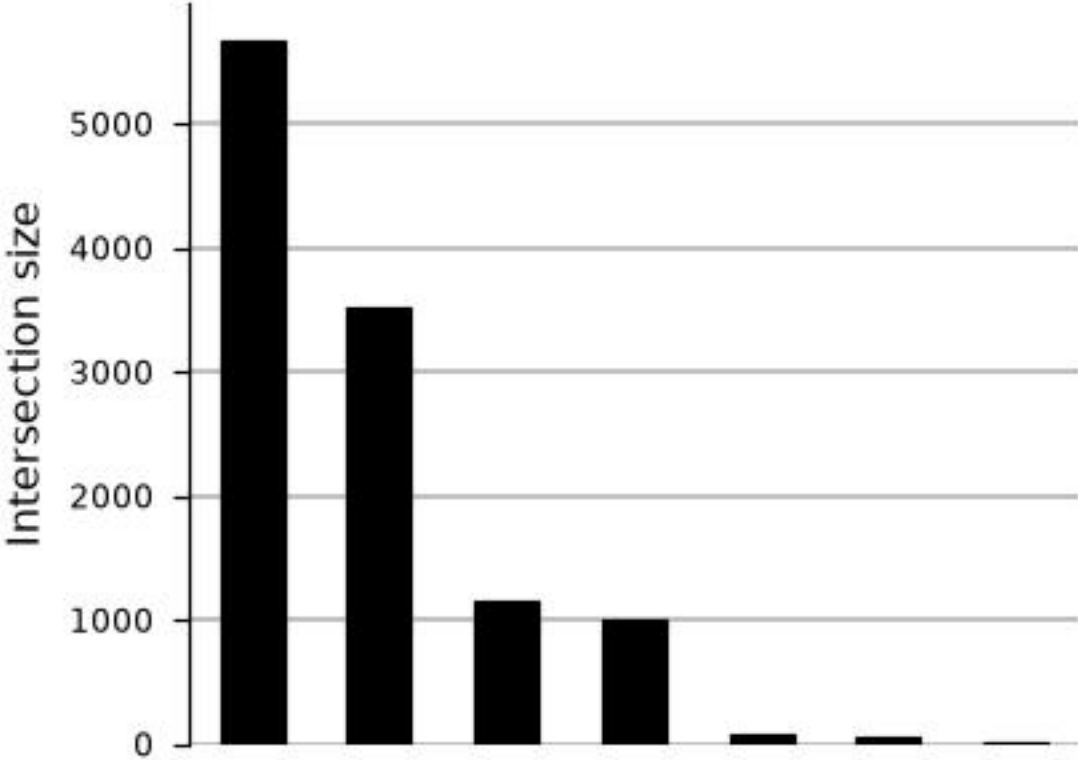


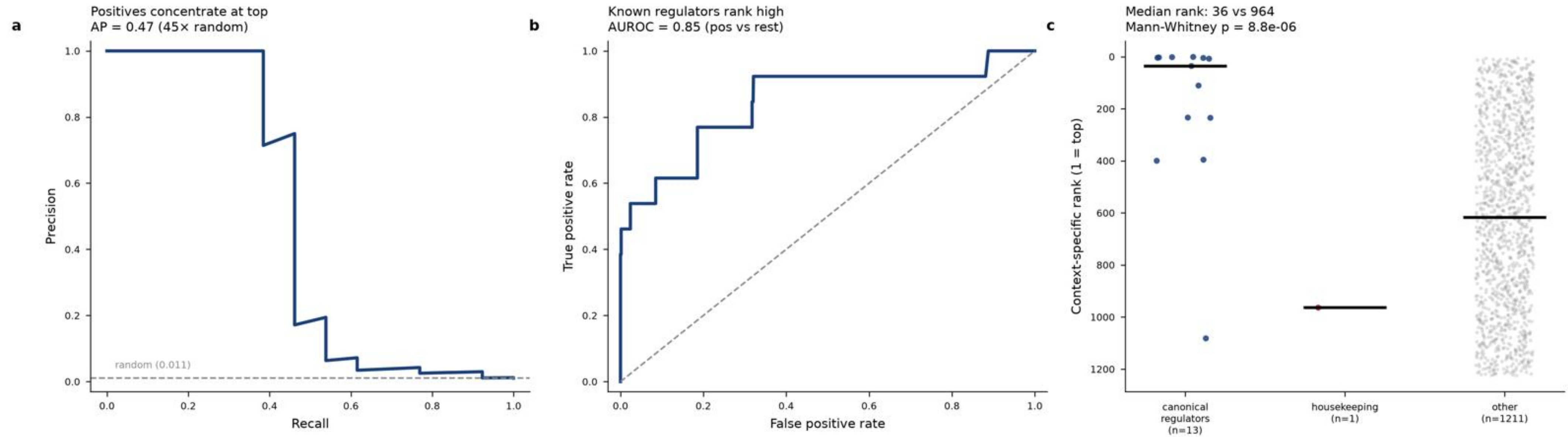
Chart gallery - Batch 6: distributions + matrices (completion)



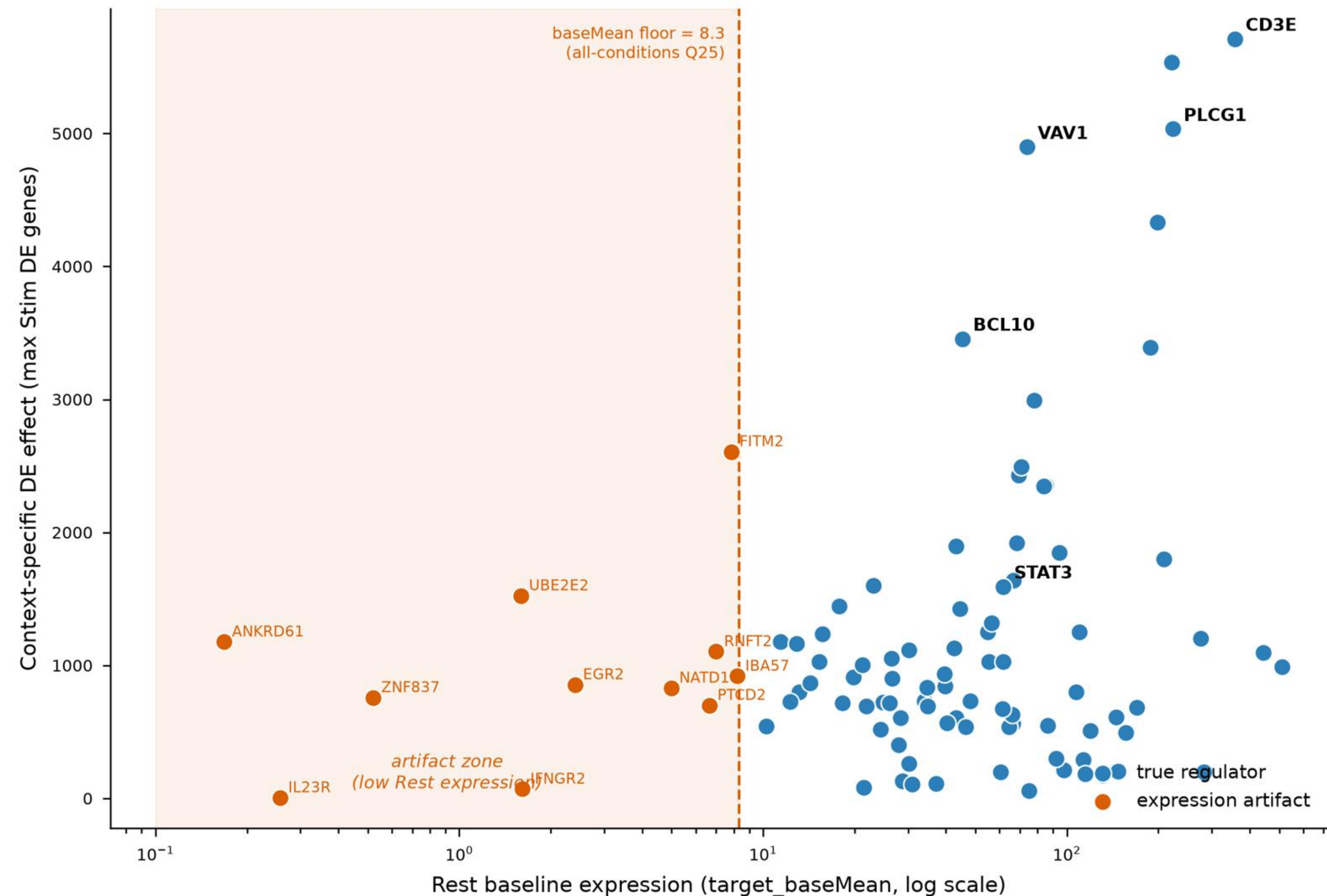
UpSet plot - evidence set intersections (11,526 targets)



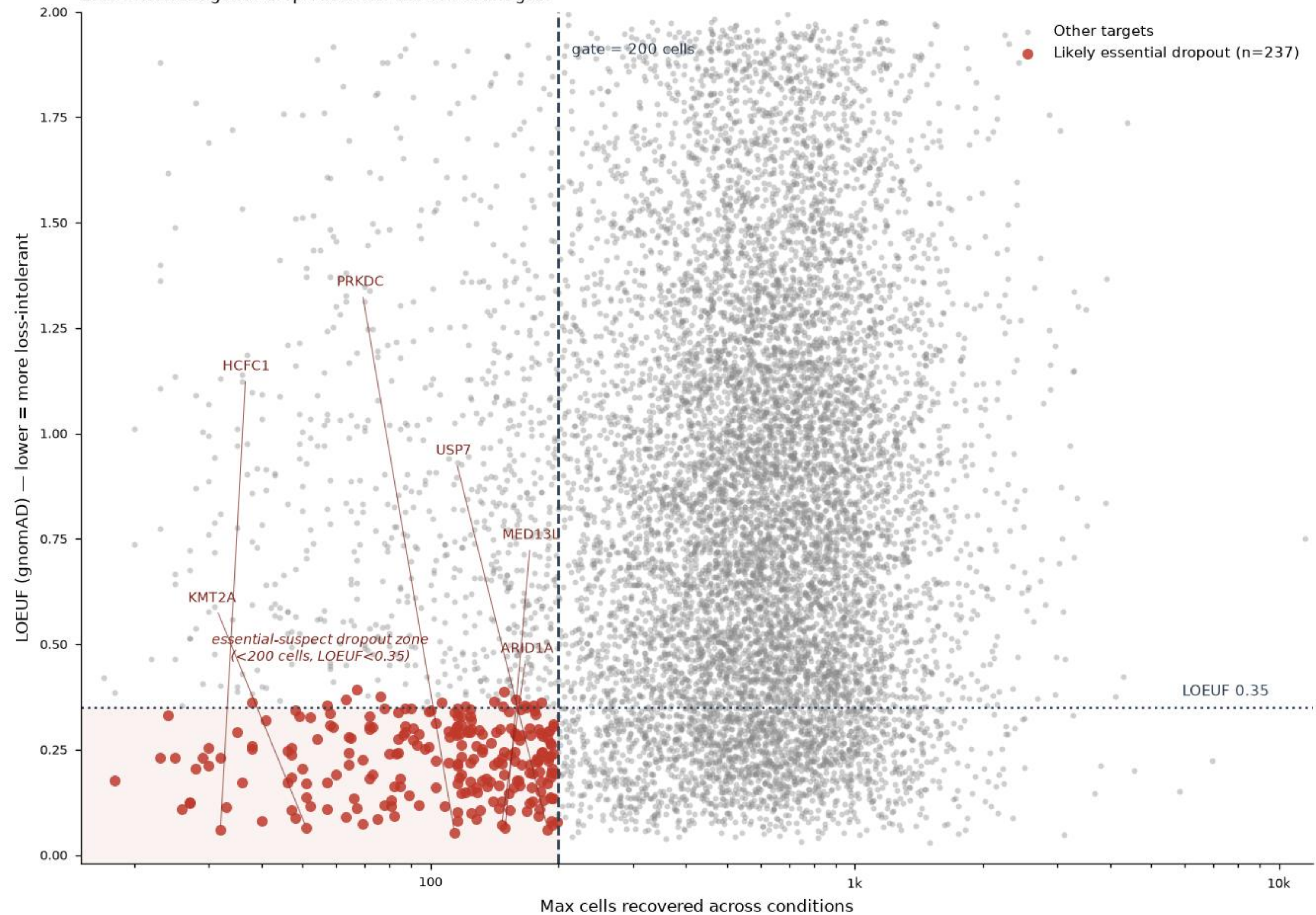
Ranking benchmark against canonical CD4 T-cell regulators (ground truth independent of concept modules)



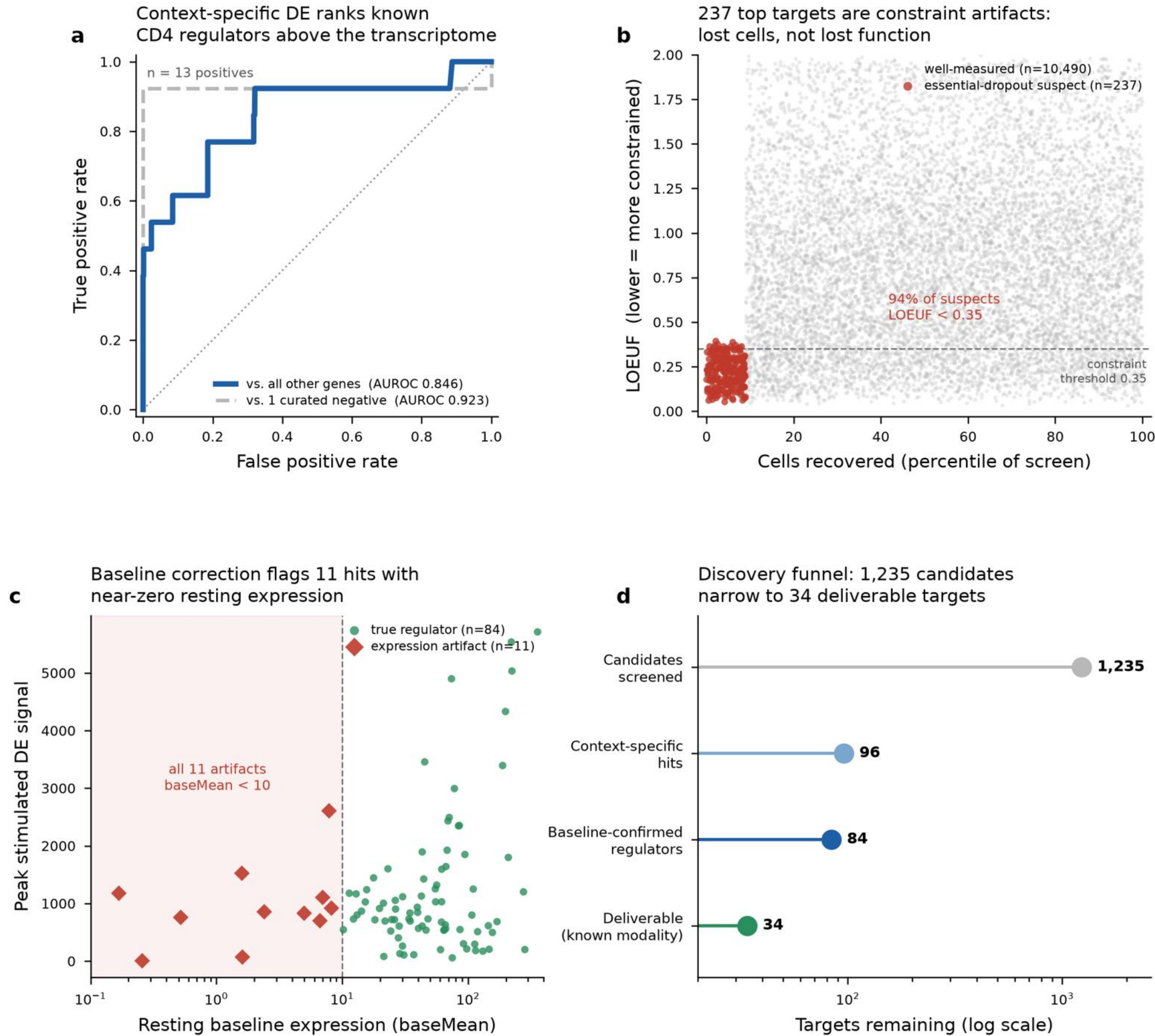
Baseline expression separates true context-specific regulators from artifacts

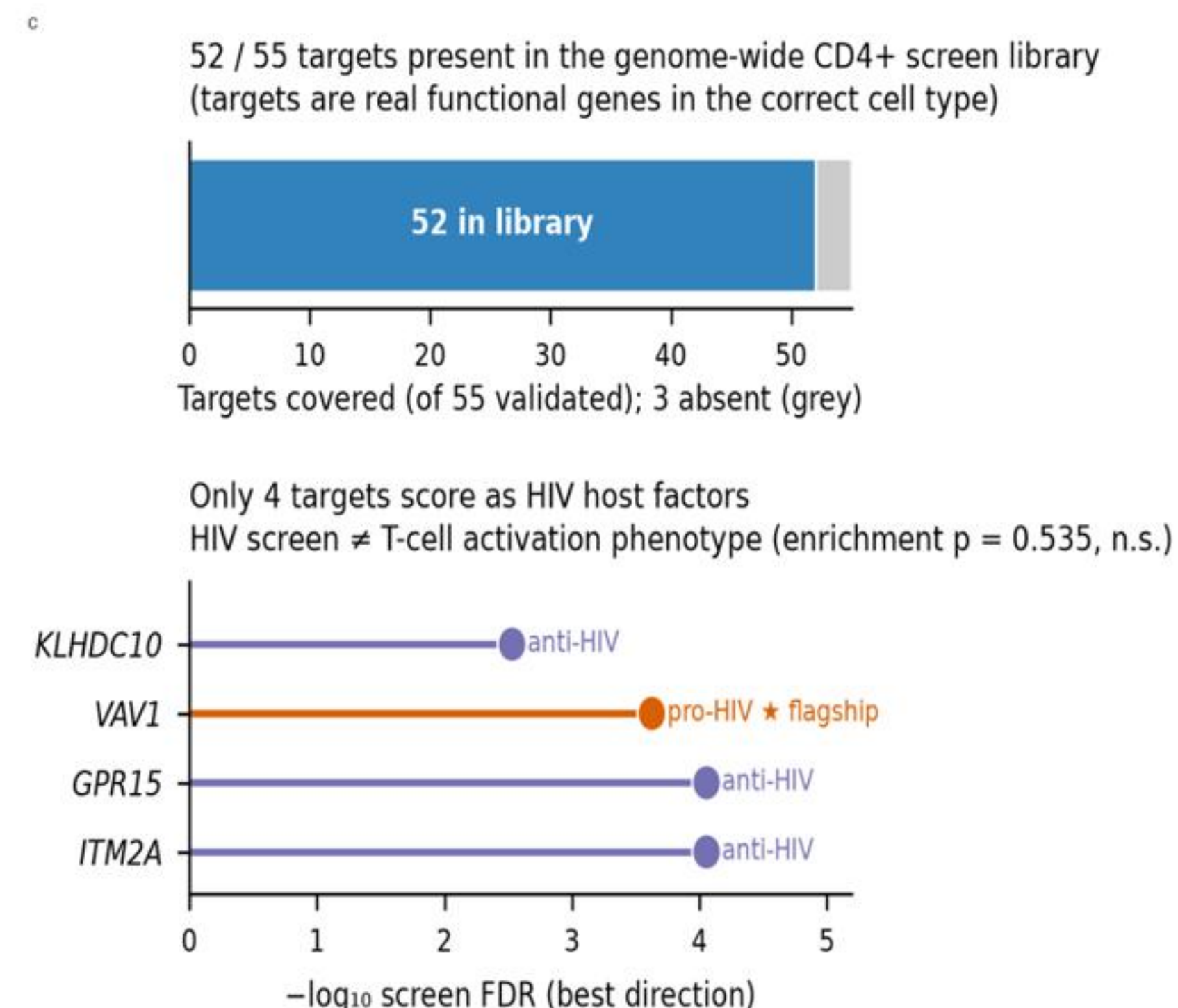
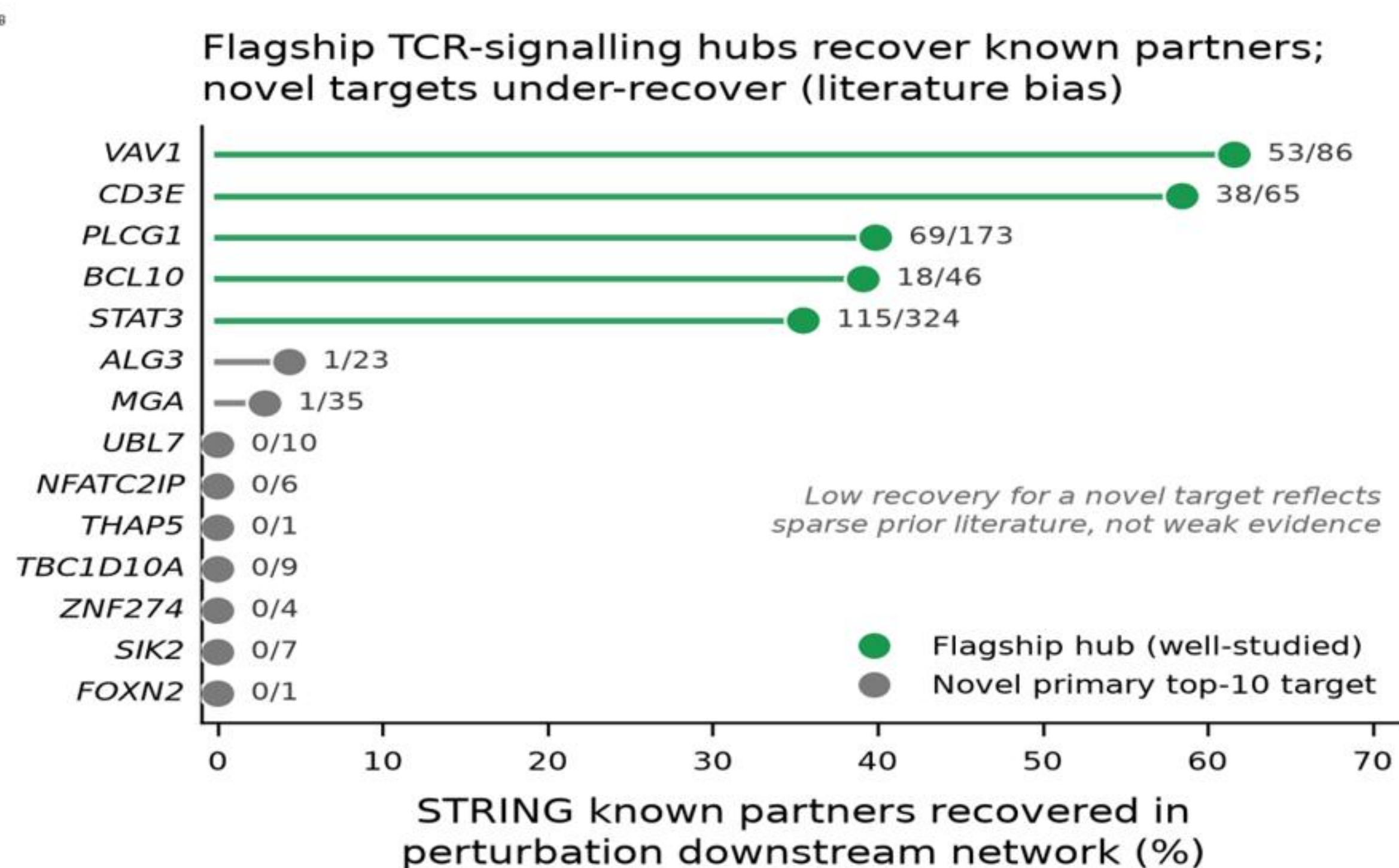
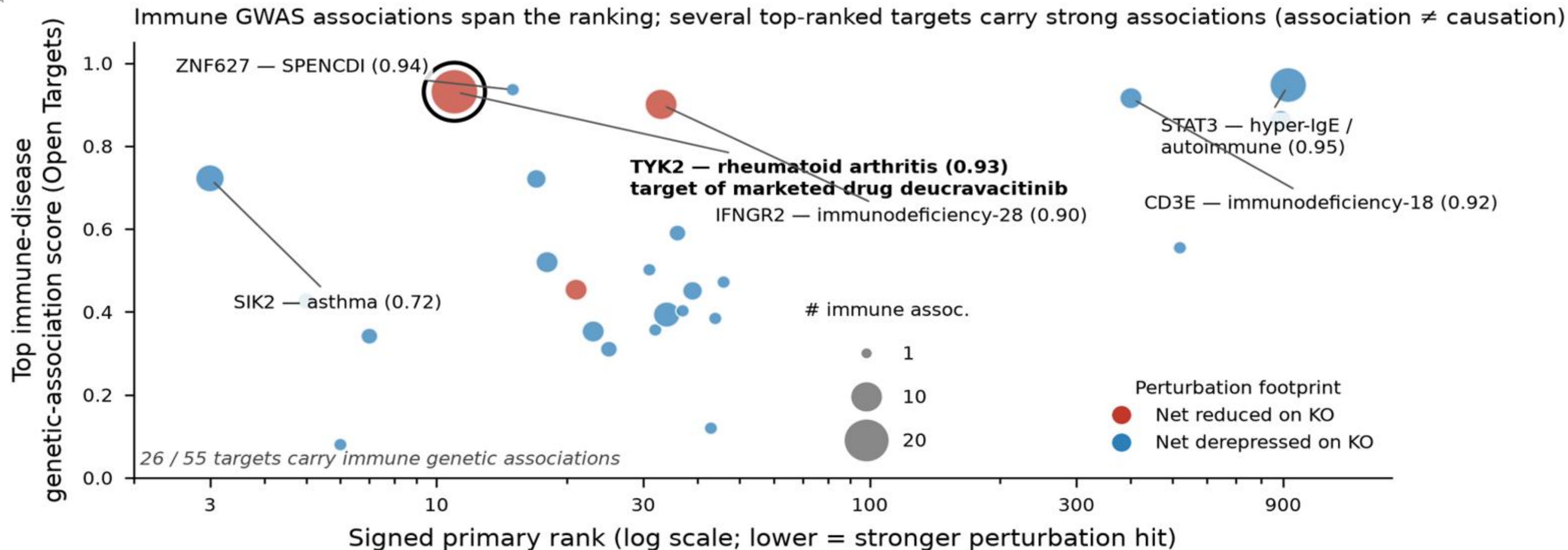


Loss-intolerant genes drop out below the cell-count gate



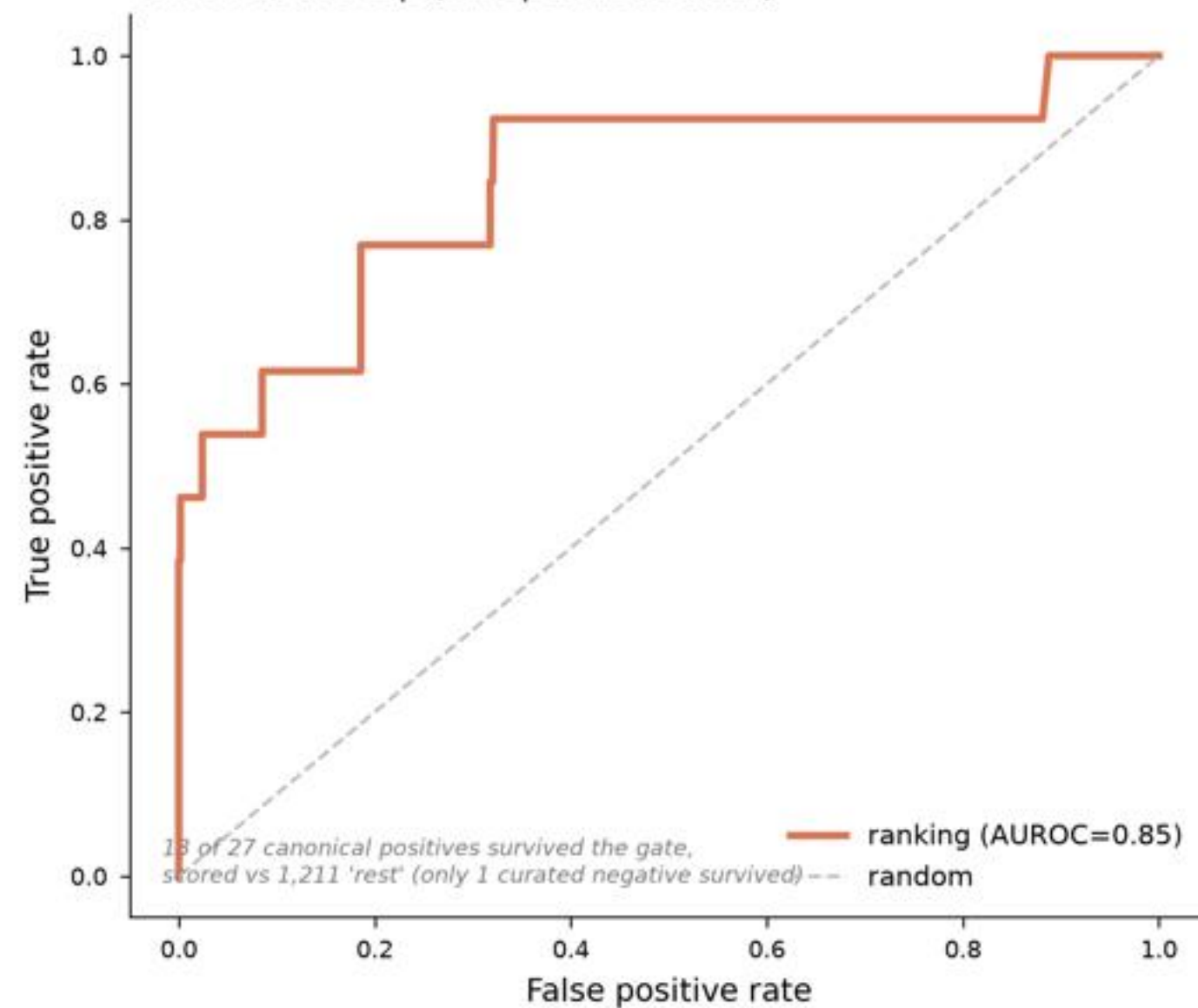
How we know a CD4⁺ T-cell Perturb-seq target is real



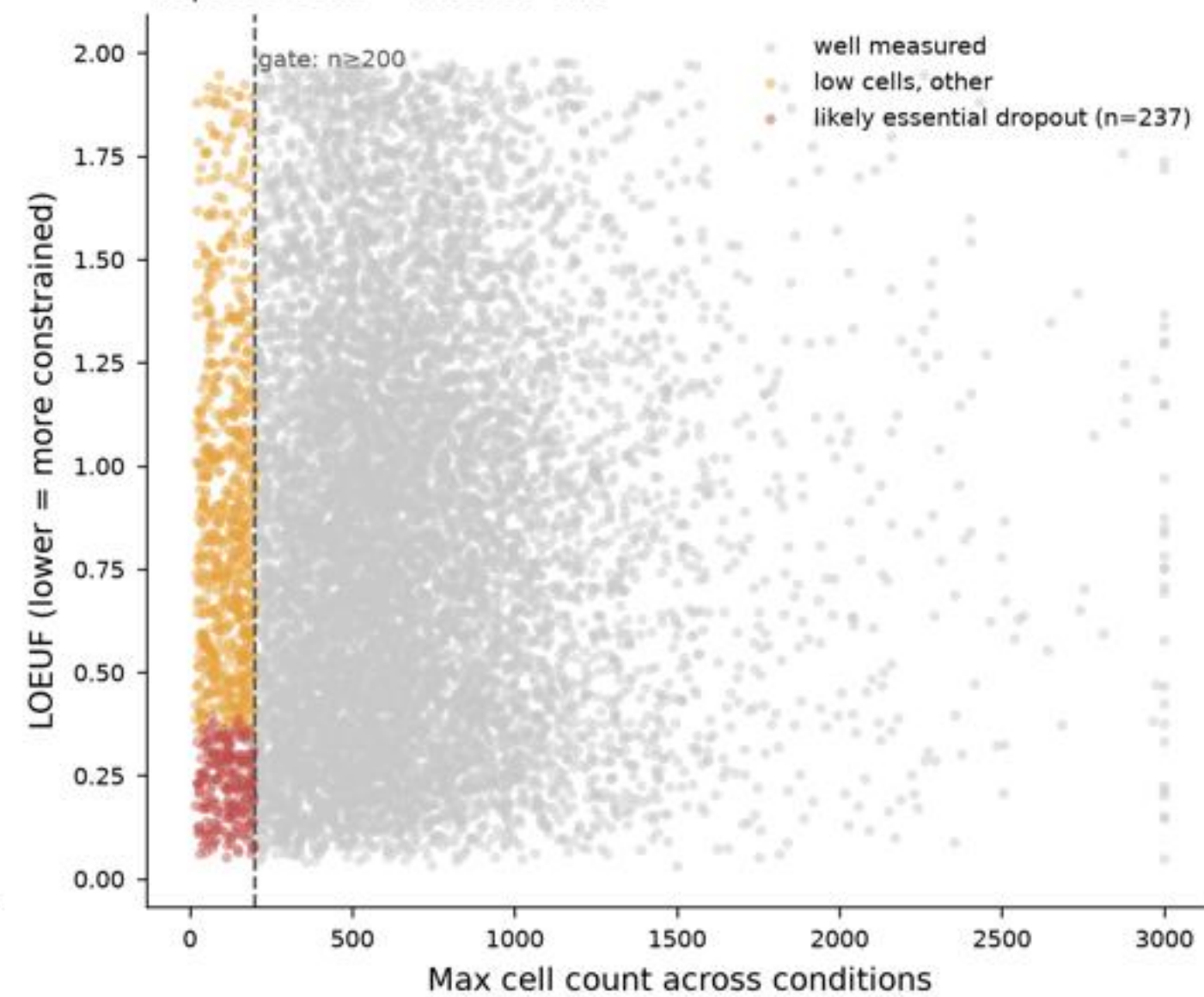


Methodological validation of the target ranking (all GB10-free)

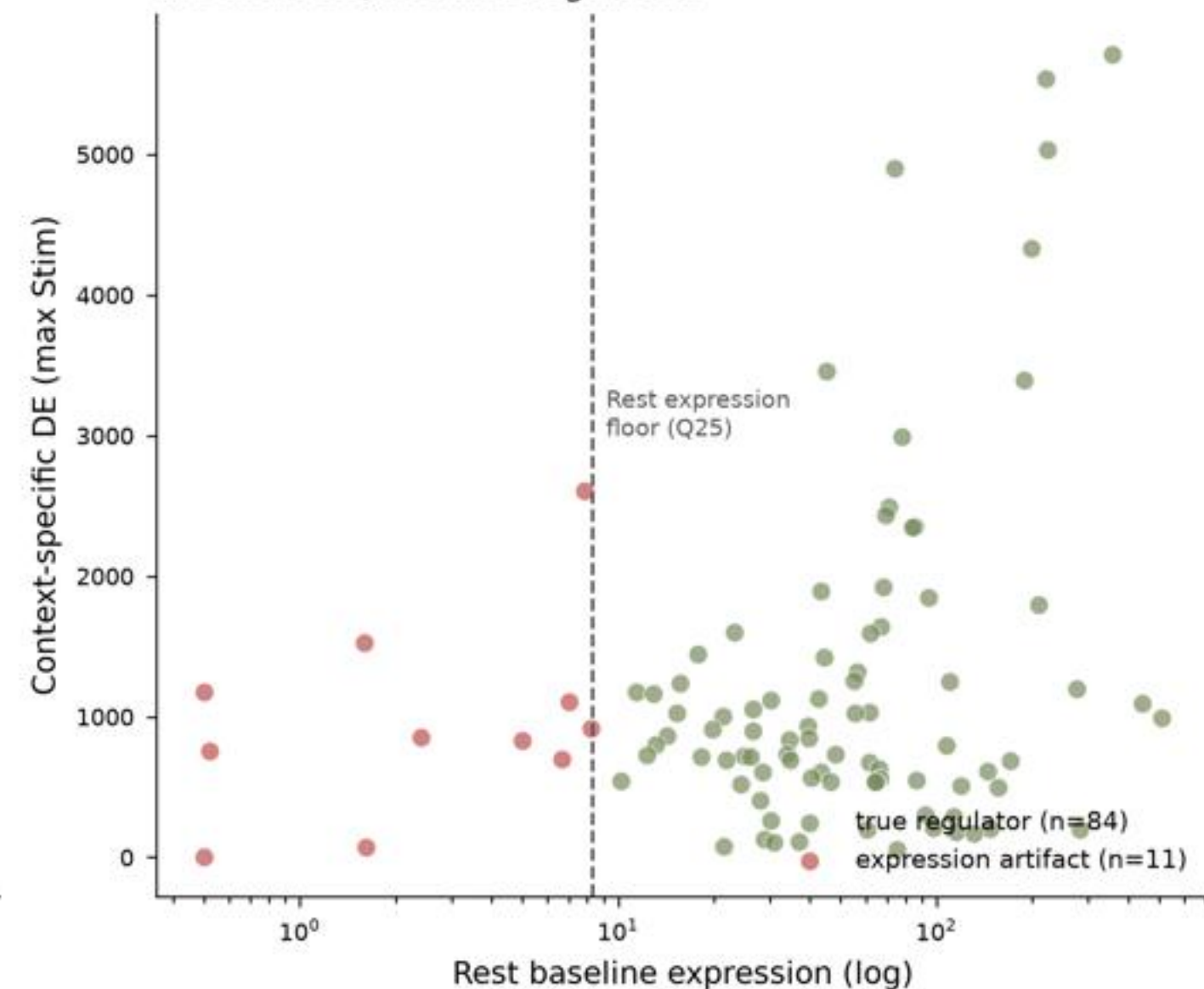
A · Benchmark: known regulators enrich at the top (independent truth)



B · Survivorship bias: essential genes deplete cells → filtered out

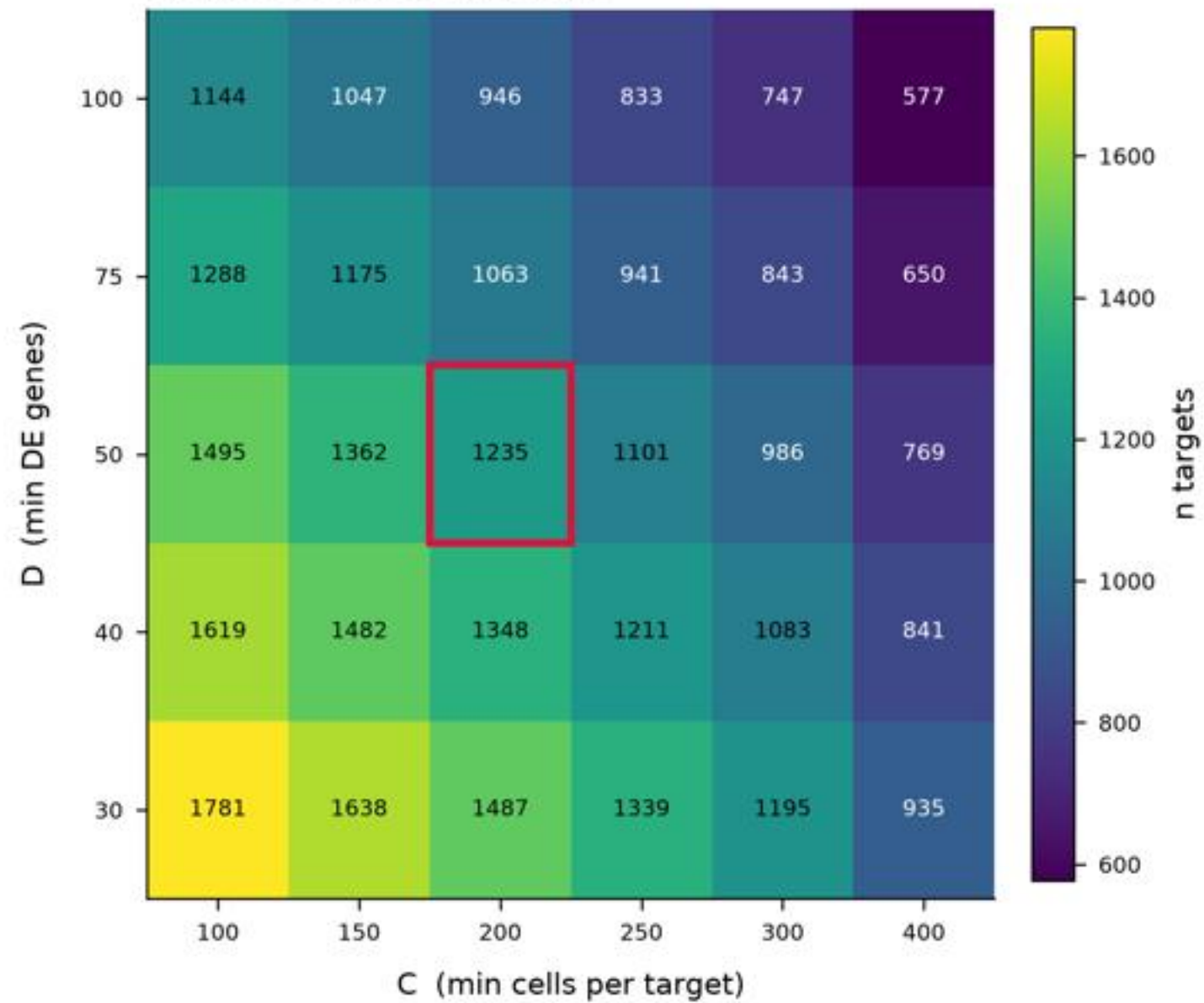


C · Expression control: 11/96 candidates are artifacts, not true regulators

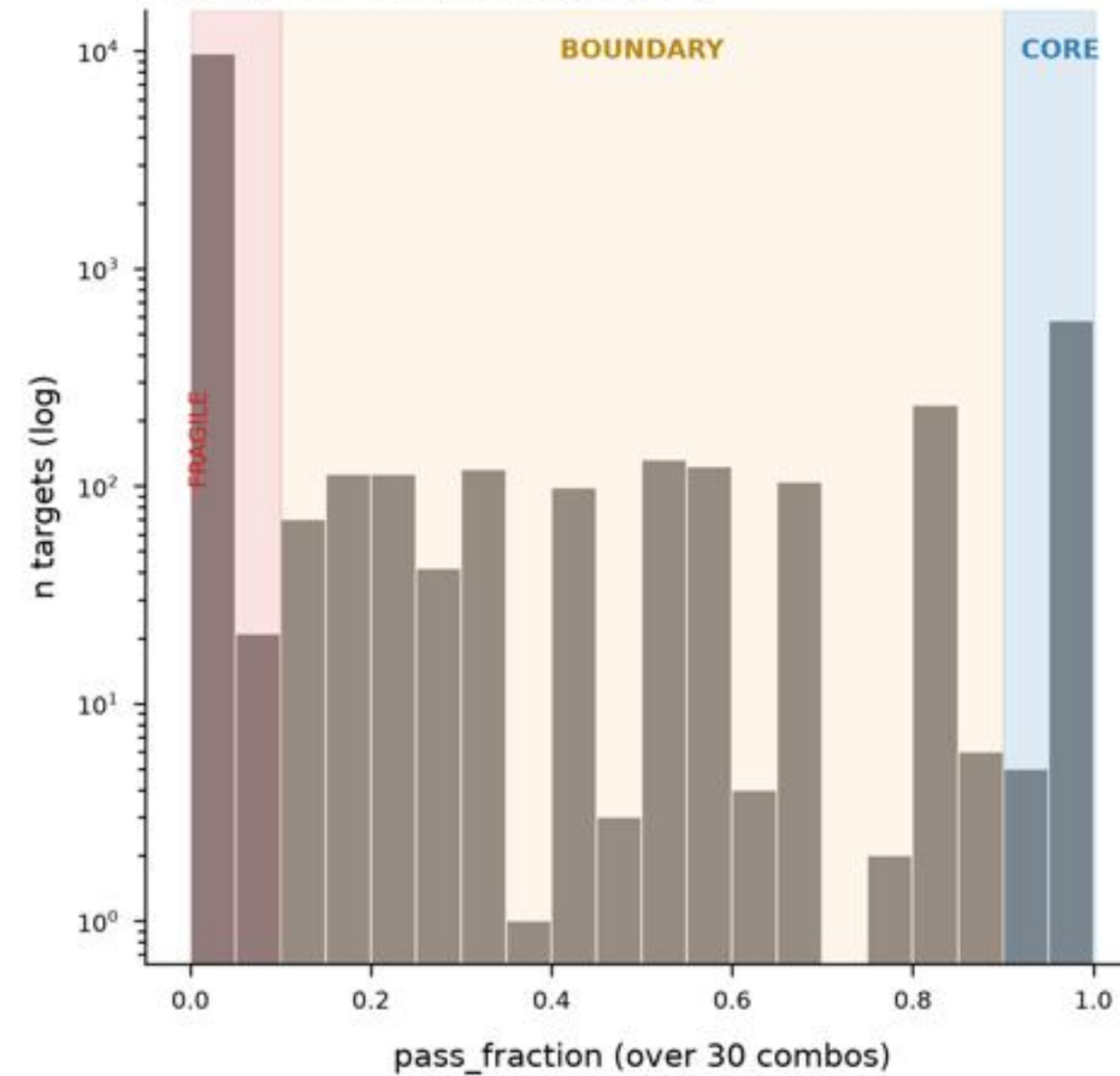


a

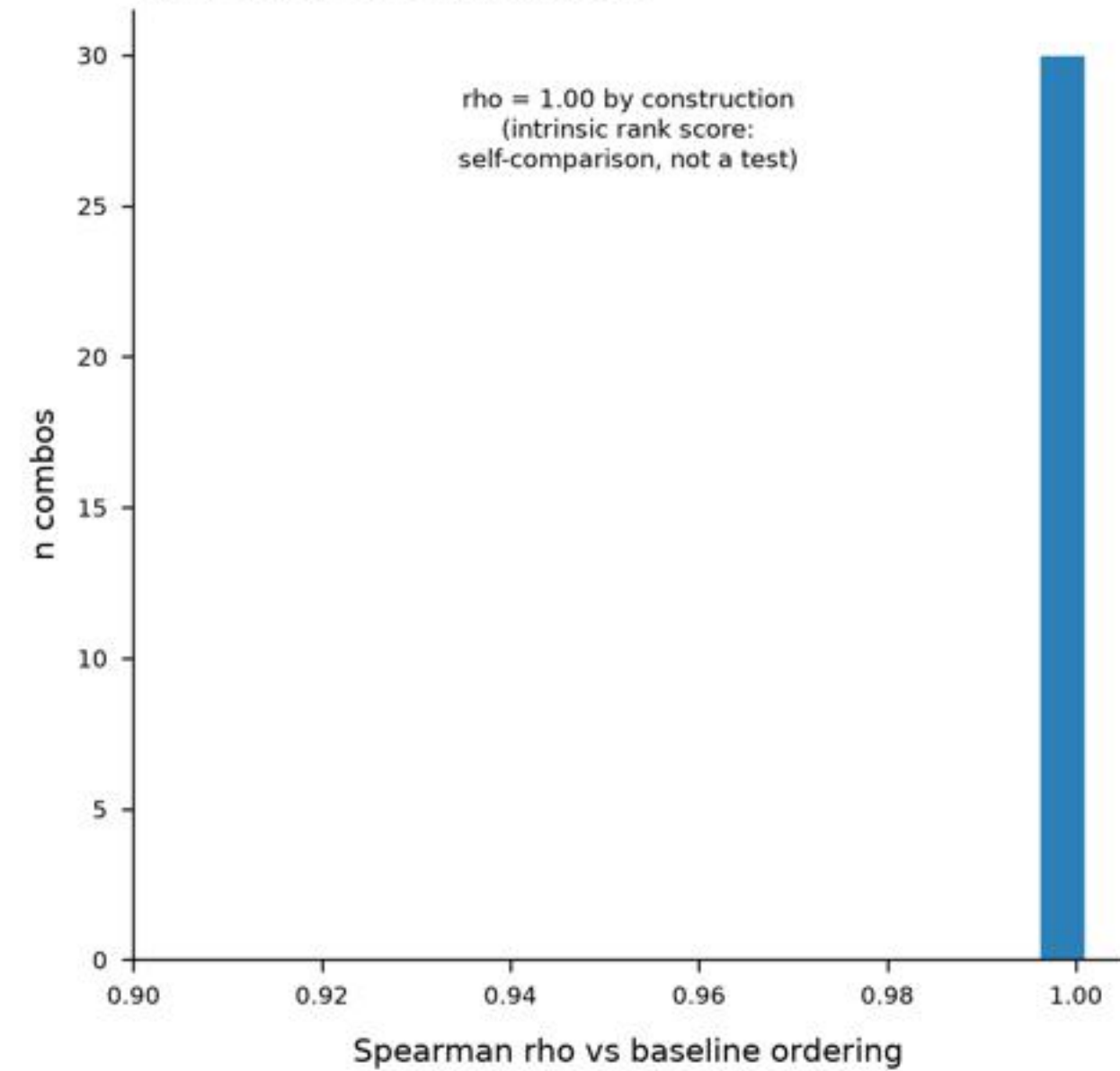
Shortlist size across thresholds

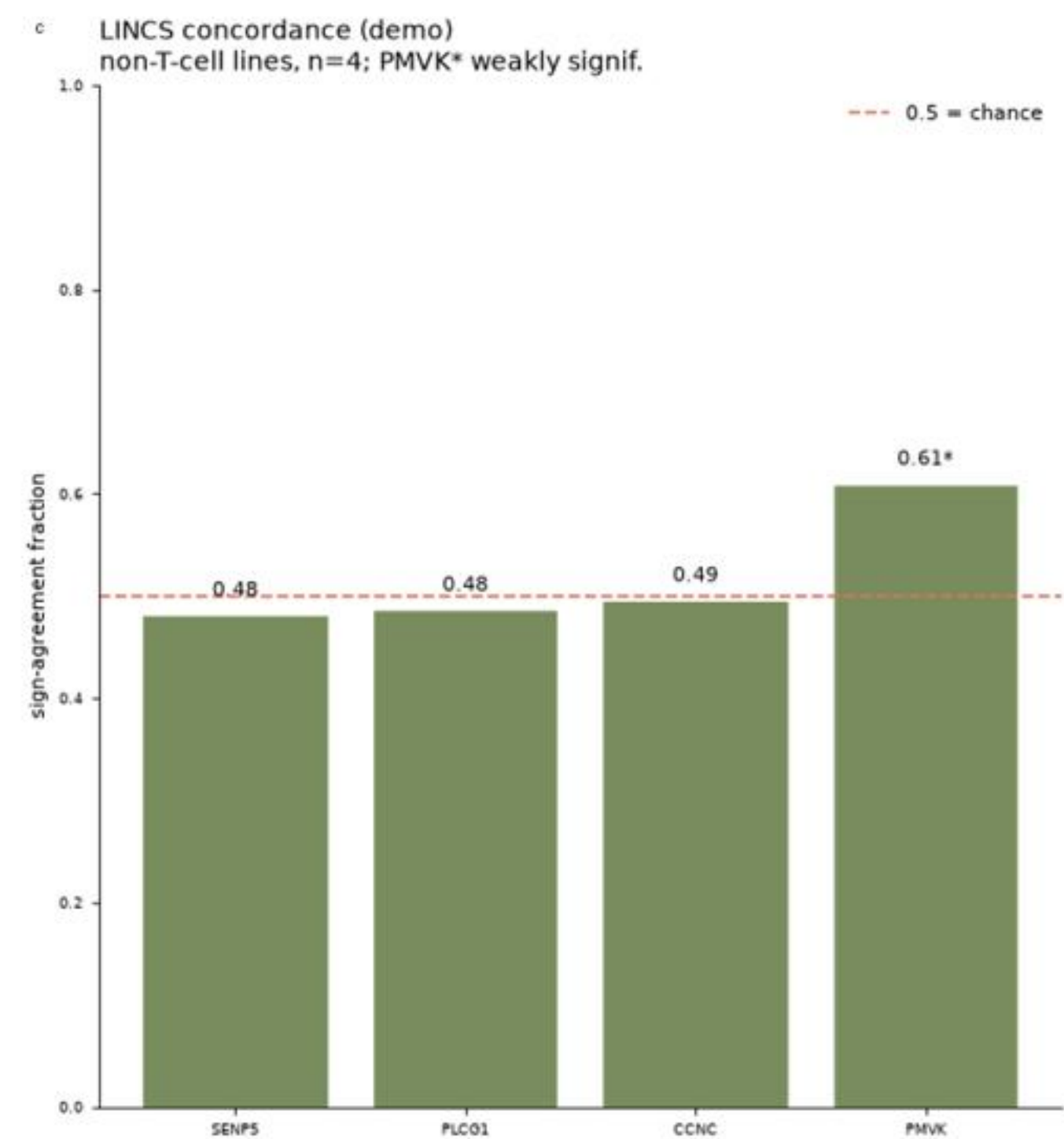
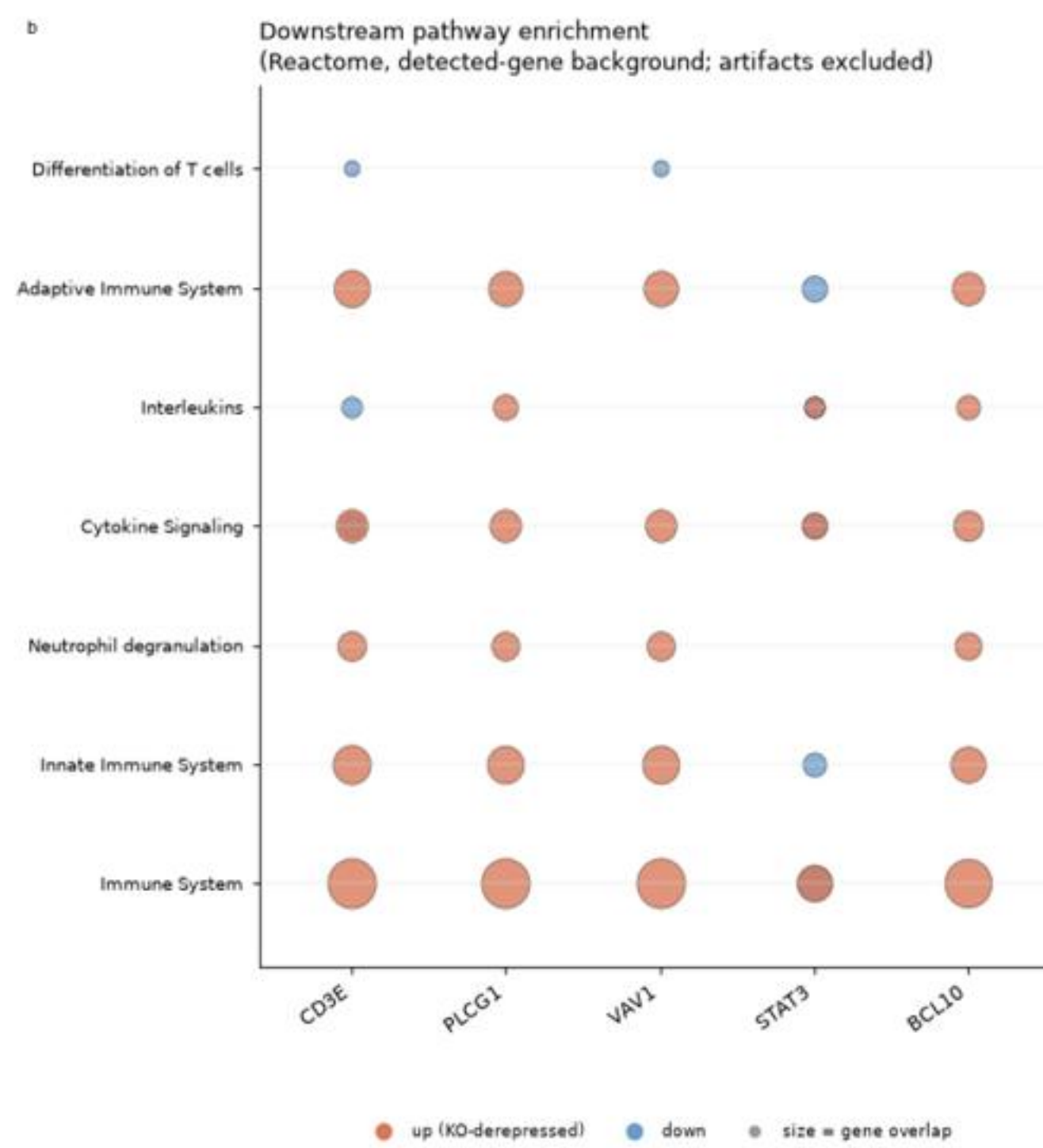
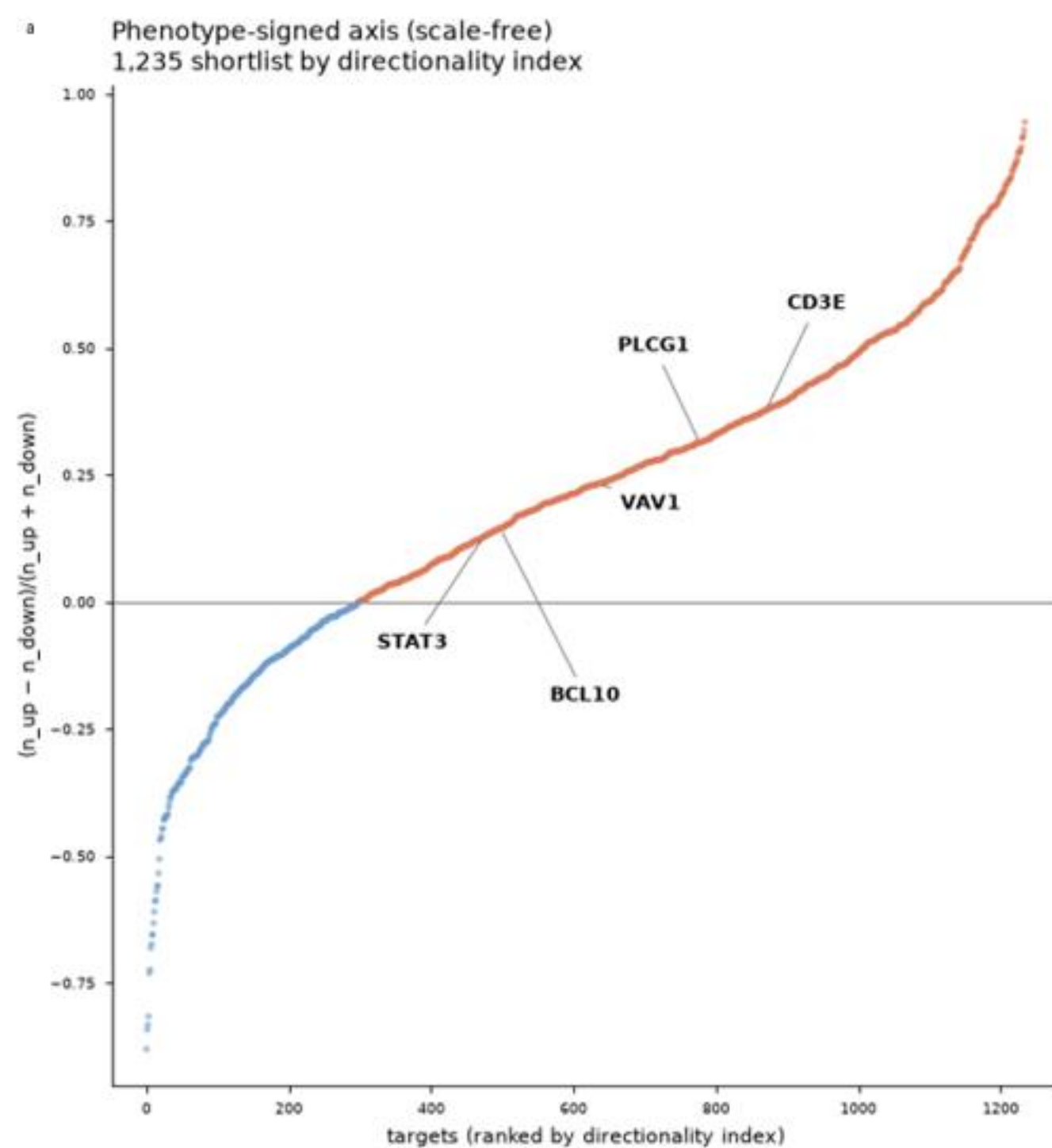


Stability over full universe (11,526)

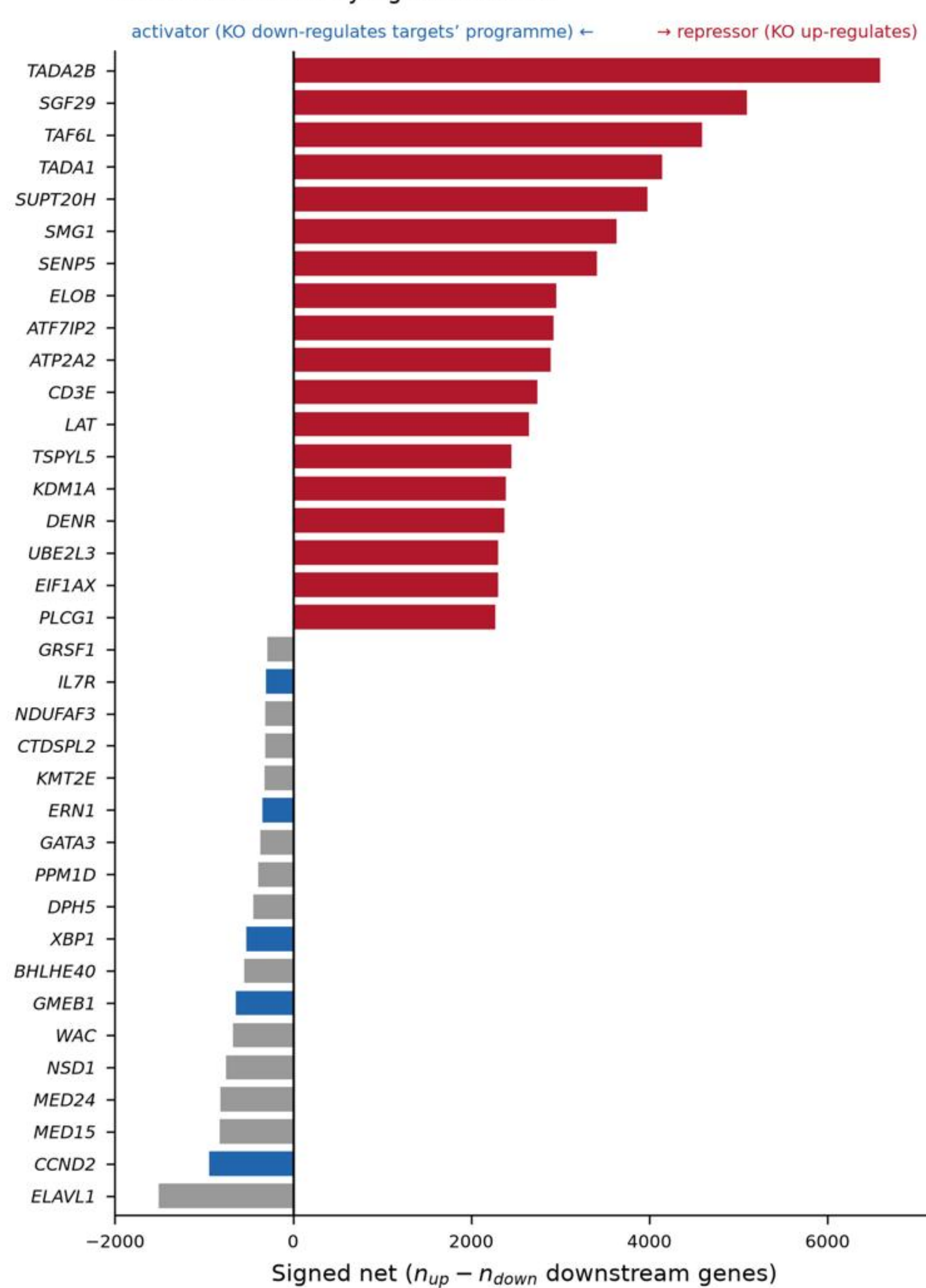


Rank score is threshold-invariant

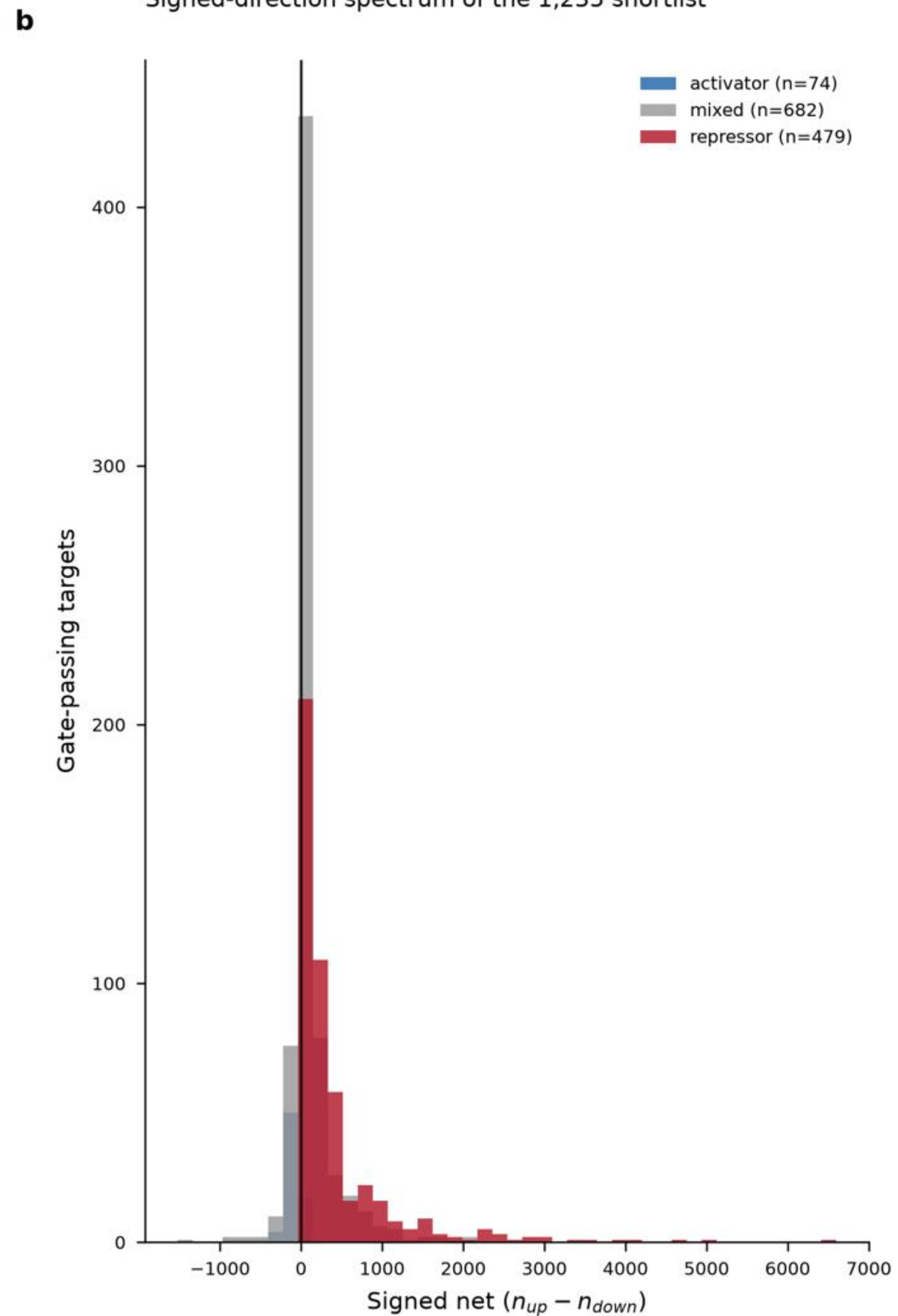




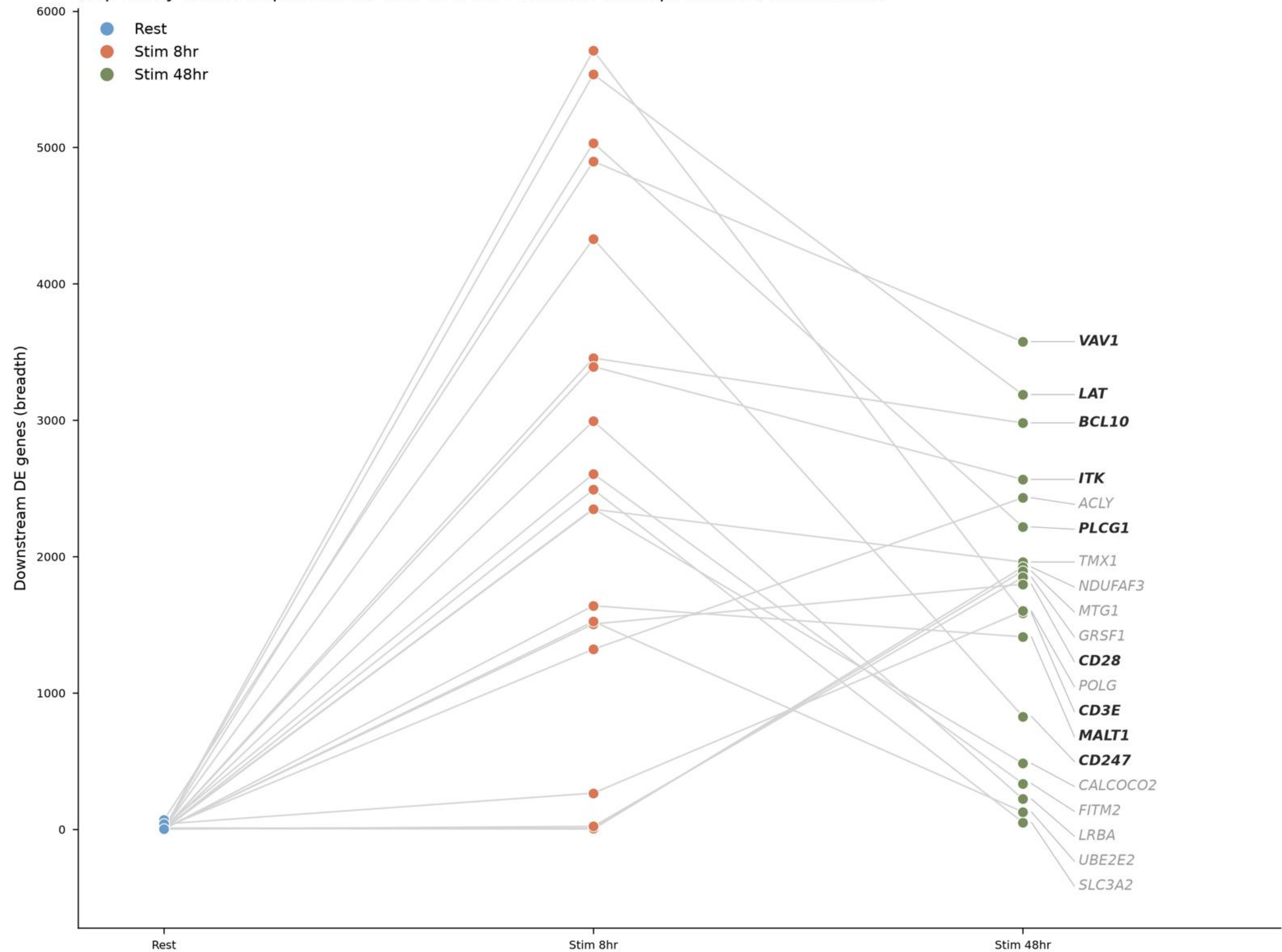
Shortlist extremes by signed direction



Signed-direction spectrum of the 1,235 shortlist



Context-specific shortlist: knockdown phenotype is silent at rest, surges on stimulation
(top 20 by context-specific DE; 9/20 in a CD4 immune concept module, shown bold)

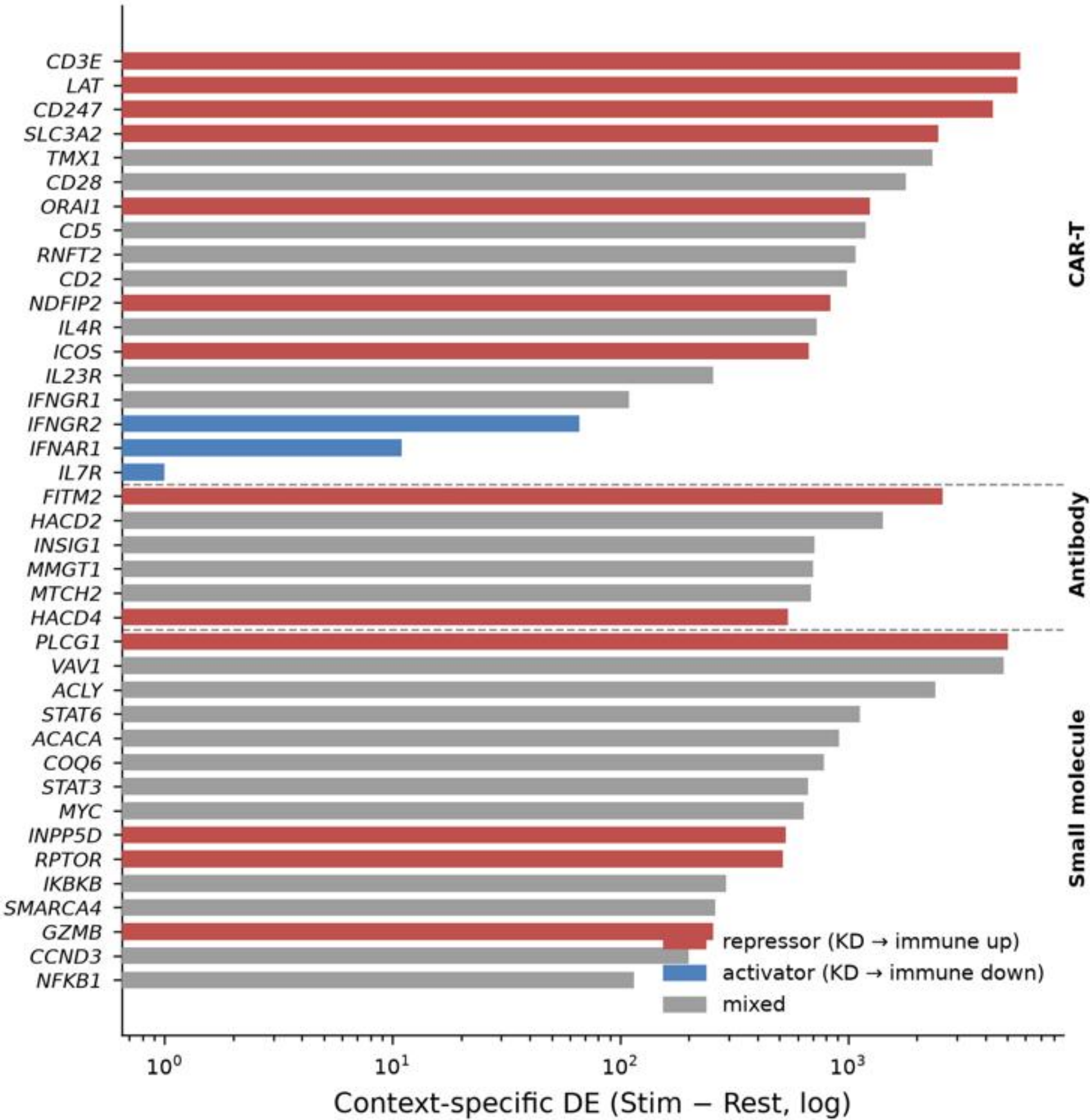


Immune drug-delivery decision layer: turning "important" into "which drug can we make today"

Filter funnel: important → context-specific → deliverable

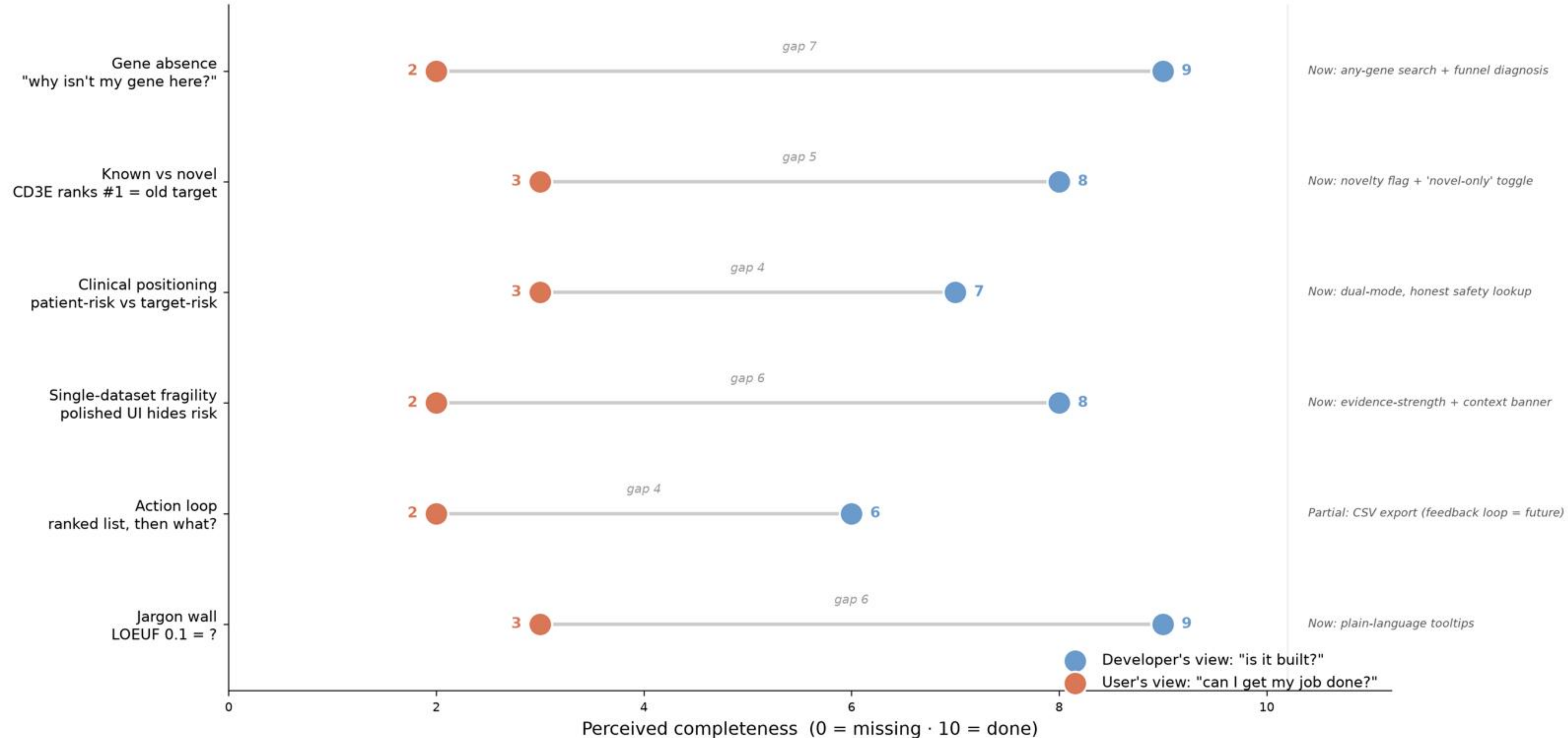


39 deliverable targets: modality (grouped) × polarity (colour)

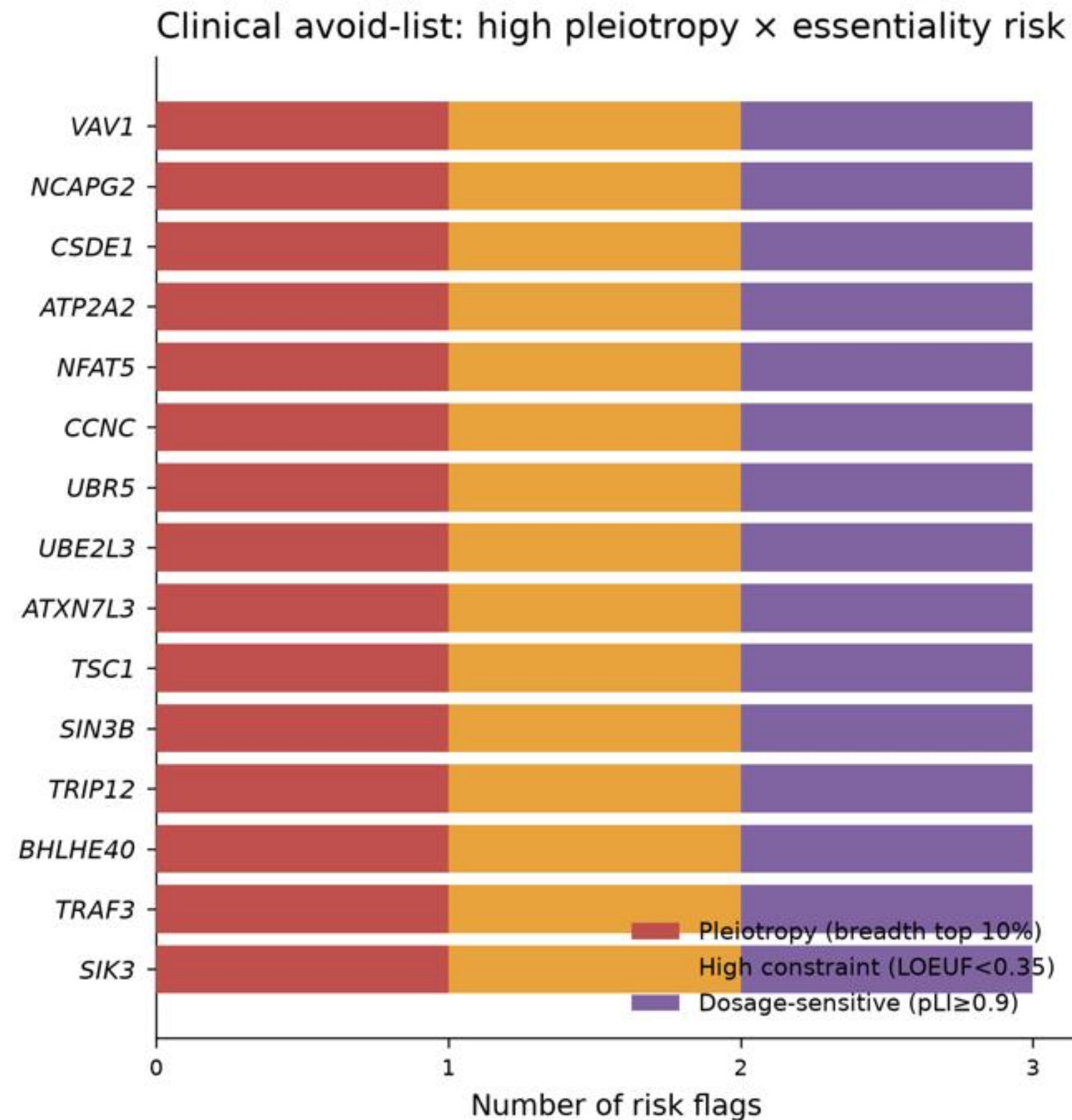
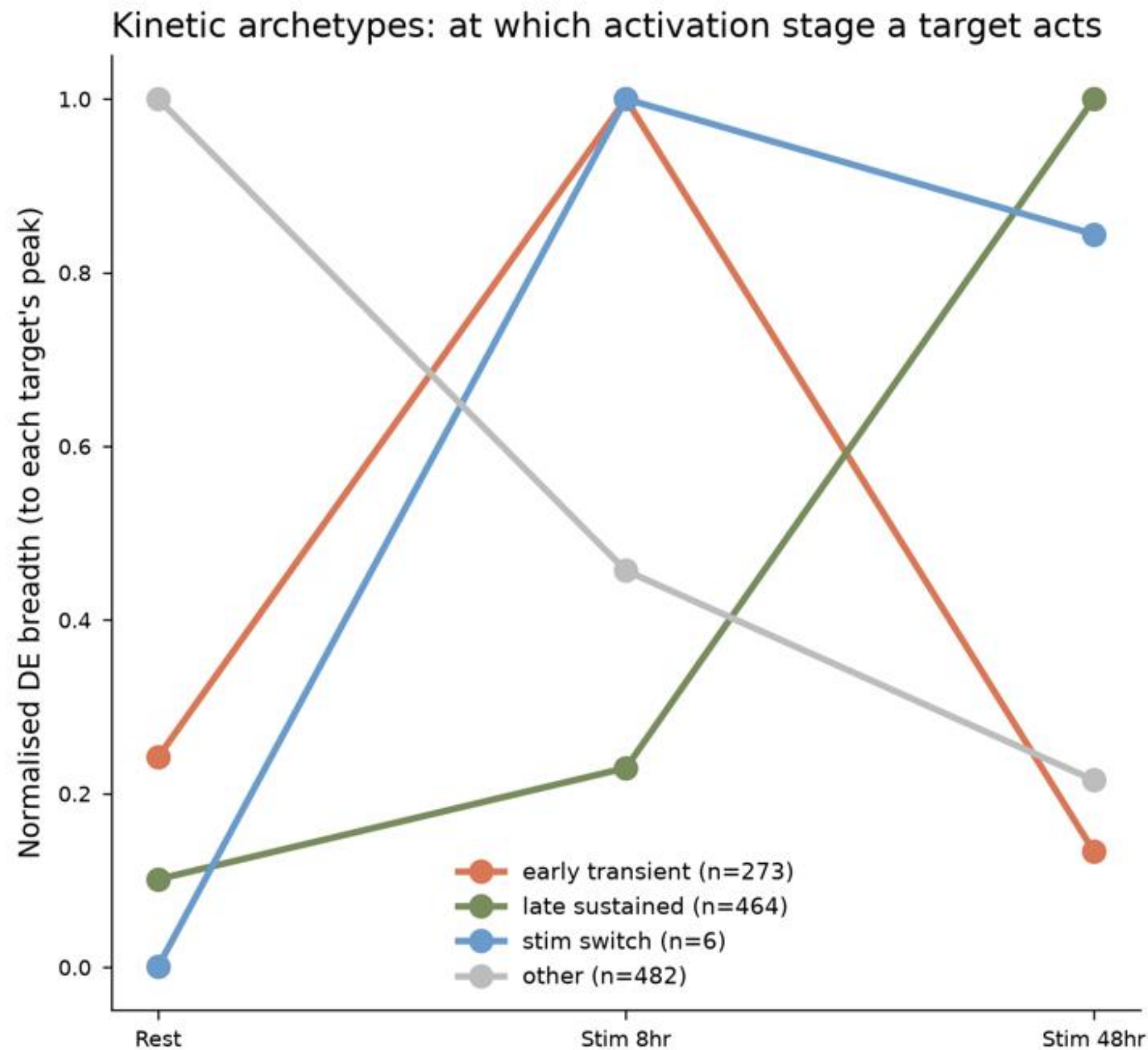


The developer-user gap: six blindspots

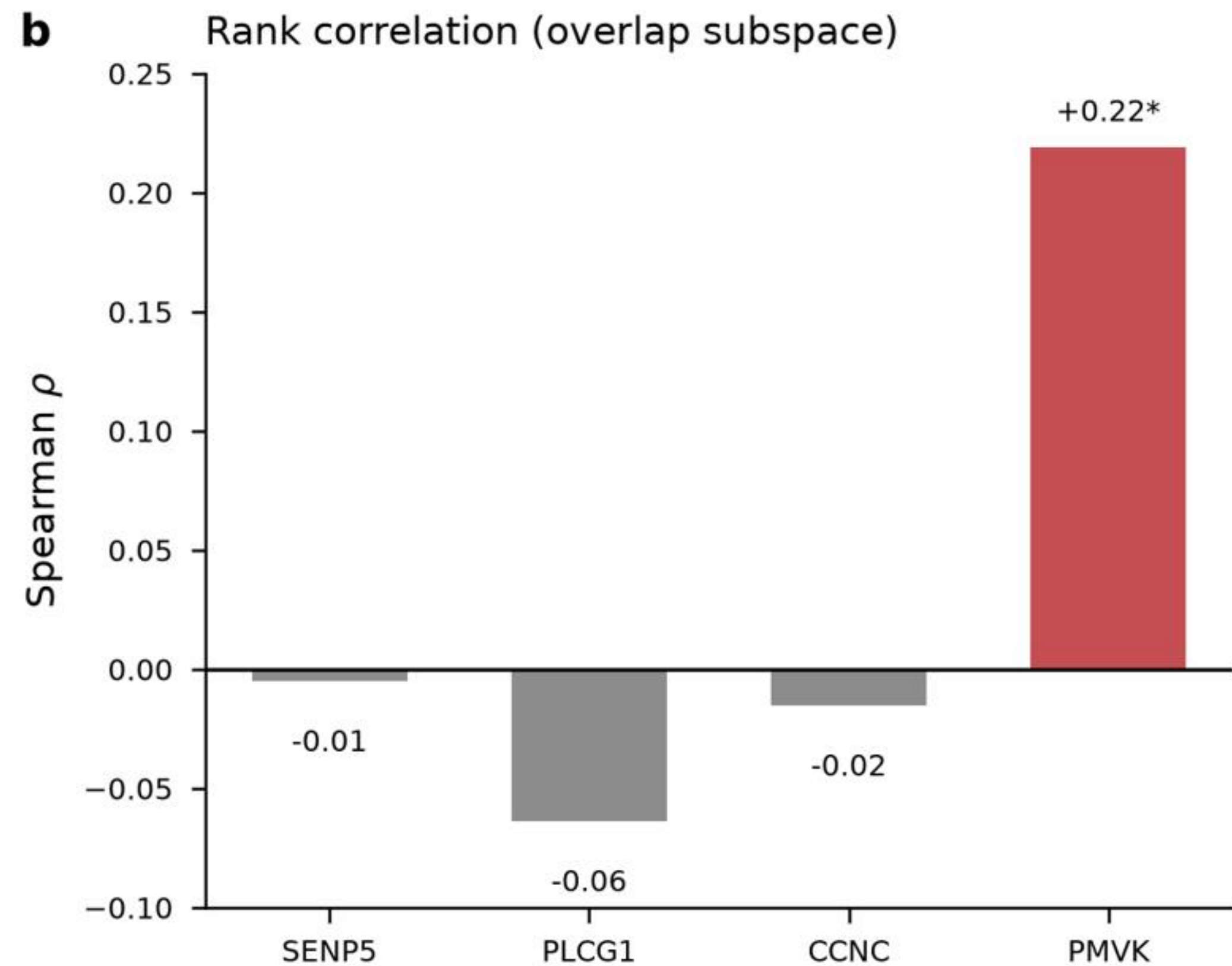
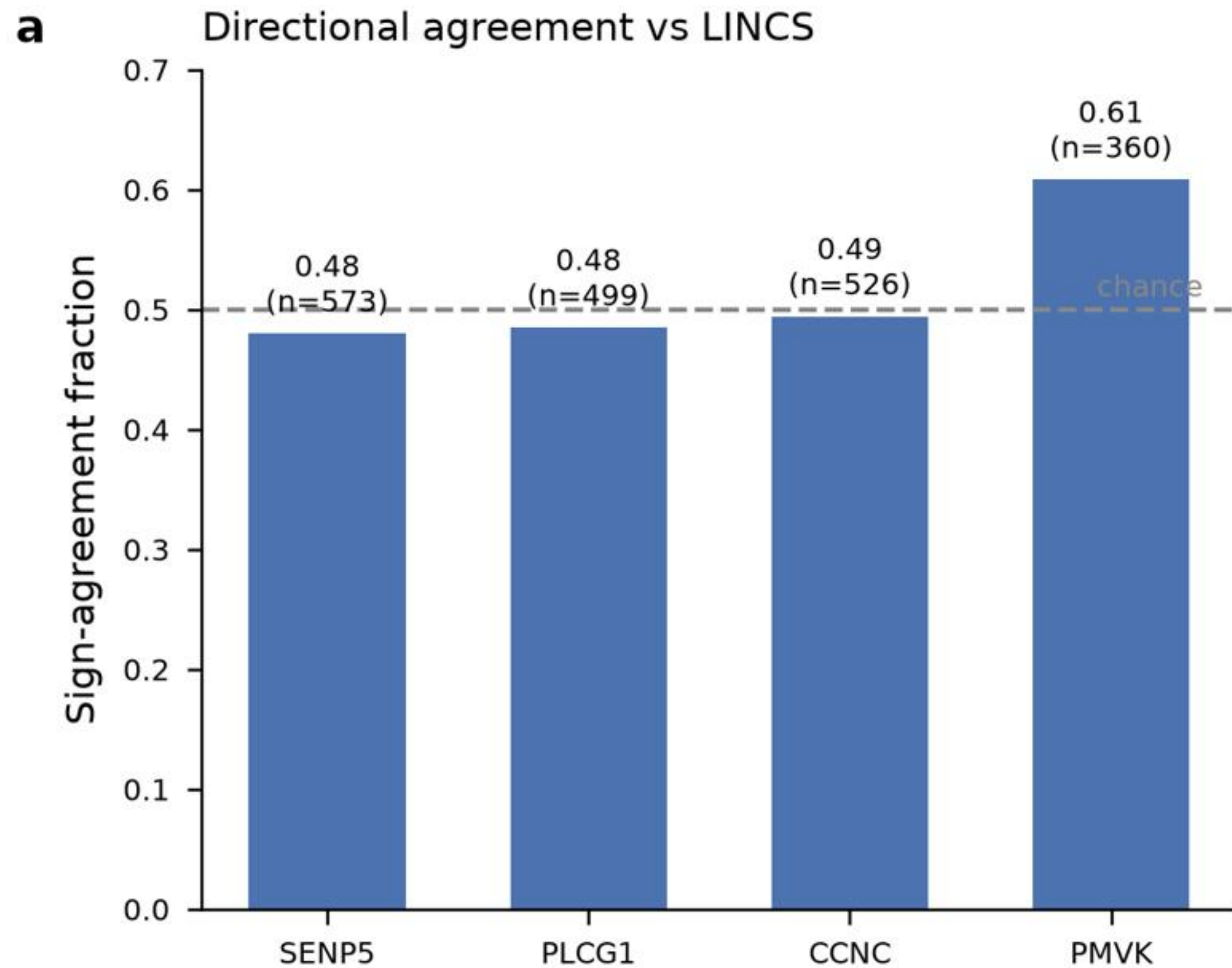
"Built" and "a user can actually use it" are different questions — the gap is where users feel the tool is incomplete



Kinetic archetypes (idea 3) + clinical avoid-list (idea 4) — both GB10-free

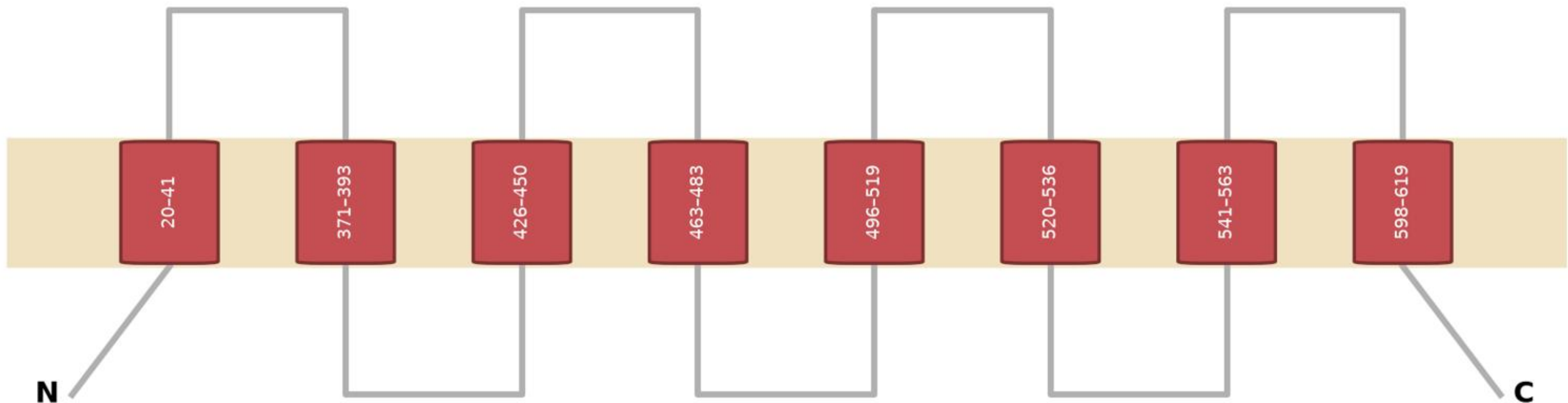


GB10 signed matrix vs LINCS — DEMO directional check (4 targets, non-T-cell lines)



GPAA1 (O43292) — GPI-anchor transamidase component GPAA1 · 8 TM

Extracellular / Luminal



Cytoplasmic

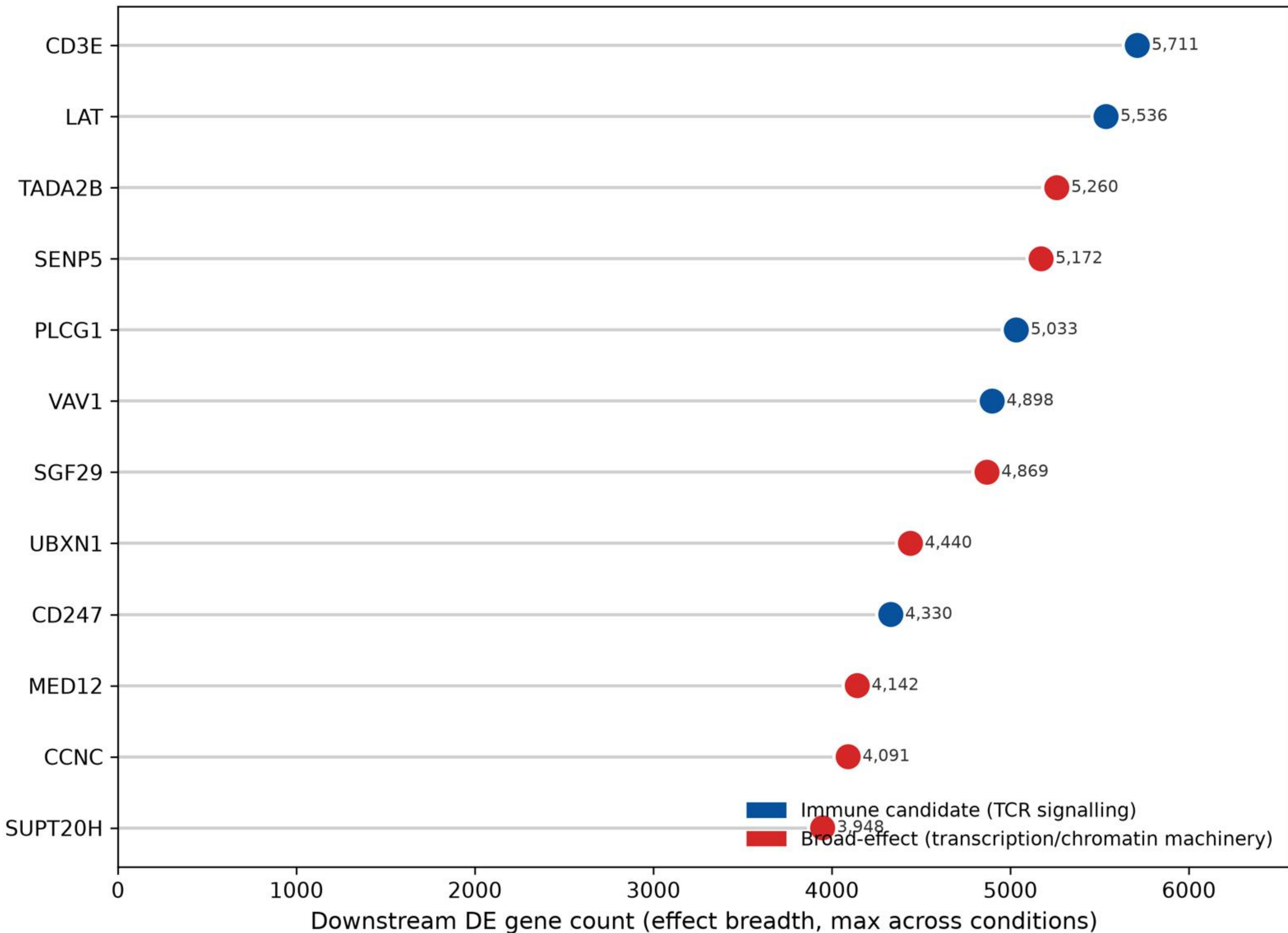
TRIM39 (Q9HCM9) — E3 ubiquitin-protein ligase TRIM39 · soluble (no predicted TM)



518 aa

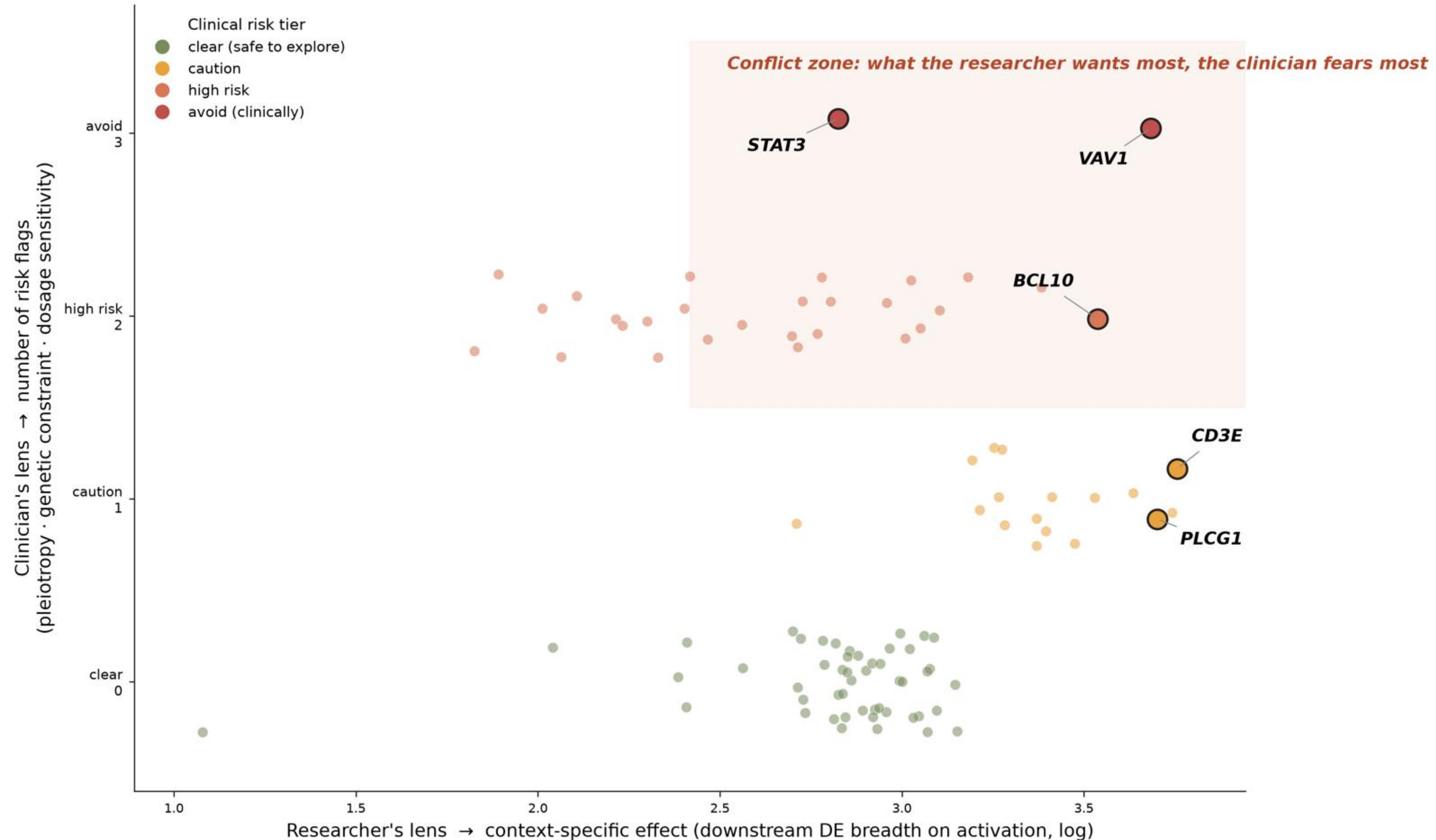
CD4 T-cell Perturb-seq target ranking

Top 12 targets · blue = immune candidate, red = broad-effect (pan-effect, needs isolation)



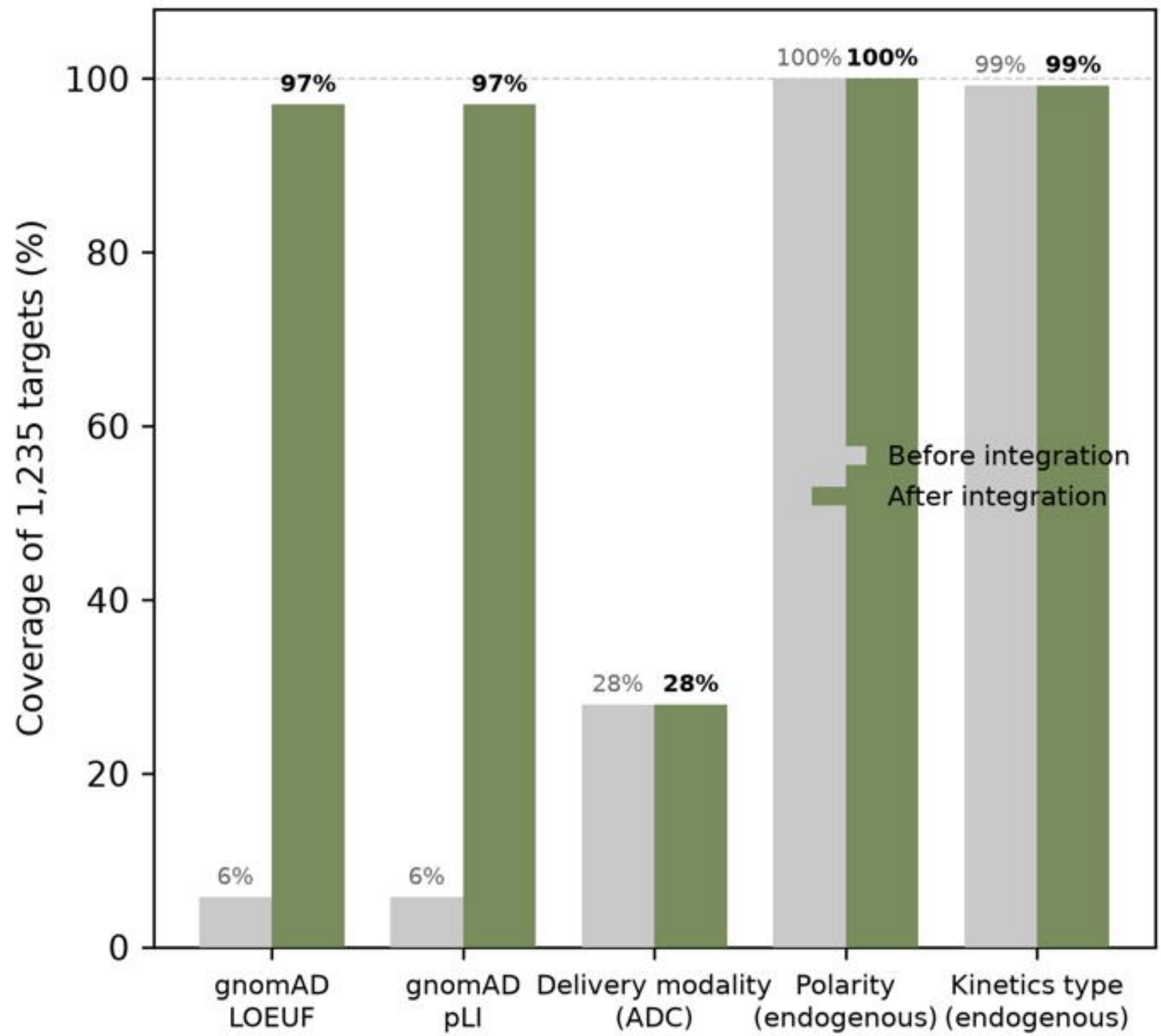
One dataset, two opposing first impressions

CD4⁺ T-cell Perturb-seq target-discovery platform · researchers want novel mechanism, clinicians want safety — the most interesting targets are often the most dangerous

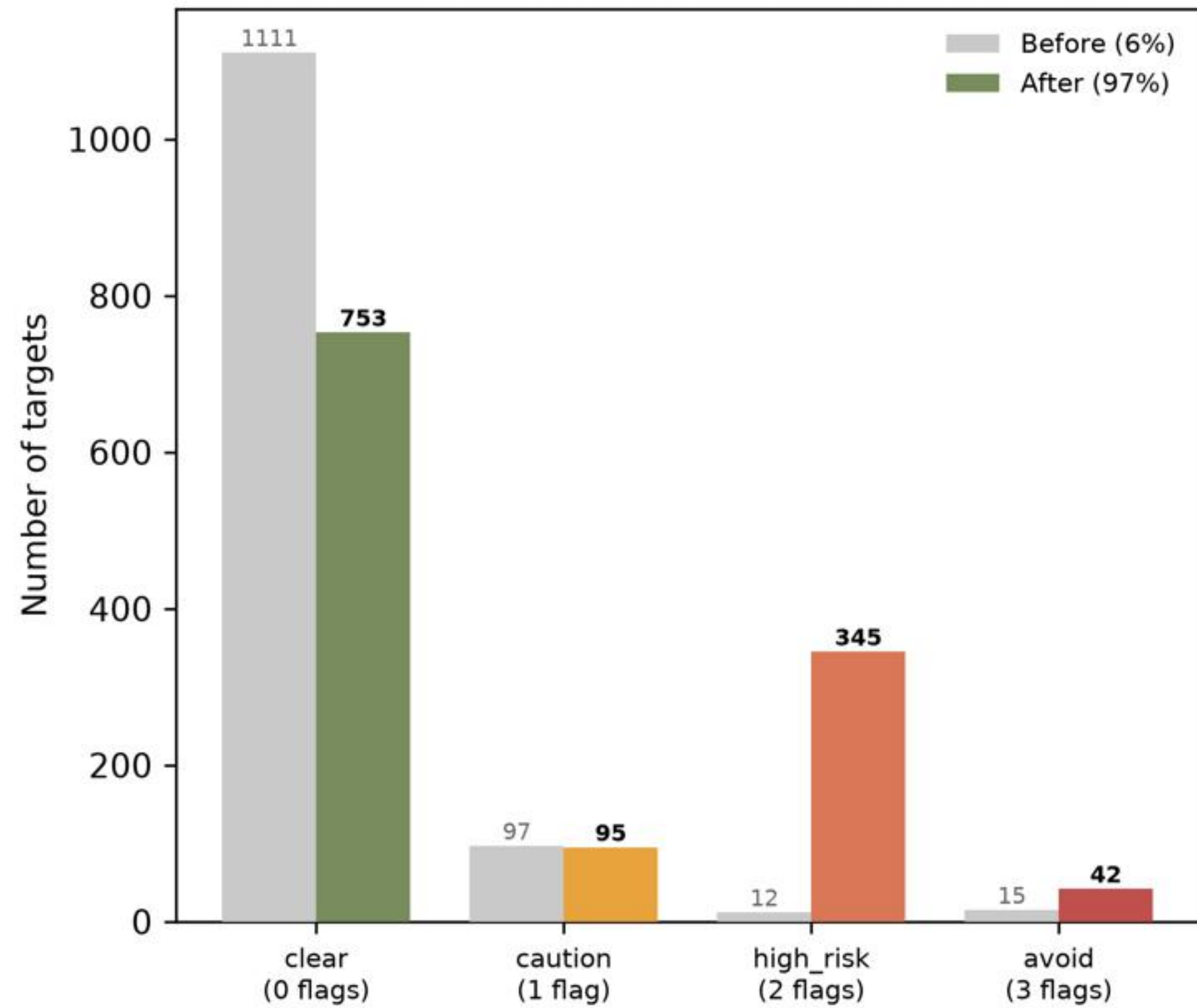


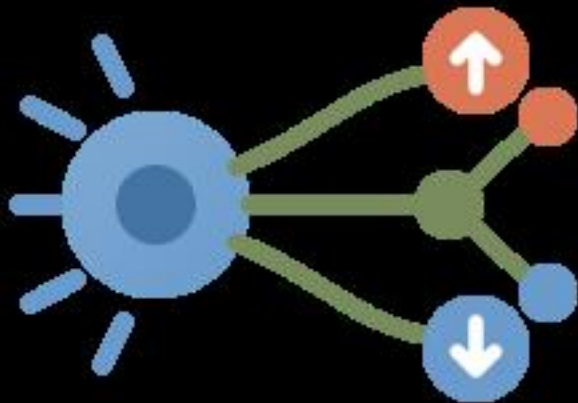
External evidence coverage completion — one bulk gnomAD download resolves the per-gene crawl ceiling

Evidence coverage completion: gnomAD 6% → 97% (bulk download)



Coverage gain reveals hidden risk: avoid tier 27 → 387







Pentabase